

Lattice-based Threshold Blind Signatures

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Abstract

Blind signatures are a central tool for privacy-preserving protocols. They allow users to obtain signatures from a signer without the signer seeing the signed message. For instance, it enables electronic cash: signatures correspond to coins which can be issued by the bank in a privacy-preserving manner via blind signing. To mitigate the risk of key compromise, *threshold* blind signatures allow the distribution of the signing key amongst N parties. While recent works have focused on improving the security of this primitive in the classical setting, no construction is known to date in the post-quantum setting.

We present the first construction of a threshold blind signature secure in the post-quantum setting, based on lattices. We prove its security under an interactive variant of the SIS assumption introduced in [Agrawal et al., CCS'22]. Our construction has a reasonable overhead of a factor of roughly $1.4\times$ to $2.5\times$ in signature size over comparable non-threshold blind signatures over lattices under heuristic but natural assumptions.

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1 Introduction

Blind signatures (BS), introduced by Chaum [Cha82], are a central building block for many privacy-preserving applications like Apple’s iCloud Private Relay [App21], voting systems [Cha82, CFN90], or Privacy Pass [DGS⁺]. What makes them so useful for privacy-preserving applications is their *blindness* property, *i.e.*, the guarantee that the signed message remains hidden from the signer and that a signer cannot trace back a signature to a signing session. At the same time, *one-more unforgeability* ensures that the user must interact with the signer to obtain a valid signature.

In recent years, blind signatures gained more attention with the growing interest in digital money. As of July 2025, central banks in 49 countries or territories are running pilot projects for so-called central bank digital currencies (CBDC) [AC25], including the European Central Bank [Ban25], the People’s Bank of China [Arr21], the Bank of Japan [oJ25], and the Reserve Bank of Australia [oA25]. In three countries, digital currency are already launched [AC25]. When rolled out by more countries, CBDC could potentially become an important pillar of how millions of small businesses and private individuals transact on a daily basis. One approach to building such a digital currency is to start from classical work on electronic cash [Cha88, Kuc10]. For example, the Swiss National Bank is currently investigating this approach for their CBDC [CCT21]. Roughly speaking, in this type of system, the central bank takes the role of the signer in the blind signature. To obtain a “coin”, the user chooses a random serial number and lets the bank blindly sign the serial number. To spend the coin, a user can present the serial number together with the valid signature¹. Intuitively, blindness guarantees that the bank cannot track the users’ spending behavior, and unforgeability prevents counterfeit money. In such a system, protecting the bank’s signing key is vital. If an adversary obtains the signing key, they can issue arbitrary amounts of coins (corresponding to *real money*) which could have dramatic effects on the whole economy of a country. A good way to eliminate this single point of failure is to distribute the trust to a consortium of bank servers. Threshold blind signatures (TBS) allow exactly this, *i.e.*, the signing key is shared among N servers, requiring a threshold T to sign collaboratively. Even if up to $T - 1$ servers are compromised, it is infeasible to forge coins.

However, the development of quantum computers threatens classical cryptographic assumptions underlying existing TBS constructions [CKM⁺23b, VZK03, LNÖ25]. Migrating to post-quantum (PQ) secure cryptography is essential, and in fact, in the US, the National Security Memorandum 10 (NSM-10) already mandates all government bodies to transition to PQ KEMs/signatures by 2035 [MPR⁺24]. Given the high criticality of a functioning monetary infrastructure for our daily lives, we think that investigating PQ building blocks for digital money is timely and important. Yet currently, no constructions for threshold blind signatures exist based on PQ assumptions.

One might consider leveraging generic techniques like secure multi-party computation (MPC) or Fully Homomorphic Encryption (FHE) [BGG⁺18, CCK23] to transform signature schemes into threshold variants. However, these approaches incur significant overhead and appear complex and inefficient to adapt to *interactive* blind signatures. Direct constructions from specific PQ assumptions are therefore highly desirable.

1.1 Our Contributions

This paper fills the aforementioned gap by presenting the first threshold blind signature, hereafter TBS, from lattices.

Security of TBS is based on One-More ISIS, an interactive variant of the Short Integer Solution (SIS) assumption introduced by Agrawal et al. [AKSY22]. We also give a less efficient variant proven secure under a weaker assumption, recently introduced by Baldimtsi et al. [BCGY24]. These assumptions allow for efficient blind signatures and are plausibly quantum secure, although they are not as well-studied as assumptions like MSIS and MLWE. For our security proof, we also generalize classical results on lattices to our setting (cf. Section F).

Two practical instantiations. We provide two instantiations of our construction with different trade-

¹Additional security measures must prevent spending the same coin twice.

offs², summarized in Table 1. Both instantiations rely on lattice assumptions and are therefore plausibly post-quantum secure.

1. The **compact** instantiation leverages the optimized proofs from LaBRADOR [BS23], combined with the zero-knowledge proof LNP [LNP22], to achieve highly compact signatures. This approach uses a more conservative parameter choice (following our proofs), but LaBRADOR’s superior proof efficiency more than compensates for the overhead. Although previous works used LaBRADOR [dPKN+25], it remains an open problem to efficiently implement these proofs with zero-knowledge.
2. The **easy to deploy** instantiation directly relies on the zero-knowledge proof LNP [LNP22], implemented by the Lazer library [LSS24]. This approach is designed for practical deployment and we provide a full implementation demonstrating its feasibility. This variant enjoys better computational efficiency as it does not rely on running both LaBRADOR and LNP.

Table 1: Parameters for our threshold blind signature, up to a threshold $T = 16$, where Q_S is the max. number of signatures issued, and $|\text{Sig}|$ is the signature size. We include in parentheses the signature size when not relying on the online-extraction heuristic for the NIZKs.

Set	Q_S	communication/party	$ \text{Sig} $
Compact	2^{50}	30.7 KB	~ 66 KB (~ 188 KB)
Easy to deploy	2^{50}	24.6 KB	115 KB (306 KB)

In comparison, the blind signature in [AKSY22]—which our threshold blind signature is based on—has signatures of size 45.19 KB³. Our compact (resp. deployable) instantiation has an overhead of a factor $1.4\times$ (resp. $2.5\times$) in signature size over the construction in [AKSY22], while providing support for distributed signing for up to $T = 16$ signers.

Implementation and evaluation. To demonstrate the practical viability of our approach, we provide a complete implementation of our easy-to-deploy instantiation, and benchmark it both on a single machine, and with geographically distributed machines. Our implementation builds upon the Lazer library [LSS24]. Our benchmarks show practical performance with threshold signing completing in 1.5 seconds (distributed) / 0.3 seconds (local), signature finalization in 2.3 seconds, and verification in 1.3 seconds. The source code for our implementations is made publicly available at <https://github.com/GuilhemN/Blind-Threshold-Signatures-Lattices>.

Security model. Our security model is based on the recent work by Lehmann, Nazarian, and Oezbay [LNÖ25] which strengthens the guarantees of one-more unforgeability for threshold blind signatures. While we do not achieve the strongest security notion in their hierarchy, our notion also gives strong guarantees, *e.g.*, it ensures that honest signers agree on the session and set of co-signers (cf. Section 2.2 for a discussion). We believe that it is an interesting open problem to provide a proof or construction under the strongest one-more unforgeability notion in [LNÖ25].

Denial-of-Service Measures. We note that our scheme does not support measures to protect against denial-of-service attacks within signing sessions. Common measures to remedy these are robustness [GJKR96] or identifiable aborts [AL07]. Our scheme supports neither, however, this is a common limitation for lattice-based threshold signatures. We also note that it is likely possible to extend our scheme to support these properties with known techniques ([ENP24] for robustness, [PKN+25] for identifiable aborts). We note that denial-of-service attacks are detectable if the signing session fails to output a valid signature, but the cheating party cannot be identified.

²In both our instantiations, we make heuristic but natural assumptions on the NIZKs, *i.e.*, that they are online-extractable. Without this assumption, the signature size increases by 120-190 KB. We refer to Section 6 for a discussion.

³We note that [AKSY22] does not rely on our online-extraction heuristic (cf. Section 6).

1.2 Technical Overview

We give a brief technical overview and discuss our design. To start, let us recall the design of classical threshold blind signatures (TBS).

Classical TBS. Known TBS constructions are based either on BLS [BLS04] or on Schnorr-variants [TZ22] (cf. [VZK03, CKM+23b, LNÖ25]).

Let us illustrate the design principles for TBS based on BLS [VZK03]. A *normal* BLS signature is computed as $\sigma = H(\mu)^{sk}$, where H is a random oracle mapping into a pairing-friendly prime-order group, and μ is the message.

- *Blind Signing.* To obtain a signature on μ , the user chooses $r \leftarrow \mathbb{Z}_p$ and sends $a = H(\mu)^r$ to the signer. The signer then answers with $b = a^{sk}$ and the user removes the blinding to obtain $\sigma = b^{1/r}$.
- *Threshold Signing.* The key sk can be distributed among N parties into shares $(sk_i)_{i \in [N]}$ with Shamir’s secret sharing. To sign, the i -th party computes $H(\mu)^{sk_i}$. The user obtains the final signature as $\sigma = \prod_i \sigma_i^{L_i}$, where L_i are the corresponding Lagrange coefficients.

Observe that in the case of BLS signatures, these two techniques are *compatible*. That is, by both blinding with a random exponent r and secret-sharing sk , the design of a TBS is immediate. Unfortunately, it turns out that the structure of existing lattice-based signatures makes the combination of blindness and threshold signing much more challenging than in the above example. We elaborate below.

Choosing a suitable signature. Our first step is to choose a suitable lattice-based signature scheme. However, no scheme is known to have both a threshold and blind signing protocol to the best of our knowledge.

Most lattice-based blind signatures [AKSY22, dPK22, BLNS23, JS24] follow the generic template by Fischlin [Fis06] instantiated with hash-and-sign signatures based on trapdoor sampling [GPV08, MP12]. While lattice trapdoors can be shared [BKP13], this comes at the cost of heavy interactive computation. A recent preprint improves efficiency [ALLW25], yet the parameters still scale with threshold T . While other blind signatures based on Fiat-Shamir exist [Rüc10, AEB20a, AEB20b, AHJ21, HKLN20, LNP22], these are either insecure [KLR24, DHP24] or heavily limit the number of issuable signatures [HKLN20, LNP22].

On the other hand, threshold signatures mostly follow the Fiat-Shamir template [DKM+24, KRT24, BKL+24, DOTT21, DOTT22]. Other candidates are known but are less efficient due to their reliance on heavier cryptographic tools such as threshold fully homomorphic encryption [BGG+18, ASY22].

An exception to the above is the recent scheme Plover [EEN+24]. Plover follows the hash-and-sign template and employs a variant of the trapdoor sampling from Eagle [YJW23]. Furthermore, the signing procedure can be thresholdized [ENP24]. Therefore, Plover is our choice to move forward. However, providing a blind issuance protocol for Plover is more challenging. We discuss our solution and challenges encountered along the way below.

Initial Design of our TBS. We combine Plover [EEN+24] with the techniques from Threshold Raccoon [DKM+24] and the blind signature of Agrawal et al. [AKSY22], as the blind signature [AKSY22] offers a good tradeoff between communication and signature size and [DKM+24] offers an efficient threshold signing protocol. Unfortunately, a naive combination of both techniques does not yield a provably secure scheme. To see this, let us give a simplified overview of the signing protocol in [AKSY22]. (Note that we present a simplified variant that closer resembles our final scheme for exposition.) The verification key is a matrix $\mathbf{A}$ and the secret key is a trapdoor $\mathbf{td}$ for $\mathbf{A}$.

1. $\text{User}_0(\mu)$: The user samples a short $\mathbf{x}'$ to mask the hash $H(\mu)$ of message μ . It sends $\mathbf{t}' = \mathbf{A}\mathbf{x}' + H(\mu)$ to the signer.
2. $\text{Signer}(\mathbf{td})$: The signer employs trapdoor sampling to compute $\mathbf{z}'$ such that $\mathbf{A}\mathbf{z}' = \mathbf{t}'$ via trapdoor $\mathbf{td}$. It sends $\mathbf{z}'$ to the user.
3. $\text{User}_1(\mathbf{z}')$: The user derives its signature on μ by computing a proof π certifying that $\mathbf{z} = \mathbf{z}' - \mathbf{x}'$ satisfies $\mathbf{A}\mathbf{z} = H(\mu)$.

As eluded to before, we replace the trapdoor sampling in *Signer* with *Plover*. Then, following the techniques from Threshold Raccoon [DKM⁺24], we obtain a blind threshold issuance protocol. We note that this is possible as *Plover*’s trapdoor can be shared via Shamir’s sharing as observed in [ENP24].

Proof issues for one-more unforgeability and our solution. While this protocol is simple and efficient, our use of the sampler from *Plover* introduces challenges to prove that it is secure, and we require additional techniques.

Indeed, the preimages $\mathbf{z}'$ output by *Plover* do *not* quite follow a discrete Gaussian distribution required in [AKSY22] to invoke the OM-ISIS assumption. Instead, *Plover* leverages a “noise flooding” technique to argue that although they output a distribution dependent on the secret trapdoor, the secret is sufficiently masked to still prove security under the LWE with hints assumption for uniform targets. Let us sketch the security one-more unforgeability (OMUF) proof in [AKSY22] based on the OM-ISIS assumption to see why this requirement appears. In OM-ISIS, the adversary is given $\mathbf{A}$ and obtains oracle access to syndrome queries and preimage queries which output a target $\mathbf{t}_i$ and a short Gaussian $\mathbf{A}$ -preimage of chosen value $\mathbf{t}$, respectively. The goal of the adversary is to find $\ell + 1$ short preimages for targets $\mathbf{t}_i$ making only ℓ preimage queries. Also, recall that for OMUF, the adversary is tasked to find $\ell + 1$ valid signatures making only ℓ signing queries.

In [AKSY22]’s OMUF proof, the reduction embeds $\mathbf{A}$ obtained from the OM-ISIS challenger into the verification key. To simulate signing queries, on the reception of the user’s target $\mathbf{t}'$, the reduction queries the preimage oracle with $\mathbf{t}'$ and sends the short preimage $\mathbf{z}'$ to the user. Further, on each $\mathbf{H}$ query μ , the reduction queries the syndrome oracle and programs the obtained $\mathbf{t}_i$ into $\mathbf{H}(\mu) = \mathbf{t}_i$. Notice that if the adversary provides $\ell + 1$ signatures π_i , the reduction can extract $\ell + 1$ short preimages $\mathbf{z}_i$ for syndromes $\mathbf{H}(\mu_i)$. As at most ℓ signing queries, and therefore at most ℓ preimage queries, were made, this breaks OM-ISIS. However, with *Plover*, this simulation strategy fails as $\mathbf{z}'$ does not follow a Gaussian distribution.

We solve this with a crucial modification to the protocol. The signer adds an additional matrix $\mathbf{D}$ to its verification key. Then, we modify the signer as follows.

- **Signer(td):** The signer samples short $\mathbf{x}$ and sets $\mathbf{t} = \mathbf{t}' + \mathbf{D}\mathbf{x}$, then it samples $\mathbf{z}'$ via trapdoor $\mathbf{td}$ such that $\mathbf{A}\mathbf{z}' = \mathbf{t}$.

For a properly distributed $\mathbf{x}$, the target $\mathbf{t}$ is randomized by $\mathbf{x}$ and appears pseudo-random. We rely on the fact that when $\mathbf{t}$ is uniformly distributed, *Plover* proves that the pre-images $\mathbf{z}'$ can be simulated without the knowledge of the trapdoor. Formally, this allows us to move the proof to a simulated world where we start by sampling the preimage $\mathbf{z}'$ without access to the *Plover* trapdoor and then sample $\mathbf{t}$.

The key insight is that by embedding a trapdoor for $\mathbf{D}$ in the signing key, we can still compute a consistent $\mathbf{x}$ for the adversary’s target $\mathbf{t}'$. Crucially, the trapdoor for $\mathbf{D}$ is only required in the security proof and does not need to be threshold-friendly and can output Gaussian pre-images, matching that of OM-ISIS, and we can employ the OM-ISIS oracle to simulate signing. Finally, observe that in our proof, we require that OM-ISIS finds preimages for $\mathbf{D}$ to simulate, however, the one-more unforgeability adversary provides us with $\mathbf{A}$ -preimages. This is readily solved by embedding $\mathbf{D}$ into $\mathbf{A}$.

Furthermore, we will let $\mathbf{x}$ be chosen by a random oracle in the final protocol to ensure honest sampling without a contribution from each party.

The naive choice of the trapdoor sampler for $\mathbf{D}$ is [MP12], however it would require increasing the width of the public matrix $\mathbf{A}$ by at least two ring elements (taking as basis $b = \sqrt{q}$ in [MP12]), leading to an increase of 25% in the width of the final public matrix, and roughly the same increase in signature size when avoiding succinct proof systems. Instead, we choose to leverage the compact gadget-based trapdoor sampler from [YJW23], and with careful design manage to only increase the width of $\mathbf{A}$ by a single ring element, while ensuring that $\mathbf{A}$ remains statistically close to uniform when we reduce from OM-ISIS (as needed for the reduction). This saves about 30 KB in signature size for our “easy to deploy” instantiation (cf. Table 1).

While the above sketch provides the intuition, the formal proof involves further technical details, such as reconciling the *approximate* sampling from Eagle [YJW23] with the *exact* sampling of the OM-ISIS game (see Theorem 2.8), and arguing statistical uniformity under the composite q required by the Eagle’s sampler (see Theorems 3.1 and 3.2). We refer to the main body for more details.

Blindness. Compared to OMUF, blindness is rather straightforward: $\mathbf{t}' = \mathbf{A}\mathbf{x}' + \mathbf{H}(\mu)$ is uniform under the (normal form) LWE assumption (over a sub-matrix of $\mathbf{A}$ that is not controlled by the signer) and the proof π can be simulated by zero-knowledge.

1.3 Related Work

To the best of our knowledge, there are only two other constructions of Threshold Blind Signatures: Snowblind [CKM⁺23b] and Threshold Blind BLS [VZK03]. In contrast to our work, both are based on classical hardness assumption and are thus susceptible to quantum attacks. Snowblind signatures consist of one group element and four scalars and each signer has to send two group elements and six scalars. Instantiating this with Curve25519 [Ber06], yields $|\text{Sig}| = 160$ bytes and 256 bytes of communication per party. Threshold Blind BLS signatures rely on pairings; a signature is one group element and each signer communicates one group element. Instantiating this with BLS12-381 [BLS03, LKWL24] yields $|\text{Sig}| = 96$ bytes and 96 bytes of communication/party.

2 Preliminaries

We write $x \leftarrow S$ for sampling x uniformly at random from the set S . For a distribution $\mathcal{D}$, we write $x \leftarrow \mathcal{D}$ for sampling x according to $\mathcal{D}$. By $\Delta(\mathcal{X}, \mathcal{Y})$ we denote the statistical distance of $\mathcal{X}$ and $\mathcal{Y}$. If one of the algorithms parses a message and parsing fails, we implicitly assume that the algorithm immediately outputs $\perp$.

2.1 Lattices

A *lattice* $\Lambda = \Lambda(\mathbf{B})$ of rank k and dimension n is the set of all integer linear combinations of the columns of $\mathbf{B} \in \mathbb{R}^{n \times k}$. We define the dual of Λ as $\hat{\Lambda} = \{c \in \mathbb{R}^n \mid \forall i, \langle c, B_i \rangle \in \mathbb{Z}\}$. For integers $n, q \in \mathbb{N}$ such that n is a power-of-two, we define $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$ and $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$. We write $\mathcal{R}_q^\times$ for the group of units in $\mathcal{R}_q$. For $\mathbf{a} \in \mathcal{R}_q^m$, set $\mathbf{a}^\perp := \{\mathbf{t} \in \mathcal{R}^m : \sum_i \mathbf{t}_i \mathbf{a}_i = 0 \pmod{q}\}$.

Definition 2.1. Let $\mathbf{x} \in \mathbb{R}^m$, $\Lambda \subseteq \mathbb{R}^m$ be a lattice, and $\sigma > 0$. Then, we set $\rho_\sigma(\mathbf{x}) := \exp(-\|\mathbf{x}\|_2^2/(2\sigma^2))$ and $\rho_\sigma(\Lambda + \mathbf{x}) := \sum_{\mathbf{y} \in \Lambda} \rho_\sigma(\mathbf{y} + \mathbf{x})$. The discrete Gaussian distribution with parameter σ over the lattice coset $\Lambda + \mathbf{x}$ is defined as $\mathcal{D}_{\Lambda + \mathbf{x}, \sigma}(\mathbf{y}) := \rho_\sigma(\mathbf{y})/\rho_\sigma(\Lambda + \mathbf{x})$.

We write $\mathcal{D}_\sigma$ for short if $\Lambda + \mathbf{x}$ is clear by context.

Definition 2.2. Let $\Lambda \subset \mathbb{R}^m$ be a lattice and $\hat{\Lambda}$ be its dual lattice. The δ -smoothing parameter $\eta_\delta(\Lambda)$ of Λ is defined as $\eta_\delta(\Lambda) := \inf s > 0 : \rho_{1/(\sqrt{2\pi}s)}(\hat{\Lambda} \setminus \{0\}) \leq \delta$. We also define a scaled variant $\eta'_\delta(\Lambda) := \frac{1}{\sqrt{2\pi}}\eta_\delta(\Lambda)$.

Lemma 2.3 (Simplified convolution lemma [Pei10, Thm. 4.5]). Let positive reals σ_1, σ_2 such that $\sigma_1^{-2} + \sigma_2^{-2} \leq \eta'_\delta(\mathbb{Z}^n)^{-2}$ for some positive $\delta \leq 1/4$. Then the distribution $\mathcal{D}_{\mathbb{Z}^n, \sigma_1} + \mathcal{D}_{\mathbb{Z}^n, \sigma_2}$ is within statistical distance 8δ of $\mathcal{D}_{\mathbb{Z}^n, \sqrt{\sigma_1^2 + \sigma_2^2}}$.

Like Plover [EEN⁺24], the hardness of our scheme is based on Hint-RLWE [KLSS23]. It is as hard as RLWE for appropriate parameters, see Section G.

Definition 2.4. Let q, Q be integers, χ, ξ be probability distributions over $\mathcal{R}_q^2$ and $\mathcal{C}$ be a probability distribution over $\mathcal{R}_q$. For an adversary $\mathcal{A}$ and security parameter λ , the advantage against the Hint Ring Learning With Errors problem is defined as

$$\text{Adv}_{\mathcal{A}, q, Q, \chi, \xi, \mathcal{C}}^{\text{Hint-RLWE}}(\lambda) := \left| \Pr[\mathcal{A}(\mathbf{a}, [\mathbf{a} \ 1] \cdot \mathbf{s}, ((\mathbf{c}_i, \mathbf{z}_i))_{i \in [Q]}) = 1] - \Pr[\mathcal{A}(\mathbf{a}, \mathbf{u}, ((\mathbf{c}_i, \mathbf{z}_i))_{i \in [Q]}) = 1] \right|,$$

where $\mathbf{a}, \mathbf{u} \leftarrow \mathcal{R}_q$, $\mathbf{s} \leftarrow \chi$ and for all $i \in [Q]$, $\mathbf{c}_i \leftarrow \mathcal{C}$, $\mathbf{r}_i \leftarrow \xi$ and $\mathbf{z}_i := \mathbf{c}_i \cdot \mathbf{s} + \mathbf{r}_i$. The $\text{Hint-RLWE}_{q, Q, \chi, \xi, \mathcal{C}}$ assumption states that for all PPT adversaries $\mathcal{A}$, $\text{Adv}_{\mathcal{A}, q, Q, \chi, \xi, \mathcal{C}}^{\text{Hint-RLWE}}(\lambda)$ is negligible.

For discrete Gaussians error, *i.e.*, $\chi = \mathcal{D}_\sigma$ and $\xi = \mathcal{D}_{\sigma'}$, we write $\text{Hint-RLWE}_{q,Q,\sigma,\sigma',\mathcal{C}}$. One recovers RLWE (without hint) by setting $Q := 0$. For RLWE, we write $\text{Adv}_{A,q,\chi}^{\text{RLWE}}(\lambda)$ short for the advantage term. Similarly, we define

Definition 2.5. Let ℓ, k, q be integers and χ be a probability distribution over $\mathcal{R}_q$. The advantage of the adversary $\mathcal{A}$ against the Module LWE problem is defined as

$$\text{Adv}_{A,q,\ell,k,\chi}^{\text{MLWE}}(\lambda) := |\Pr[1 \leftarrow \mathcal{A}(\mathbf{A}, \mathbf{A}\mathbf{s} + \mathbf{e})] - \Pr[1 \leftarrow \mathcal{A}(\mathbf{A}, \mathbf{b})]|,$$

where $(\mathbf{A}, \mathbf{b}, \mathbf{s}, \mathbf{e}) \leftarrow \mathcal{R}_q^{k \times \ell} \times \mathcal{R}_q^k \times \chi^\ell \times \chi^k$. The $\text{MLWE}_{q,\ell,k,\chi}$ assumption states that any efficient $\mathcal{A}$ has negligible advantage.

We render the one-more ISIS assumption, introduced by Agrawal et al. [AKSY22] and the randomized version of one-more ISIS, introduced by Baldimtsi et al. [BCGY24]. Randomized one-more ISIS works like normal one-more ISIS, but additionally, the boxed terms in the experiment is executed. Note that Agrawal et al. render the assumption over $\mathbb{Z}_q$ but later instantiate their system in the ring-setting. We therefore directly render the ring-version of the assumption with fixed dimension 4 and degree n . In particular, we interpret vectors in $\{\pm 1\}^{4n}$ as 4 polynomials with n coefficients that are ± 1 .

Definition 2.6. Let q, ℓ, β be functions of the security parameter λ . Let χ be a probability distribution over $\mathcal{R}^4$. The $\text{OM-ISIS}_{q,\chi,\ell,\beta}$ $\text{rOM-ISIS}_{q,\chi,\ell,\beta}$ assumption is defined using the following experiment.

1. The challenger $\mathcal{C}$ uniformly samples $\mathbf{A} := [1 \quad \mathbf{a} \quad \mathbf{b} \quad \mathbf{d}] \in \{1\} \times \mathcal{R}_q^3$ and $\mathbf{B} \in \{1\} \times \mathcal{R}_q^3$ and sends $\mathbf{A}$ and $\mathbf{B}$ to the adversary $\mathcal{A}$.
2. The adversary adaptively makes queries of the following types to the challenger in any order.
 - **Syndrome queries.** The adversary $\mathcal{A}$ requests $\mathcal{C}$ for a challenge, to which $\mathcal{C}$ replies with a uniformly random $\mathbf{t} \leftarrow \mathcal{R}_q$. We write S for the set of received challenges.
 - **Preimage queries.** The adversary $\mathcal{A}$ queries an element $\mathbf{t}' \in \mathcal{R}_q$, to which $\mathcal{C}$ replies with a short element $\mathbf{x} \leftarrow \chi$ such that $\mathbf{A} \cdot \mathbf{x} + \mathbf{B} \cdot \mathbf{y} = \mathbf{t}'$, where $\mathbf{y} \leftarrow \{\pm 1\}^{4n}$. We denote by ℓ the total number of preimage queries.
3. In the end, the adversary $\mathcal{A}$ outputs $\ell + 1$ tuples of the form $(\mathbf{x}_j, \mathbf{y}_j, \mathbf{t}_j)_{j \in [\ell+1]}$ with distinct $\mathbf{t}_j$.
4. The adversary wins if $\mathbf{A} \cdot \mathbf{x}_j + \mathbf{B} \cdot \mathbf{y}_j = \mathbf{t}_j, \|\mathbf{x}_j\|_2 \leq \beta$ and $\mathbf{t}_j \in S$ and $\mathbf{y}_j \in \{\pm 1\}^{4n}$ for all $j \in [\ell + 1]$.

We denote by $\text{Adv}_{A,q,\chi,\ell,\beta}^{\text{OM-ISIS}}$ $\text{Adv}_{A,q,\chi,\ell,\beta}^{\text{rOM-ISIS}}$ the advantage of $\mathcal{A}$ of winning the above game.

The rOM-ISIS assumption strengthens OM-ISIS against attacks from [AKSY22], which combine oracle preimages of basis vectors to obtain small preimages for other targets. Although these attacks require very short oracle outputs which our parameters already avoid, the randomization technique of [BCGY24] prevents them entirely.

2.1.1 Trapdoor Sampling

We will use the compact lattice gadgets from Yu, Jia, and Wang [YJW23]. We say that $(\mathbf{A}, \mathbf{T}) \in \mathbb{Z}_q^{n \times m} \times \mathbb{Z}^{m \times n}$ is a *matrix-approximate trapdoor pair* if there exists $\mathbf{P}, \mathbf{Q} \in \mathbb{Z}^{n \times n}$ such that $\mathbf{A}\mathbf{T} = \mathbf{P} \pmod{q}$ and $\mathbf{P}\mathbf{Q} = q\mathbf{I}_n$. For our purpose, it is sufficient to know that there exists an efficient algorithm ApproxPreSamp that takes as input such a matrix-approximate trapdoor pair $(\mathbf{A}, \mathbf{T}) \in \mathbb{Z}_q^{n \times m} \times \mathbb{Z}^{m \times n}$, a uniformly random vector $\mathbf{u} \in \mathbb{Z}_q^n$, and $r, s \in \mathbb{R}_{>0}$ and outputs a preimage $\mathbf{x}$ of $\mathbf{u}$ up to some small error $\mathbf{e}$. More formally, we have:

Theorem 2.7 ([YJW23, Theorem 2]). Let $\mathbf{P}, \mathbf{Q} \in \mathbb{Z}^{n \times n}$ such that $\mathbf{PQ} = q\mathbf{I}_n$. Let $(\mathbf{A}, \mathbf{T})$ be a matrix-approximate trapdoor pair, (r, s) satisfying $s^2 \geq (r^2 + \eta'_\delta(\mathbb{Z}^n)^2) \cdot (s_1(\mathbf{T})^2 + 1)$ and $r \geq \eta'_\delta(\Lambda(\mathbf{Q}))$, where $s_1(\mathbf{T})$ denotes the largest singular value of $\mathbf{T}$. Then if $\delta = \text{negl}(\lambda)$, We define $\varepsilon_{\text{sam}}^{\text{p}}$ as

$$\Delta \left(\left\{ (\mathbf{A}, \mathbf{x}, \mathbf{u}, \mathbf{e}) : \begin{array}{l} \mathbf{u} \leftarrow \mathbb{Z}_q^n \\ \mathbf{x} \leftarrow \text{ApproxPreSamp}(\mathbf{A}, \mathbf{T}, \mathbf{u}, r, s) \\ \mathbf{e} = \mathbf{u} - \mathbf{Ax} \pmod{q} \end{array} \right\}, \left\{ (\mathbf{A}, \mathbf{x}, \mathbf{u}, \mathbf{e}) : \begin{array}{l} \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, s} \\ \mathbf{e} \leftarrow \chi_{\mathbf{e}} \\ \mathbf{u} := \mathbf{Ax} + \mathbf{e} \pmod{q} \end{array} \right\} \right) \leq \text{negl}(\lambda),$$

with $\chi_{\mathbf{e}}$ the distribution of a (efficiently computed) representative of $\mathbf{v} \pmod{\Lambda(\mathbf{P})}$ when $\mathbf{v} \leftarrow \mathbb{Z}_q^n$.

Corollary 2.8. If $\mathbf{A} \cdot \mathbf{x}$ for $\mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, s}$ is statistically close to uniform, then the above distributions are statistically close to:

$$\{(\mathbf{A}, \mathbf{x}, \mathbf{u}, \mathbf{e}) : \mathbf{u} \leftarrow \mathbb{Z}_q^n, \mathbf{e} \leftarrow \chi_{\mathbf{e}}, \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, s} \text{ s.t. } \mathbf{Ax} = \mathbf{u} - \mathbf{e}\}$$

Proof. First, we can equivalently express $\{(\mathbf{A}, \mathbf{x}, \mathbf{u}, \mathbf{e}) : \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, s}, \mathbf{e} \leftarrow \chi_{\mathbf{e}}, \mathbf{u} := \mathbf{Ax} + \mathbf{e} \pmod{q}\}$ as $\{(\mathbf{A}, \mathbf{x}, \mathbf{u}, \mathbf{e}) : \mathbf{x}_1 \leftarrow \mathcal{D}_{\mathbb{Z}^m, s}, \mathbf{e} \leftarrow \chi_{\mathbf{e}}, \mathbf{u} := \mathbf{Ax}_1 + \mathbf{e} \pmod{q}, \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, s} \text{ s.t. } \mathbf{Ax} = \mathbf{u} - \mathbf{e}\}$.

Under the corollary assumption, $\mathbf{Ax}_1$ is close to uniform and thus $\mathbf{u}$ is close to uniform. Hence, we deduce that the previous distribution is close to

$$\{(\mathbf{A}, \mathbf{x}, \mathbf{u}, \mathbf{e}) : \mathbf{u} \leftarrow \mathbb{Z}_q^n, \mathbf{e} \leftarrow \chi_{\mathbf{e}}, \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m, s} \text{ s.t. } \mathbf{Ax} = \mathbf{u} - \mathbf{e}\}$$

□

As usual, elements $\mathbf{a} \in \mathcal{R}_q$ can be identified by their coefficient vector $\mathcal{V}(\mathbf{a}) \in \mathbb{Z}_q^n$ and we write $\mathcal{M}(\mathbf{a})$ for the anticirculant matrix $[\mathcal{V}(\mathbf{a}) \mathcal{V}(\mathbf{a} \cdot x) \dots \mathcal{V}(\mathbf{a} \cdot x^{n-1})] \in \mathbb{Z}_q^{n \times n}$. For brevity we will write $\text{ApproxPreSamp}([1 \ \mathbf{a} \ \mathbf{b} \ \mathbf{d}], [\mathbf{s}_1 \ \mathbf{s}_2 \ \mathbf{s}_3 \ 1]^\top, \mathbf{u}, r, s)$ for ring elements $\mathbf{a}, \mathbf{b}, \mathbf{d}, \mathbf{u}, \mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3 \in \mathcal{R}_q$ when we actually mean

$$\text{ApproxPreSamp}([\mathbf{I}_n \ \mathcal{M}(\mathbf{a}) \ \mathcal{M}(\mathbf{b}) \ \mathcal{M}(\mathbf{d})], \begin{bmatrix} \mathcal{M}(\mathbf{s}_1) \\ \mathcal{M}(\mathbf{s}_2) \\ \mathcal{M}(\mathbf{s}_3) \\ \mathbf{I}_n \end{bmatrix}, \mathcal{V}(\mathbf{u}), r, s),$$

and we can interpret the output $\tilde{\mathbf{x}} \in \mathbb{Z}^{4n}$ as vector $\mathbf{x} \in \mathcal{R}_q^4$.

2.2 Threshold Blind Signatures

We recall threshold blind signatures and discuss their security notions. We refer to Section B.4 for formal security definitions. Note that blind signatures can be seen as a special case of threshold blind signature with $T = N = 1$. For the sake of completeness, we also give a full definition of blind signatures in Section B.3.

We introduce threshold blind signatures following the definition of Lehmann et al. [LNÖ25] for $r = 3$ rounds of interaction.

Definition 2.9. A three-round Threshold Blind Signature is a tuple of PPT algorithms $\text{TBS} = (\text{Setup}, \text{KeyGen}, \text{User}_0, \text{User}_1, \text{User}_2, \text{User}_3, \text{Signer}_1, \text{Signer}_2, \text{Signer}_3)$, that work as follows:

- $\text{pp} \leftarrow \text{Setup}(1^\lambda)$: Given the security parameter as input, it outputs public parameters pp . We implicitly assume that all other algorithms take pp as input.
- $(\text{vk}, (\text{sk}_i)_{i \in [N]}, \text{aux}) \leftarrow \text{KeyGen}(N, T)$: Given the threshold parameters $N, T \in \mathbb{N}$ with $T < N$, it outputs a public key vk , N secret keys $(\text{sk}_i)_{i \in [N]}$ and some auxiliary information aux .
- $(\text{st}_0^U, \text{pm}_0^U) \leftarrow \text{User}_0(\text{vk}, \text{aux}, \mu, \text{SS})$: Given a public key vk , auxiliary information aux , a message to be signed μ and the set of signers SS that are going to sign the message, it outputs a first message of the user pm_0^U and a new user state st_0^U . The auxiliary information comes from key generation and is public.

- $(st_1^{(S,i)}, pm_1^{(S,i)}) \leftarrow \text{Signer}_1(i, sk_i, aux, ctx, pm_0^U)$: Given the index $i \in [N]$ of the signer, the i -th secret key sk_i , the auxiliary information aux , the context information ctx and the user's first message pm_0^U , it outputs a new state for the i -th signer $st_1^{(S,i)}$ and a message from that signer $pm_1^{(S,i)}$.
- $(st_1^U, pm_1^U) \leftarrow \text{User}_1(st_0^U, (pm_1^{(S,i)})_{i \in SS})$: Given the user's state from the previous round st_0^U and the messages $(pm_1^{(S,i)})_{i \in SS}$ of all signers in the signer set SS , it outputs a new state st_1^U and a new message pm_1^U .
- $(st_2^{(S,i)}, pm_2^{(S,i)}, ssid_i, SS_i) \leftarrow \text{Signer}_2(i, st_1^{(S,i)}, pm_1^U)$: Given the old state of the i -th signer $st_1^{(S,i)}$ and the second message from the user pm_1^U , it outputs a new state $st_2^{(S,i)}$, a next message $pm_2^{(S,i)}$, a subsession identifier⁴ $ssid_i$ and a signer set SS_i that the i -th signer thinks it is running the protocol for.
- $(st_2^U, pm_2^U) \leftarrow \text{User}_2(st_1^U, (pm_2^{(S,i)})_{i \in SS})$: Given the previous state of the user st_1^U and the messages from all the signers $(pm_2^{(S,i)})_{i \in SS}$, it outputs an updated state st_2^U and a next user message pm_2^U .
- $(pm_3^{(S,i)}, aux_{sig}) \leftarrow \text{Signer}_3(i, st_2^{(S,i)}, pm_2^U)$: Given an index $i \in [N]$, the previous state of the signer $st_2^{(S,i)}$ and the user's message pm_2^U it outputs the final message $pm_3^{(S,i)}$ of the signer with auxiliary information aux_{sig} . The auxiliary information is only needed in the security proof of the protocol and unforgeability holds even when the adversary sees aux_{sig} .
- $sig/\perp \leftarrow \text{User}_3(st_2^U, (pm_3^{(S,i)})_{i \in SS})$: Given the previous state of the user st_2^U and the messages $pm_3^{(S,i)}$ from all signers $i \in SS$ it outputs either $\perp$ to indicate failure or a signature sig if no failure occurred.
- $1/0 \leftarrow \text{Vfy}(vk, \mu, sig)$: Given the public key vk , a message μ and a signature sig , it outputs 1 if the signature verifies for that message and else it outputs 0.

We remark that in our definition Signer_2 outputs $(ssid_i, SS_i)$ instead of Signer_3 . The values $(ssid_i, SS_i)$ are relevant for the definition of one-more unforgeability which differs from the definition in [LNÖ25]. We discuss this modification below.

2.2.1 Security Properties

We sketch correctness, blindness and one-more unforgeability.

Correctness If all signers and the user follow the protocol honestly, the user obtains a signature that verifies with respect to its message and the signer's public key.

One-more Unforgeability Defining unforgeability for threshold blind signatures is rather complex since the message is hidden and the signing protocol is distributed amongst T signers, it is non-trivial to count the number of issued signatures. We sketch one-more unforgeability following the strongest definition OMUF-3 within the hierarchy given in [LNÖ25] except for some small modifications. Intuitively, OMUF-3 as defined in [LNÖ25] states that a malicious user cannot produce more than ctr valid signatures for distinct signature-message pairs, where ctr is the number of signatures that can be obtained from honest runs of the protocol. In OMUF-3, the game keeps track of ctr through a counter: the counter is incremented whenever $T - C$ honest shares are issued through the signing oracle for the same session $ssid$ and intended set SS of co-signers, where C is the number of corrupted users.

Let us discuss our modification:

1. In [LNÖ25], the session $ssid_i$ and set of co-signers SS_i with respect to which signer i signs is fixed in the final round. If $T - C$ signers ran the last-round signing protocol for identical $ssid$ and SS , then ctr is incremented. In our definition, these steps happen in round 2 instead of round 3. That is, the

⁴The $ssid$ will typically contain a part of the transcript together with context information that identifies the running session.

attacks that are not prevented are those where an adversary aborts the session at the end of round 2 and uses the information learned to compute a forgery.

While this weakens the definition slightly, counting signatures one round early is a common limitation in classical and post-quantum threshold signatures (e.g., [DKM⁺24, CKM23a]). Here, this weakening is required because the signature is already determined after round 2 in the security proof, as we move to a game where the signature is sampled first to show that it does not leak secret information, and other information for the last round is derived from that signature. We remark that our definition still ensures *session agreement* (i.e., which blind signing sessions go together) and *signer set agreement* (i.e., signers agree on the signing set $\mathbf{SS}$ and can, e.g., choose to avoid co-signing with suspicious parties). We refer to [LNÖ25] for more details.

2. Our definition ensures that it is hard to produce more than ctr valid signatures for *distinct messages*, but not distinct signature-message pairs. This definition is standard for lattice-based blind signatures [Fis06, AKSY22, dPK22, BLNS23, JS24]. That is, we do achieve standard unforgeability but not strong unforgeability. The reason is that all the above schemes output a NIZK as final signature. Re-computing the NIZK for the same statement but with different randomness trivially yields a different signature for the same message.
3. Our definition allows the signers to output some auxiliary information $\mathbf{aux}_{\text{sig}}$ along with their last message. This value is only needed in the security proof. The underlying Plover [EEN⁺24] signature has the same property. Unforgeability holds even when the adversary sees $\mathbf{aux}_{\text{sig}}$.

Blindness Finally, blindness ensures that an adversary cannot link signatures (obtained through the threshold blind signing protocol) to its issuance session.

Non-Interactive Zero Shares

We make use of the ZeroShare notation introduced in [KRT24] for the masking technique of Raccoon [DKM⁺24, EKT24]. Let $H_{\text{mask}} : \{0, 1\}^\lambda \rightarrow \mathcal{R}_q$ be modeled as a random oracle. The i -th signer is given a tuple of random strings $(\text{seed}_{i,j}, \text{seed}_{j,i})$ that it shares with the j -th other signer. That is, the i -th signer gets $\vec{\text{seed}}_i := (\text{seed}_{i,j}, \text{seed}_{j,i})_{j \in [N]}$. and we write $\vec{\text{seed}}_{[\mathbf{SS}]}$ for $(\text{seed}_{i,j}, \text{seed}_{j,i})_{j \in \mathbf{SS}}$. We define

$$\text{ZeroShare}(\vec{\text{seed}}_i[\mathbf{SS}], x) := \sum_{j \in \mathbf{SS} \setminus \{i\}} (\mathbf{m}_{i,j} - \mathbf{m}_{j,i}), \quad (1)$$

where $\mathbf{m}_{i,j} = H_{\text{mask}}(\text{seed}_{i,j}, x)$ for $i, j \in \mathbf{SS}$. Observe that

$$\sum_{i \in \mathbf{SS}} \text{ZeroShare}(\vec{\text{seed}}_i[\mathbf{SS}], x) = 0.$$

2.3 Non-interactive Zero-Knowledge Proofs

We recall the notion of non-interactive zero-knowledge proofs (NIZK) in the random oracle model (ROM) with the common reference string $\text{crs} \in \{0, 1\}^\ell$. Let $\mathcal{R}$ be an NP-relation and denote $\mathcal{L}_{\mathcal{R}} = \{\mathbf{x} \in \{0, 1\}^* \mid \exists \mathbf{w} \text{ s.t. } (\mathbf{x}, \mathbf{w}) \in \mathcal{R}\}$ the language induced by $\mathcal{R}$.

Definition 2.10. An NIZK Π for the NP-relation $\mathcal{R}$ consists of PPT algorithms (Setup, Prove, Verify) with access to random oracle H defined as follows:

- $\text{crs} \leftarrow \text{Setup}^H(1^\lambda)$: Outputs a common reference string.
- $\pi \leftarrow \text{Prove}^H(\text{crs}, \mathbf{x}, \mathbf{w})$: Given $\text{crs} \in \{0, 1\}^\ell$, statement $\mathbf{x}$ and witness $\mathbf{w}$ with $(\mathbf{x}, \mathbf{w}) \in \mathcal{R}$, it outputs a proof π .

- $0/1 \leftarrow \text{Verify}^H(\text{crs}, \mathbb{x}, \pi)$: Given $\text{crs} \in \{0, 1\}^\ell$, statement $\mathbb{x}$ and proof π , it outputs a bit.

We sometimes omit H and crs as parameters if it is clear by context. Below, we recall the correctness, zero-knowledge, and straightline extractability properties of NIZKs.

Definition 2.11 (Correctness). An NIZK Π for a relation $\mathcal{R}$ is correct, if for any $(\mathbb{x}, \mathbb{w}) \in \mathcal{R}$, we have

$$\Pr[\pi \leftarrow \text{Prove}^H(\text{crs}, \mathbb{x}, \mathbb{w}) : \text{Verify}^H(\text{crs}, \mathbb{x}, \pi) = 1] = 1 - \text{negl}(\lambda).$$

Definition 2.12 (Zero-knowledge). An NIZK Π for a relation $\mathcal{R}$ is zero-knowledge if there exists a PPT simulator $\text{Sim} = (\text{Sim}_0, \text{Sim}_1)$, where Sim_0 and Sim_1 have a shared state, such that for any PPT adversary $\mathcal{A}$, we have

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{ZK}}(\lambda) := |\Pr[\mathcal{A}^{\text{H}, \text{Prove}}(\text{crs}) = 1] - \Pr[\mathcal{A}^{\text{Sim}_0, \mathcal{S}}(\text{crs}) = 1]|,$$

where $\text{crs} \leftarrow \{0, 1\}^\ell$, and Prove and $\mathcal{S}$ are oracles that on input $(\mathbb{x}, \mathbb{w})$ return $\perp$ if $(\mathbb{x}, \mathbb{w}) \notin \mathcal{R}$ and otherwise return $\text{Prove}^H(\text{crs}, \mathbb{x}, \mathbb{w})$ or $\text{Sim}_1(\text{crs}, \mathbb{x})$, respectively.

Definition 2.13 (Straightline Extractability). An NIZK is straightline-extractable if for all PPT adversaries $\mathcal{A}$, there exists a PPT simulator ExtSetup and extractor Ext , such that the following notions hold.

CRS Indistinguishability. For any PPT adversary $\mathcal{A}$,

$$\begin{aligned} \text{Adv}_{\Pi, \mathcal{A}}^{\text{CRS}}(\lambda) &= |\Pr[\text{crs} \leftarrow \{0, 1\}^\ell : \mathcal{A}^H(\text{crs}) = 1] \\ &\quad - \Pr[(\overline{\text{crs}}, \text{td}) \leftarrow \text{ExtSetup}(1^\lambda) : \mathcal{A}^H(\overline{\text{crs}}) = 1]| = \text{negl}(\lambda). \end{aligned}$$

Straightline Extractability. For any $Q_H = \text{poly}(\lambda)$ and PPT adversary $\mathcal{A}$ that makes at most Q_H random oracle queries with

$$\Pr \left[\begin{array}{c} (\overline{\text{crs}}, \text{td}) \leftarrow \text{ExtSetup}(1^\lambda), \\ ((x_i, \pi_i))_{i \in [Q_S]} \leftarrow \mathcal{A}^H(\overline{\text{crs}}) \end{array} : \begin{array}{c} \forall i \in [Q_S] : \\ \text{Verify}^H(\text{crs}, x_i, \pi_i) = 1 \end{array} \right] \geq \mu(\lambda), \quad (2)$$

it holds that

$$\Pr \left[\begin{array}{c} (\overline{\text{crs}}, \text{td}) \leftarrow \text{ExtSetup}(1^\lambda), ((x_i, \pi_i))_{i \in [Q_S]} \leftarrow \mathcal{A}^H(\overline{\text{crs}}), \\ (w_i \leftarrow \text{Ext}(\overline{\text{crs}}, \text{td}, x_i, \pi_i))_{i \in [Q_S]} \end{array} : \begin{array}{c} \forall i \in [Q_S] : (x_i, w_i) \in \mathcal{R} \\ \wedge \text{Verify}^H(\overline{\text{crs}}, x_i, \pi_i) = 1 \end{array} \right] \geq \frac{\mu(\lambda) - \text{negl}(\lambda)}{\text{poly}(\lambda, Q_H)(3)}.$$

3 Supporting Results on Lattices

In this section, we present some simple results and generalizations of statements that we need later. The results might be of independent interest.

Regularity Bounds

We provide a generalization of the regularity bound from Stehlé and Steinfeld [SS11] for composite q .

Theorem 3.1 (Generalization of [SS11, Theorem 3.1] to composite q). Let $n \geq 8$ be a power-of-two, and $q = q_1 \cdot \dots \cdot q_\ell$, where $q_1, \dots, q_\ell$ are distinct primes ≥ 5 . Let $m \geq 2, \varepsilon > 0, \delta \in (0, 1/2)$ and $\mathbf{s} \leftarrow \mathcal{D}_{\mathbb{Z}^{mn}, \sigma}$, with $\sigma \geq 1/\sqrt{2\pi} \cdot \ln(2mn(1 + 1/\delta))/\pi \cdot \sqrt{n} \cdot q^{1/m+\varepsilon}$. Then we have for $\delta' := 2^{\ell n}(5/4)^{nm\ell}q^{-nm\varepsilon}$

$$\Delta \left(\left(\mathbf{a}_1, \dots, \mathbf{a}_m, \sum_{i \leq m} \mathbf{s}_i \mathbf{a}_i \right), \mathcal{U}((\mathcal{R}_q^\times)^m \times \mathcal{R}_q) \right) \leq 2\delta + \delta'.$$

We provide a proof in Section F.

Ensuring Invertibility

In our construction, we will not pick $\mathbf{a} \in \mathcal{R}_q$ but as invertible element $\mathbf{a} \in \mathcal{R}_q^\times$. Similarly, we will rejection-sample $\mathbf{b}$ to ensure that it is invertible. The following lemma ensures that Hint-MLWE with invertible $\mathbf{a}$ and OM-ISIS with invertible $[1 \ \mathbf{a} \ \mathbf{b}]$ is still hard and that $\mathbf{b}$ is not resampled too often:

Lemma 3.2 (Adaptation of [LPR13, Claim 2.25] for $f = x^n + 1$). *Let $q = \prod_{i=1}^k p_i^{f_i}$ the prime factorization of q . The fraction of invertible elements in $\mathcal{R}_q$ is at least $\prod_{i=1}^k (1 - (1/p_i))^n$.*

Proof. By the Chinese Remainder theorem, we deduce that the proportion of units in $\mathcal{R}_q$ is equal to the product of the proportion of units in $\mathcal{R}_{p^{f_i}}$. Let us thus consider one prime $p \mid q$, and its multiplicativity $f \geq 1$.

Now, we can factor the polynomial $x^n + 1 \bmod p^f$ as a product of distinct irreducible polynomials: $x^n + 1 \bmod p^f = \prod_{j=1}^\ell \phi_j$. We apply again the Chinese Remainder Theorem to deduce that the proportion of units in $\mathcal{R}_{p^f} = \mathbb{Z}_{p^f}/(x^n + 1)$ is the product of the proportion of units in the $\mathbb{Z}_{p^f}/(\phi_j)$.

This proportion is thus $\prod_{i=1}^\ell \frac{(p^f)^{\deg(\phi_i)} - (p^{f-1})^{\deg(\phi_i)}}{(p^f)^{\deg(\phi_i)}} = \prod_{i=1}^\ell (1 - \frac{1}{p^{\deg(\phi_i)}})$. In particular, it is $\geq (1 - \frac{1}{p})^n$.

Going back to q , we deduce that the proportion of units in $\mathcal{R}_q$ is at least $\prod_{i=1}^k (1 - \frac{1}{p_i})^n$ $\square$

We will write $\text{Adv}^{\text{Hint-RLWE-}\times}$ and $\text{Adv}^{\text{RLWE-}\times}$ for the advantages in the respective experiments where $\mathbf{a}$ is no longer sampled from $\mathcal{R}_q$ but instead from $\mathcal{R}_q^\times$. It is easy to see that these assumptions are implied by their analog without invertibility condition due to Theorem 3.2.

Rounding and decomposition

We define decomposition similar to [EEN⁺24], except that we fix a small issue with the distribution of the composed values.

Let $\beta \in \mathbb{N}, \beta \geq 2$ be an even number. Any integer $x \in \mathbb{Z}^+$ can be decomposed uniquely as $x = \beta \cdot x_1 + x_2$, where $0 \leq x_1 \leq \lfloor x/\beta \rfloor, 0 \leq x_2 < \beta$.

We use this decomposition over $\mathbb{Z}$ to decompose elements in $\mathbb{Z}_q$, for odd q . Note $\text{Decomp}_\beta : \mathbb{Z}_q \rightarrow \mathbb{Z} \times \mathbb{Z}$ the randomized function which takes as input $y \in \mathbb{Z}_q$, samples $\varepsilon \leftarrow \{-1, 1\}$. Then define $x = \varepsilon \cdot y + \lfloor (q-1)/2\beta \rfloor \cdot \beta + \beta/2$. We consider its unique representative $\bar{x} \in \{0, \dots, q-1\}$ and decompose it as $\bar{x} = \beta \cdot x_1 + x_2$ as described above. Finally, we return the pair $(y_1 = \varepsilon \cdot (x_1 - \lfloor (q-1)/2\beta \rfloor), y_2 = \varepsilon \cdot (x_2 - \beta/2)) \in \mathbb{Z} \times \mathbb{Z}$. It verifies $y = \beta \cdot y_1 + y_2 \bmod q$.

We extend Decomp_β to polynomials in $\mathbb{Z}_q[x]$, by applying the function to each coefficient. For $c \leftarrow \mathbb{Z}_q$ and $(c_1, c_2) := \text{Decomp}_\beta(c)$, we have $|c_1| \leq \lfloor (q-1)/2\beta \rfloor$, $\mathbf{E}[c_1] = 0$ and $\mathbf{E}[c_1^2] \leq (M^2 - 1)/12$ for $M = 2\lfloor (q-1)/2\beta \rfloor + 1$.

Note that in [EEN⁺24], $\mathbf{E}[c_1]$ was not 0. This is solved by introducing the randomization ε . This was in particular flawing their reduction of Hint-RLWE to RLWE, see Section G.

4 Blind Signature

We first present our non-thresholdized blind signature BS_B , where we denote by B the norm-bound used in the NIZK. We will sometimes omit B when it's clear from the context. Later in Section 5, we will show how to lift the blind signature BS_B to a threshold blind signature TBS_B . Looking ahead, we will prove security of TBS_B by reducing to security of BS_B .

Variants We provide two variants of our blind signature:

1. Our main variant BS_B which we prove secure under the one-more ISIS assumption. This is the most efficient construction.
2. Our second variant $\text{BS}_B^{\$}$ which we prove secure under the weaker randomized one-more ISIS assumption at the cost of a slight increase in signature size and communication.

In the following, BS_B (resp. $\text{BS}_B^\S$) is obtained by ignoring (resp. considering) all $\boxed{\text{boxed terms}}$.

Parameters and Preparations We recall some preliminaries and summarize the parameters of our construction. An overview of all used parameters of the construction is given in Section 4. We instantiate the probability distributions $\chi_{\text{sk}}, \chi_{\text{S},\text{x}}, \chi_{\text{U},\text{x}}, \chi_{\text{rnd}}, \chi_{\text{e}}, \chi_{\text{s}'}$ as discrete Gaussian distributions over $\mathcal{R}_q$ with respective standard deviations $\sigma_{\text{sk}}, \sigma_{\text{S},\text{x}}, \sigma_{\text{U},\text{x}}, \sigma_{\text{rnd}}, \sigma_{\text{e}}, \sigma_{\text{s}'} > 0$.

We select concrete standard deviations as follow:

- $\delta = \text{negl}(\lambda)$, ε such that $2^{\ell n}(5/4)^{nm\ell}q^{-nm\varepsilon} = \text{negl}(\lambda)$ for $m = 3$, and ℓ the number of prime factors in q
- $\sigma_{\text{s}'} \geq 1/\sqrt{2\pi} \cdot \ln(2mn(1+1/\delta))/\pi \cdot \sqrt{n} \cdot q^{1/m+\varepsilon}$, i.e., it fulfills the condition of the regularity Theorem 3.1.
- $\sigma_{\text{S},\text{x}}^2 \geq \sigma_{\text{s}'}^2 + \eta'_\delta(\mathbb{Z}^n)^2 \cdot (q^2/\tilde{p}^2 + 1) \cdot (6n(\lambda \log(2) + \log(6n))\sigma_{\text{s}'}^2 + 1)$, for the conditions of Theorem 2.7.

We provide an overview of the parameters used in our threshold blind signature TBS, with a detailed explanation of each parameter, in Table 2. For a comprehensive discussion on parameter selection and security analysis, please refer to Section 6 and Section H.

NIZK and Oracles Let Π be a NIZK with random oracle H_Π for the relation $\mathcal{R}$ defined as

$$\mathcal{R} = \left\{ (\mathbb{x}, \mathbb{w}) \mid \mathbf{Ax} + \mathbf{By} = \mathbf{u}, \|\mathbf{x}\|_2 \leq B, \mathbf{y} \in \{\pm 1\}^{4n} \right\}, \quad (4)$$

where $\mathbb{w} = (\mathbf{x}, \mathbf{y}) \in \mathcal{R}_q^4$ and $\mathbb{x} = (\mathbf{A}, \mathbf{B}, \mathbf{u})$ with $\mathbf{A}, \mathbf{B} \in \mathcal{R}_q^{1 \times 4}$ and $\mathbf{u} \in \mathcal{R}_q$. In addition, let $\text{H}_\mu : \{0, 1\}^* \rightarrow \mathcal{R}_q$ be a random oracle.

4.1 Overview and Construction

The construction is in Figure 1. We present BS_B first, then discuss the modifications to obtain $\text{BS}_B^\S$. The user first hashes the message μ to sign with the verification key and obtains $\mathbf{u} \in \mathcal{R}_q$. Next, the user blinds this hash by sampling a short $\mathbf{x}'$ and sends $\mathbf{t}' = \mathbf{u} + \mathbf{Ax}'$ to the signer. The RLWE sample $\mathbf{Ax}'$ blinds $\mathbf{u}$ so the signer does not learn anything about μ .

The signer shifts the user's message $\mathbf{t}'$ by adding $\mathbf{Ax} + \tilde{\mathbf{e}}$ and computes a Plover-style [EEN⁺24] signature on the shifted user message $\mathbf{t}$ (this corresponds to the randomization $\mathbf{Dx}$ presented in the technical overview). Adding the shift $\mathbf{Ax} + \tilde{\mathbf{e}}$ to the user's $\mathbf{t}'$ may look superfluous as $\mathbf{t}'$ itself is already the shift of the hashed message $\mathbf{u}$ and the Plover-style signing also shifts $\mathbf{t}'$ by $\mathbf{w}$. However, this $\mathbf{Ax} + \tilde{\mathbf{e}}$ gives the reduction from OM-ISIS enough control over the signed value $\mathbf{t}$ in the security proof. Intuitively, this corresponds to controlling the random oracle in Plover's security proof. But note that the user's blinding $\mathbf{Ax}'$ makes it impossible for a reduction to link messages $\mathbf{t}'$ with corresponding hash queries $\text{H}_\mu(\mu, \text{vk})$. That makes it necessary to give the reduction more leeway using $\mathbf{Ax} + \tilde{\mathbf{e}}$.

The signer then computes the decomposition $\mathbf{c}_1, \mathbf{c}_2$ of $\mathbf{t} - \mathbf{w}$, where $\mathbf{w}$ is some random shift. Next, it sets $\mathbf{z}$ such that $\mathbf{Az} = \mathbf{t}$ using the signing key $(\mathbf{s}, \mathbf{e})$. Note that $\mathbf{z}_2$ and $\mathbf{c}_1$ uniquely define $\mathbf{z}_1$ so the signer does not have to explicitly compute it. The signer outputs the short $\mathbf{x}, \tilde{\mathbf{e}}$ that was for shifting, together with the Plover-style signature $(\mathbf{z}_2, \mathbf{c}_1)$.

Finally, the user computes a witness $\mathbb{w} = [\mathbf{z}_1' \ \mathbf{z}_2 \ \mathbf{c}_1 \ 0]^\top - (\mathbf{x} + \mathbf{x}')$, where $\mathbf{z}_1 = \mathbf{u} + \mathbf{A}(\mathbf{x} + \mathbf{x}') - \mathbf{az}_2 - \mathbf{bc}_1$. The user checks if the computed witness has a small norm and if that is the case it computes the final signature as a proof π for the statement that $\mathbb{w}$ is short and satisfies $\mathbf{Aw} = \text{H}_\mu(\mu, \text{vk})$.

Variant $\text{BS}_B^\S$. In our second variant $\text{BS}_B^\S$, the parameters pp contain an additional matrix $\mathbf{B} \in \mathcal{R}_q^{1 \times 4}$. Furthermore, the signer computes $\mathbf{t} = \mathbf{t}' + \mathbf{Ax} + \mathbf{By} + \tilde{\mathbf{e}}$ for $\mathbf{y} \leftarrow \{0, 1\}^{4n}$ instead and also outputs $\mathbf{y}$ to the user. The user computes its proof using $\mathbf{y}$ as additional witness. Looking ahead, this modification allows to simulate signing via an randomized one-more ISIS oracle in the OMUF proof.

Table 2: Overview of parameters of our blind signature.

Parameter	Explanation
$\mathcal{R}_q$	Polynomial ring $\mathcal{R}_q = \mathbb{Z}[X]/(q, X^n + 1)$
χ_{sk}	Distribution used for the secret key $(\mathbf{s}, \mathbf{e})$, instantiated as $\mathcal{D}_{\mathcal{R}_q, \sigma_{\text{sk}}}$.
$\chi_{\text{S}, \mathbf{x}}$	Distribution used for Signer-side blinding $\mathbf{t}$, instantiated as $\mathcal{D}_{\mathcal{R}_q, \sigma_{\text{S}, \mathbf{x}}}$.
$\chi_{\text{U}, \mathbf{x}}$	Distribution used for User-side blinding $\mathbf{t}'$, instantiated as $\mathcal{D}_{\mathcal{R}_q, \sigma_{\text{U}, \mathbf{x}}}$.
χ_{rnd}	Distribution used for hiding $(\mathbf{s}, \mathbf{e})$ in $\mathbf{z}_1, \mathbf{z}_2$, instantiated as $\mathcal{D}_{\mathcal{R}_q, \sigma_{\text{rnd}}}$.
β	Divider between high/low order bits of $\mathbf{c}$
$\chi_{\mathbf{e}}$	Error distribution of the approximate trapdoor sampler instantiated as $\mathcal{D}_{\mathcal{R}_q, \sigma_{\mathbf{e}}}$.
$\mathcal{C}_1$	Distribution of $\mathbf{c}_1$ in the Hint-RLWE experiment
r, s	Parameters of the approximate trapdoor sampler
B	Two-norm bound from the NIZK statement Equation (4)
δ, ε	Parameters of the regularity lemma (only used in the proof)
$\chi_{\mathbf{s}'}$	Distribution of $\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3$ (only used in the proof.), instantiated as $\mathcal{D}_{\mathcal{R}_q, \sigma_{\mathbf{s}'}}$.
Q_{S}	Number of signatures issued
$\tilde{p}$	$\tilde{p}$ such that $\tilde{p} q$

Remark. One can observe that the signer-user message can be further compressed. In Figure 1, the user receives $\text{pm}^{\text{S}} = (\mathbf{x}, \mathbf{z}_2, \mathbf{c}_1, \tilde{\mathbf{e}}) \in \mathcal{R}_q^7$ and computes the witness as

$$\mathbb{w} = \begin{bmatrix} \mathbf{u} + \mathbf{A}(\mathbf{x}' + \mathbf{x}) - \mathbf{a}\mathbf{z}_2 - \mathbf{b}\mathbf{c}_1 - \mathbf{x}_1 - \mathbf{x}'_1 \\ \mathbf{z}_2 - \mathbf{x}_2 - \mathbf{x}'_2 \\ \mathbf{c}_1 - \mathbf{x}_3 \\ -\mathbf{x}_4 \end{bmatrix}.$$

This, means equivalently the signer could send one vector $\mathbf{x}^* \in \mathcal{R}_q^4$ with the user computing the witness as

$$\mathbf{x}^* := \begin{bmatrix} \mathbf{x}_1 + \mathbf{a}\mathbf{z}_2 + \mathbf{b}\mathbf{c}_1 - \mathbf{A}\mathbf{x} \\ \mathbf{x}_2 - \mathbf{z}_2 \\ \mathbf{x}_3 - \mathbf{c}_1 \\ \mathbf{x}_4 \end{bmatrix}; \quad \mathbb{w}^* := \begin{bmatrix} \mathbf{u} + \mathbf{A}\mathbf{x}' - \mathbf{x}_1^* - \mathbf{x}'_1 \\ -\mathbf{x}_2^* - \mathbf{x}'_2 \\ -\mathbf{x}_3^* \\ -\mathbf{x}_4^* \end{bmatrix}.$$

Setup (1^λ)	Vfy ($\text{vk}, \mu, \text{sig}$)
1: $\text{crs} \leftarrow \Pi.\text{Setup}(1^\lambda)$ 2: $\mathbf{a} \leftarrow \mathcal{R}_q^\times, \mathbf{d} \leftarrow \mathcal{R}_q$ 3: $\mathbf{B} \leftarrow \{1\} \times \mathcal{R}_q^3$ 4: $\text{pp} := (\text{crs}, \mathbf{a}, \mathbf{d}, \boxed{\mathbf{B}})$ 5: return pp	1: parse ($\text{crs}, \mathbf{a}, \mathbf{d}, \boxed{\mathbf{B}}$) $\leftarrow$ pp 2: parse ($\mathbf{A}, \mathbf{b}$) $\leftarrow$ ($[1 \ \mathbf{a} \ \mathbf{b} \ \mathbf{d}]$, vk) 3: $\mathbf{u} := H_\mu(\mu, \text{vk})$ 4: $\mathbb{x} := (\mathbf{A}, \boxed{\mathbf{B}}, \mathbf{u})$ 5: return $\Pi.\text{Verify}(\text{crs}, \mathbb{x}, \text{sig})$
KeyGen (pp)	User₀ (vk, μ)
1: $(\mathbf{s}, \mathbf{e}) \leftarrow \chi_{\text{sk}}^2$ 2: $\mathbf{b} := \beta - (\mathbf{a}\mathbf{s} + \mathbf{e})$ 3: if $\mathbf{b} \notin \mathcal{R}_q^\times$: 4: restart 5: $\text{vk} := \mathbf{b}, \text{sk} := (\mathbf{s}, \mathbf{e})$ 6: return (vk, sk)	1: parse ($\text{crs}, \mathbf{a}, \mathbf{d}, \boxed{\mathbf{B}}$) $\leftarrow$ pp 2: parse ($\mathbf{A}, \mathbf{b}$) $\leftarrow$ ($[1 \ \mathbf{a} \ \mathbf{b} \ \mathbf{d}]$, vk) 3: $\mathbf{u} := H_\mu(\mu, \text{vk})$ 4: $\mathbf{x}' \leftarrow \chi_{\mathbf{u}, \mathbf{x}}^2 \times \{0\}^2 \in \mathcal{R}_q^4$ 5: $\mathbf{t}' := \mathbf{A}\mathbf{x}' + \mathbf{u}$ 6: $\text{st}^{\mathbf{u}} := (\mathbf{x}', \mathbf{u}, \mu, \text{vk})$ 7: return ($\text{st}^{\mathbf{u}}, \text{pm}^{\mathbf{u}} := \mathbf{t}'$)
Signer (sk, $\text{pm}^{\mathbf{u}}$)	User₁ ($\text{st}^{\mathbf{u}}, \text{pm}^{\mathbf{S}}$)
1: parse ($\mathbf{t}', \mathbf{b}$) $\leftarrow$ ($\text{pm}^{\mathbf{u}}, \text{vk}$) 2: parse ($\text{crs}, \mathbf{a}, \mathbf{d}, \boxed{\mathbf{B}}$) $\leftarrow$ pp 3: parse $\mathbf{A} \leftarrow [1 \ \mathbf{a} \ \mathbf{b} \ \mathbf{d}]$ 4: $\mathbf{x} \leftarrow \chi_{\mathbf{S}, \mathbf{x}}^4$ 5: $\tilde{\mathbf{e}} \leftarrow \chi_{\mathbf{e}}$ 6: $\tilde{\mathbf{y}} \leftarrow \{\pm 1\}^{4n} \in \mathbb{Z}^{4n}$ 7: $\mathbf{t} := \mathbf{A}\mathbf{x} + \boxed{\mathbf{B}\tilde{\mathbf{y}}} + \tilde{\mathbf{e}} + \mathbf{t}'$ 8: $(\mathbf{r}, \mathbf{p}) \leftarrow \chi_{\text{rnd}}^2$ 9: $\mathbf{w} := \mathbf{a}\mathbf{r} + \mathbf{p}$ 10: $(\mathbf{c}_1, \mathbf{c}_2) \leftarrow \text{Decomp}_\beta(\mathbf{t} - \mathbf{w})$ 11: $\mathbf{z}_2 := \mathbf{c}_1\mathbf{s} + \mathbf{r}$ 12: $\text{aux}_{\text{sig}} := \mathbf{w}$ 13: $\text{pm}^{\mathbf{S}} := (\mathbf{x}, \mathbf{z}_2, \mathbf{c}_1, \tilde{\mathbf{e}}, \boxed{\tilde{\mathbf{y}}})$ 14: return ($\text{pm}^{\mathbf{S}}, \text{aux}_{\text{sig}}$)	1: parse ($\mathbf{x}, \mathbf{z}_2, \mathbf{c}_1, \tilde{\mathbf{e}}, \boxed{\tilde{\mathbf{y}}}$) $\leftarrow$ $\text{pm}^{\mathbf{S}}$ 2: parse ($\mathbf{x}', \mathbf{u}, \mu, \text{vk}$) $\leftarrow$ $\text{st}^{\mathbf{u}}$ 3: $\mathbf{z}'_1 := \mathbf{u} + \mathbf{A}(\mathbf{x}' + \mathbf{x}) - \mathbf{a}\mathbf{z}_2 - \mathbf{b}\mathbf{c}_1$ 4: $\mathbf{y} := (\mathbf{z}'_1, \mathbf{z}_2, \mathbf{c}_1, 0)^\top$ 5: $\mathbf{w} := (\mathbf{y} - (\mathbf{x} + \mathbf{x}'), \boxed{\tilde{\mathbf{y}}})$ 6: $\mathbb{x} := (\mathbf{A}, \boxed{\mathbf{B}}, \mathbf{u})$ 7: if $\ \mathbf{w}\ _2 > B$ then return $\perp$ 8: if $\tilde{\mathbf{y}} \notin \{\pm 1\}^{4n}$ then return $\perp$ 9: $\pi \leftarrow \Pi.\text{Prove}(\text{crs}, \mathbb{x}, \mathbf{w})$ 10: return sig := π

Figure 1: The algorithms Setup, KeyGen, and Vfy and the signing protocol (User₀, User₁, Signer) of our blind signature BS_B (resp. $\text{BS}_B^{\$}$) are obtained by ignoring (resp. considering) boxed terms. As $\text{aux} = \perp$ in our case, we omit it from the description.

We chose to present the scheme without this compression for clarity. This makes the similarities between our scheme and Plover [EEN⁺24] clearer and corresponds more to the thresholdized protocol from Section 5.

4.2 Security Analysis

We analyze security of our main variant BS_B and refer to Section E for a security analysis of $\text{BS}_B^\$$. In the following, proofs are only sketched. Full proofs can be found in Section C.

Theorem 4.1 (Correctness). *Assume $2^{-2n/10} = \text{negl}(\lambda)$ and $\sigma_{\text{rnd}}^{-2} + \sigma_{S,x}^{-2} \leq \eta'_\delta(\mathbb{Z}^n)^{-2}$ for $\delta = \text{negl}(\lambda)$. Let us show that $(\mathbb{x}, \mathbb{w}) \in \mathcal{R}$ where $B = e^{1/4} \sqrt{n} \left(n\sqrt{2} \left\lceil \frac{q-1}{2\beta} \right\rceil \sigma_{\text{sk}} + \sigma_{U,x} + \sqrt{2\sigma_{\text{rnd}}^2 + 4\sigma_{S,x}^2} \right) + \sqrt{n} \frac{\beta+\bar{p}}{2}$ with overwhelming probability, then correctness follows as Π is correct.*

The bound $\frac{\beta+\bar{p}}{2}$ can be improved to the expected value $\frac{\beta+\bar{p}}{\sqrt{12}}$ up to a small factor using Hoeffding's inequality. For conciseness of the statement, we however leave it to the concrete parameters only.

Theorem 4.2 (Blindness). *If there exists an PPT adversary $\mathcal{A}$ that wins the blindness experiment from Figure 3, then there exist PPT adversaries $\mathcal{B}, \mathcal{B}'$ such that*

$$\text{Adv}_{\text{BS}_B, \mathcal{A}}^{\text{blind}}(\lambda) \leq \text{Adv}_{\Pi, \mathcal{B}}^{\text{ZK}}(\lambda) + 2\text{Adv}_{\mathcal{B}', q, \chi_{U,x}}^{\text{RLWE-}\times}(\lambda).$$

Sketch. In the first hybrid game, the zero-knowledge property of Π allows to replace the NIKZ proof with a simulated proof. In the second step, the RLWE assumption for the submatrix $[1 \ \mathbf{a}]$ is used to argue that the first user message $\mathbf{t}$ hides $\mathbf{u}$. Note that this holds even if the verification key $\mathbf{vk}$ is setup maliciously as $\mathbf{a}$ and crs are part of the public parameters. Then, no message from the user depends on the input anymore, which concludes the proof. $\square$

Theorem 4.3 (One-more Unforgeability). *If there exists an PPT adversary $\mathcal{A}$ that wins the one-more unforgeability experiment from Figure 4 and makes at most Q_S queries to the $\mathcal{O}^{\text{Signer}}$ oracle, then there exist PPT adversaries $\mathcal{B}, \mathcal{B}', \mathcal{B}'', \bar{\mathcal{B}}$ such that*

$$\begin{aligned} \text{Adv}_{\text{BS}_B, \mathcal{A}}^{\text{bs-omuf}}(\lambda) &\leq \text{Adv}_{\Pi, \mathcal{B}}^{\text{CRS}}(\lambda) + \text{Adv}_{\mathcal{B}', q, Q_S, \chi_{S,x}^2, \chi_{\text{rnd}}^2, \mathcal{C}_1}^{\text{Hint-RLWE-}\times}(\lambda) \\ &\quad + \text{Adv}_{\mathcal{B}'', q, \chi_{S,x'}}^{\text{RLWE-}\times}(\lambda) + \varepsilon_{\text{samp}} + \text{negl}(\lambda) \\ &\quad + \text{poly}(\lambda, Q_H) (e^2 \text{Adv}_{\bar{\mathcal{B}}, q, \chi_{S,x}^4, Q_S, B}^{\text{OM-ISIS-}\times} \\ &\quad + 2\delta + 2^{\ell n} (5/4)^{3n\ell} q^{-3n\varepsilon} + \varepsilon_{\text{samp}}), \end{aligned}$$

where δ, ε, n are such that Theorem 3.1 holds.

Sketch. The proof proceeds in a series of hybrid games to finally allow for a reduction from OM-ISIS. Roughly, we first embed a trapdoor $\mathbf{T}_d$ for $\mathbf{A}$ into $\mathbf{d}$, following the *compact lattice gadget* strategy from [YJW23]. Then, we simulate Plover signatures for a random target, following the strategy from [EEN+24], which we embed in the signing session via the trapdoor $\mathbf{T}_d$. At this point, signing can be simulated via $\mathbf{T}_d$, in particular, the secret key sk is not required anymore. Finally, we argue that the use of $\mathbf{T}_d$, and therefore signing, can be simulated via the OM-ISIS challenger for $\mathbf{A}$. Then, if we embed OM-ISIS syndromes into H_μ , each signature yields a short preimage, breaking OM-ISIS. In more detail, the hops proceed as follows.

- First, the public value $\mathbf{d}$ in the matrix $\mathbf{A}$ is replaced by a lattice gadget $\mathbf{d}$, ensuring that the game knows a trapdoor $\mathbf{T}_d$ for $\mathbf{A}$. Under the RLWE assumption $\mathbf{d}$ still looks random, however, this change will allow trapdoor-sampling preimages of $\mathbf{A}$ via $\mathbf{T}_d$.
- As a next step, we would like to employ the simulation strategy from Plover [EEN+24] to compute short preimages of $[1 \ \mathbf{a} \ \mathbf{b}]$. The goal is to reduce the use of sk to only $\mathbf{z}_1$ and $\mathbf{z}_2$, such that applying the Hint-RLWE assumption in the next game completely dispenses with sk entirely. We observe that Plover signatures $\mathbf{z}$ can be simulated for *uniform* targets of the form $[1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z}$, as is done in the reduction in [EEN+24]. However, their reduction relies on programming H_μ to embed the target $[1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z}$ into H_μ .

In our proof, hash queries cannot be linked to signing queries because the user blinds their message, and indeed, a malicious user could send any $\mathbf{t} \in \mathcal{R}_q$ to the signer. To that end, we can use the trapdoor $\mathbf{T}_d$ for $\mathbf{A}$ introduced in the prior game. Roughly, we first simulate the Plover signature $\mathbf{z}$ for target $[1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z}$ and afterward sample short $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} + \tilde{\mathbf{e}} + \mathbf{t}' = [1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z}$, *i.e.*, such that $\mathbf{z}$ is a Plover signature on $\mathbf{A}\mathbf{x} + \tilde{\mathbf{e}} + \mathbf{t}'$.

- Now, the secret key $\mathbf{sk} = (\mathbf{s}, \mathbf{e})$ is only used to compute $\mathbf{b}$ in $\mathbf{A}$ and the Plover signature $\mathbf{z}$, as simulated in the prior game. The latter can be simulated via the “hints” from Hint-RLWE. Therefore, we can dispense of $\mathbf{sk}$ entirely in our simulation, and after the change, only $\mathbf{T}_d$ is required to simulate signing.
- Finally, the reduction from OM-ISIS can forward hash queries as syndrome queries to the challenger and replace the public key with the uniformly random matrix $\tilde{\mathbf{A}}$ provided by the challenger. Further, instead of trapdoor-sampling approximate preimages for $\mathbf{A}$ via $\mathbf{T}_d$, the OM-ISIS challenger provides the reduction with exact preimages for $\tilde{\mathbf{A}}$ for every signing query. We show that this is sufficient to simulate signing queries, without the use of $\mathbf{T}_d$. Finally, if the adversary forged, $Q_S + 1$ signatures, then the straightline extractability of the NIZK allows the reduction to extract $Q_S + 1$ short preimages, breaking OM-ISIS.

We remark that the last step is quite subtle. For the proof to go through, we generalize the regularity lemma by Stehlé and Steinfeld [SS11] to argue that $\mathbf{d}$ is statistically close to uniform when q is composite (cf. Theorem 3.1), which is needed to use Yu, Jia and Wang’s [YJW23] trapdoor sampler. We also extend the analysis of their trapdoor sampler to show that its output preimage is distributed close to a Gaussian conditioned on the target value (cf. Theorem 2.7).

□

5 Threshold Setting

We present our threshold blind signature TBS_B proven secure under one-more ISIS and a variant $\text{TBS}_B^\$$ proven secure under randomized one-more ISIS. Analogous to Section 4, terms that are relevant only for $\text{TBS}_B^\$$ are highlighted in boxes. We may omit B if clear.

Parameters and Preparations Our construction builds on the non-threshold blind signature BS from Section 4 with the same parameters except that the bound B contains an additional $\sqrt{T}$ term for correctness (see Theorem 4.1). In addition to the NIZK Π and random oracle H_μ , we employ three additional random oracles:

In addition, we rely on 3 additional random oracles:

- H_{com} mapping into $\{0, 1\}^{2\lambda}$ to commit to ring elements $\mathbf{w}$.
- H_{mask} mapping into $\mathcal{R}_q$ to generate non-interactive zero-shares (cf. Section 2.2.1).
- H_x which outputs $(\mathbf{x}, \tilde{\mathbf{e}}, \tilde{\mathbf{y}})$ with $\mathbf{x} \leftarrow \chi_{S,x}^4$, $\tilde{\mathbf{e}} \leftarrow \chi_e$ and $\tilde{\mathbf{y}} \leftarrow \{\pm 1\}^{4n}$. We assume that H_x also fixes the randomness for Decomp_β implicitly in Signer_3 .

5.1 Overview and Construction

The protocol is depicted in Figure 2. We describe TBS_B below. We obtain $\text{TBS}_B^\$$ similar to how $\text{BS}_B^\$$ is obtained from BS_B .

To obtain a threshold version of the blind signature from Section 4, the key generation first computes the N secret keys $(\mathbf{sk}_i)_{i \in [N]}$ as Shamir secret sharing of the short vector $\mathbf{s}$ used for signing. Then, key generation computes mutual seeds for the Raccoon-style zero-share masking technique. Note that our protocol does not make use of the aux value, but we explicit $\text{aux} := \perp$ to ensure compatibility with the definition of Lehmann

et al. [LNÖ25]. The rest of the key generation follows the blind signature from Section 4. Also, the first message of the user is almost exactly computed as before, *i.e.*, the user sends a request $\mathbf{t}' = \mathbf{A}\mathbf{x}' + \mathbf{H}_\mu(\mu, \mathbf{vk})$. The only difference is that the user additionally informs the signers about the signer set $\mathbf{SS}$.

On a high level, the signer first agrees upon the perturbation randomness in $\mathbf{w}$ through a commit-and-open protocol, then derives fresh values $(\mathbf{x}, \tilde{\mathbf{e}})$ via $\mathbf{H}_x(\mathbf{vk}, \mathbf{w})$ to shift $\mathbf{t}'$ as in BS. In more detail, on request $(\mathbf{t}', \mathbf{SS})$, the signer commits to the values $\mathbf{w}_i = \mathbf{a}\mathbf{r}_i + \mathbf{p}_i$ in commitment $\mathbf{cmt}_i$. These $\mathbf{cmt}_i$ are sent to the user, which distributes all commitments $(\mathbf{cmt}_j)_{j \in \mathbf{SS}}$ to the signers. Then, each signer opens their commitments by sending $\mathbf{w}_i$ to the user. Again, the user forwards all $(\mathbf{w}_j)_{j \in \mathbf{SS}}$ to the signers. Each signer derives $(\mathbf{x}, \tilde{\mathbf{e}}) = \mathbf{H}_x(\mathbf{vk}, \mathbf{w})$, where $\mathbf{w} = \sum_{j \in \mathbf{SS}} \mathbf{w}_j$, then shifts $\mathbf{t} = \mathbf{t}' + \mathbf{A}\mathbf{x} + \tilde{\mathbf{e}}$. As in BS, $\mathbf{t} - \mathbf{w}$ is decomposed into $(\mathbf{c}_1, \mathbf{c}_2)$ and the partial signature

$$\mathbf{z}_{2,i} = \mathbf{c}_1 L_{\mathbf{SS},i} \llbracket \mathbf{s} \rrbracket_i + \mathbf{r}_i + \Delta_i$$

is output, where Δ_i is a zero-share computed as in Eq. (1) and $L_{\mathbf{SS},i}$ is the i -th Lagrange coefficient for the set of signers $\mathbf{SS}$ (cf. Theorem B.1).

Compare this to the non-threshold version of the protocol, where the signer computed $\mathbf{z}_2 = \mathbf{c}_1 \mathbf{s} + \mathbf{r}$. The Lagrange coefficients and the zero-share ensure that when the user sums up all $\mathbf{z}_{2,i}$ it will obtain the same value $\mathbf{z}_2$ as in the non-thresholdized protocol, albeit with a slightly larger norm. The zero-share ensures that no more than the summed value $\mathbf{z}_2$ is leaked which avoids lattice-related attacks.

Finally, the user sums $\mathbf{z}_{2,i}$ and computes a proof for relation $\mathcal{R}$ to derive the signature.

5.2 Security Analysis

We analyze TBS_B and refer to Section E for an analysis of $\text{TBS}_B^\$$. We sketch proofs for one-more unforgeability and blindness of TBS_B . Full proofs for security and correctness can be found in Section D.

Theorem 5.1 (Correctness). *Under the same conditions as Theorem 4.1, our TBS is correct for $B = e^{1/4} \cdot \sqrt{n} \cdot \left(n\sqrt{2} \left\lceil \frac{q-1}{2\beta} \right\rceil \sigma_{\text{sk}} + \sigma_{\mathbf{u},\mathbf{x}} + \sqrt{2T\sigma_{\text{rnd}}^2 + 4\sigma_{\mathbf{S},\mathbf{x}}^2} \right) + \sqrt{n} \frac{\beta + \bar{p}}{2}$.*

Sketch. This follows from the properties of Shamir's sharing. Notice that the bound B contains an additional $\sqrt{T}$ which stems from the fact that T values $\mathbf{z}_{2,i}$ are summed. $\square$

Theorem 5.2 (Blindness). *If there exists an PPT adversary $\mathcal{A}$ that wins the blindness experiment from Figure 6, then there exist PPT adversaries $\mathcal{B}, \mathcal{B}'$ such that*

$$\text{Adv}_{\text{TBS}_B, \mathcal{A}}^{\text{blind}}(\lambda) \leq \text{Adv}_{\Pi, \mathcal{B}}^{\text{ZK}}(\lambda) + 2\text{Adv}_{\mathcal{B}', q, \chi_{\mathbf{u}, \mathbf{x}}}^{\text{RLWE-}\times}(\lambda).$$

Sketch. Essentially, we can reuse the proof strategy from Theorem 4.2. To see this, first observe that even in the threshold case, still the signature $\mathbf{sig}$ is a NIZK, and thus, can be simulated. Second, the first message from the user to the signer is the same in the threshold protocol as in the non-threshold protocol, *i.e.*, $\mathbf{A}\mathbf{x} + \mathbf{H}_\mu(\mu, \mathbf{vk})$. Hence, also here the same argument applies as before. It remains to consider the new user messages $\mathbf{pm}_1^{\mathbf{u}}$ and $\mathbf{pm}_2^{\mathbf{u}}$. But these messages merely bundle the contributions collected from the signers and broadcast them back. Therefore, they do not provide any additional information to the adversary. $\square$

Theorem 5.3 (One-more Unforgeability). *If there exists an PPT adversary $\mathcal{A}$ that wins the one-more unforgeability experiment from Figure 5, then there exist PPT adversaries $\mathcal{B}, \mathcal{B}', \tilde{\mathcal{B}}$ such that*

Setup (1^λ) 1: $\text{crs} \leftarrow \Pi.\text{Setup}(1^\lambda)$ 2: $\mathbf{a} \leftarrow \mathcal{R}_q^\times, \mathbf{d} \leftarrow \mathcal{R}_q$ 3: $\boxed{\mathbf{B}} \leftarrow \{1\} \times \mathcal{R}_q^3$ 4: $\text{pp} := (\text{crs}, \mathbf{a}, \mathbf{d}, \boxed{\mathbf{B}})$ 5: return pp	Vfy (vk, μ , sig) 1: $\mathbf{u} := H_\mu(\mu, \text{vk})$ 2: $\mathbf{x} := (\mathbf{A}, \boxed{\mathbf{B}}, \mathbf{u})$ 3: return $\Pi.\text{Verify}(\text{crs}, \mathbf{x}, \text{sig})$
KeyGen (N, T) 1: $(\mathbf{s}, \mathbf{e}) \leftarrow \chi_{\text{sk}}^2$ 2: $\mathbf{b} := \beta - (\mathbf{a}\mathbf{s} + \mathbf{e})$ 3: if $\mathbf{b} \notin \mathcal{R}_q^\times$: 4: restart 5: for $(i, j) \in [N]^2$ do 6: $\text{rand}_{i,j} \leftarrow \{0, 1\}^\lambda; \text{seed}_{i,j} := i\ j\ \text{rand}_{i,j}$ 7: $(\vec{\text{seed}}_i)_{i \in [N]} := ((\text{seed}_{i,j})_{j \in [N]})_{i \in [N]}$ 8: $\vec{P} \leftarrow \mathcal{R}_q[X]$ with $\deg(\vec{P}) = T - 1, \vec{P}(0) = \mathbf{s}$ 9: $(\llbracket \mathbf{s} \rrbracket_i)_{i \in [N]} := (\vec{P}(i))_{i \in [N]}$ 10: $\text{vk} := \mathbf{b}, \text{aux} := \perp$ 11: $(\text{sk}_i)_{i \in [N]} := ((\llbracket \mathbf{s} \rrbracket_i, \vec{\text{seed}}_i)_{i \in [N]})$ 12: return (vk, $(\text{sk}_i)_{i \in [N]}$, aux)	User₀ (vk, aux, μ , SS) 1: $\mathbf{u} := H_\mu(\mu, \text{vk})$ 2: $\mathbf{x}' \leftarrow \chi_{0,\mathbf{x}}^2 \times \{0\}^2 \in \mathcal{R}_q^4$ 3: $\mathbf{t}' := \mathbf{A}\mathbf{x}' + \mathbf{u}$ 4: return $(\text{st}_0^U := (\mathbf{x}', \mathbf{u}, \mu, \text{vk}), \text{pm}_0^U := (\mathbf{t}', \text{SS}))$
User₂ ($\text{st}_1^U, (\text{pm}_2^{(S,i)})_{i \in \text{SS}}$) 1: $\text{parse } (\mathbf{w}_i)_{i \in \text{SS}} \leftarrow (\text{pm}_2^{(S,i)})_{i \in \text{SS}}$ 2: $\mathbf{w} := \sum_{i \in \text{SS}} \mathbf{w}_i$ 3: $(\mathbf{x}, \tilde{\mathbf{e}}, \boxed{\tilde{\mathbf{y}}}) \leftarrow H_x(\text{vk}, \mathbf{w})$ 4: $\text{st}_2^U := (\mathbf{x}', \mathbf{x}, \tilde{\mathbf{e}}, \mathbf{u}, \mu, \text{vk}, \mathbf{w}, \boxed{\tilde{\mathbf{y}}})$ 5: return $(\text{st}_2^U, \text{pm}_2^U := (\mathbf{w}_i)_{i \in \text{SS}})$	Signer₁ ($i, \text{sk}_i, \text{aux}, \text{ctx}, \text{pm}_0^U$) 1: $\text{parse } (\mathbf{t}', \text{SS}) \leftarrow \text{pm}_0^U$ 2: $(\mathbf{r}_i, \mathbf{p}_i) \leftarrow \chi_{\text{rnd}}^2$ 3: $\mathbf{w}_i := \mathbf{a}\mathbf{r}_i + \mathbf{p}_i$ 4: $\text{cmt}_i := H_{\text{com}}(i, \mathbf{w}_i)$ 5: $\text{st}_1^{(S,i)} \leftarrow (\text{SS}, \mathbf{t}', \text{cmt}_i, \mathbf{w}_i, \mathbf{r}_i)$ 6: return $(\text{st}_1^{(S,i)}, \text{pm}_1^{(S,i)} := \text{cmt}_i)$
User₃ ($\text{st}_2^U, (\text{pm}_3^{(S,i)})_{i \in \text{SS}}$) 1: $\text{parse } (\mathbf{z}_{2,i})_{i \in \text{SS}} \leftarrow (\text{pm}_3^{(S,i)})_{i \in \text{SS}}$ 2: $\mathbf{t} := \mathbf{A}(\mathbf{x}' + \mathbf{x}) + \boxed{\mathbf{B}\tilde{\mathbf{y}}} + \tilde{\mathbf{e}} + \mathbf{u}$ 3: $(\mathbf{c}_1, \mathbf{c}_2) := \text{Decomp}_\beta(\mathbf{t} - \mathbf{w})$ 4: $\mathbf{z}_2 := \sum_{i \in \text{SS}} \mathbf{z}_{2,i}$ 5: $\mathbf{z}'_1 := \mathbf{u} + \mathbf{A}(\mathbf{x}' + \mathbf{x}) + \tilde{\mathbf{e}} - \mathbf{a}\mathbf{z}_2 - \mathbf{b}\mathbf{c}_1$ 6: $\mathbf{y} := (\mathbf{z}'_1, \mathbf{z}_2, \mathbf{c}_1, 0)^\top$ 7: $\mathbf{w} := \mathbf{y} - (\mathbf{x} + \mathbf{x}'), \mathbf{x} := (\mathbf{A}, \mathbf{u})$ 8: if $\ \mathbf{w}\ _2 > B$ then return $\perp$ 9: $\pi \leftarrow \Pi.\text{Prove}(\text{crs}, \mathbf{x}, \mathbf{w})$ 10: return sig := π	User₁ ($\text{st}_0^U, (\text{pm}_1^{(S,i)})_{i \in \text{SS}}$) 1: return $(\text{st}_1^U := \text{st}_0^U, \text{pm}_1^U := (\text{pm}_1^{(S,i)})_{i \in \text{SS}})$
	Signer₂ ($i, \text{st}_1^{(S,i)}, \text{pm}_1^U$) 1: $\text{parse } (\text{cmt}_j)_{j \in \text{SS}} \leftarrow \text{pm}_1^U$ 2: $\text{SS}_i := \text{SS}$ 3: $\text{ssid}_i := (\mathbf{t}', \text{SS}_i, (\text{cmt}_j)_{j \in \text{SS}}, \text{ctx})$ 4: $\text{st}_2^{(S,i)} \leftarrow (\text{ssid}_i, \mathbf{w}_i, \mathbf{r}_i)$ 5: return $(\text{st}_2^{(S,i)}, \text{pm}_2^{(S,i)} := \mathbf{w}_i, \text{ssid}_i, \text{SS}_i)$
	Signer₃ ($i, \text{st}_2^{(S,i)}, \text{pm}_2^U$) 1: $\text{parse } (\mathbf{w}_j)_{j \in \text{SS}} \leftarrow \text{pm}_2^U$ 2: req ($\forall j \in \text{SS}, \text{cmt}_j = H_{\text{com}}(j, \mathbf{w}_j)$) 3: $\mathbf{w} := \sum_{j \in \text{SS}} \mathbf{w}_j$ 4: $(\mathbf{x}, \tilde{\mathbf{e}}, \boxed{\tilde{\mathbf{y}}}) := H_x(\text{vk}, \mathbf{w})$ 5: $\mathbf{t} := \mathbf{A}\mathbf{x} + \boxed{\mathbf{B}\tilde{\mathbf{y}}} + \tilde{\mathbf{e}} + \mathbf{t}'$ 6: $(\mathbf{c}_1, \mathbf{c}_2) \leftarrow \text{Decomp}_\beta(\mathbf{t} - \mathbf{w})$ 7: $\Delta_i := \text{ZeroShare}(\vec{\text{seed}}_i[\text{SS}], \text{ssid}_i)$ 8: $\mathbf{z}_{2,i} := \mathbf{c}_1 \cdot L_{\text{SS},i} \cdot \llbracket \mathbf{s} \rrbracket_i + \mathbf{r}_i + \Delta_i$ 9: $\text{pm}_3^{(S,i)} := \mathbf{z}_{2,i}, \text{aux}_{\text{sig}} := \mathbf{w}$ 10: return $(\text{pm}_3^{(S,i)}, \text{aux}_{\text{sig}})$

Figure 2: The algorithms Setup, KeyGen, and Vfy and the signing protocol of our blind threshold signature TBS_B (resp. TBS_B^S) are obtained by ignoring (resp. considering) boxed terms. We keep the parsing of states and public parameters implicit for readability. We also assume that parties parse $\mathbf{A} \leftarrow [1 \ \mathbf{a} \ \mathbf{b} \ \mathbf{d}]$ via $\mathbf{b} \leftarrow \text{vk}$ and $(\text{crs}, \mathbf{a}, \mathbf{b}, \mathbf{B}) \leftarrow \text{pp}$. We assume that the signers check that their previous round message is consistent with the input and that the signer set remains unchanged. We assume that in $\mathcal{O}_{\text{Sign}_1}$, the signers check that $\text{SS} \subseteq [N]$ with $|\text{SS}| = T$.

$$\begin{aligned}
\text{Adv}_{\text{BS}_B, \mathcal{A}}^{\text{bs-omuf}}(\lambda) &\leq Q_{H_{\text{com}}} \cdot Q_S \cdot (\text{Adv}_{B, q, \chi_{\text{rnd}}}^{\text{RLWE-}\times}(\lambda) + q^{-n}) \\
&\quad + Q_{H_x} Q_S (\text{Adv}_{B', q, \chi_{\text{rnd}}}^{\text{RLWE-}\times}(\lambda) + q^{-n}) \\
&\quad + \text{Adv}_{\text{BS}_B, \tilde{B}}^{\text{bs-omuf}} + \frac{(H_{\text{com}} + Q_S)^2}{2^{2\lambda}} \\
&\quad + \frac{Q_{H_{\text{mask}}}}{2^\lambda} + \frac{Q_{H_{\text{com}}}}{2^{2\lambda}}
\end{aligned}$$

Before sketching our proof, we note that the RLWE-related terms could be avoided by ensuring that $\mathbf{w}_i$ has a high min-entropy. This has a negative impact on parameters in our setting, therefore we give a computational argument instead.

Sketch. In our proof, we reduce one-more unforgeability of TBS with norm bound B to one-more unforgeability of BS_B . Roughly, we show that signing leaks no more information than a non-threshold BS_B signing session (*i.e.*, we can simulate signing via a signing oracle for BS_B) thanks to the masking of the response in Signer_3 . As signatures have the same form as BS_B signatures, finding $Q_S + 1$ valid signatures is hard by Theorem 4.3.

In more detail, we delay choosing the perturbation $\mathbf{w}_i$ for honest signers $i \in \text{HS}$ until the second round when cmt_i is opened by programming H_{com} accordingly. At the same time, we extract the malicious perturbations $\mathbf{w}_i$ from commitments cmt_i , where $i \in \text{CS}$. Further, we argue that due to masking via Δ_i , the transcript of a signing session leaks only the aggregated response $\mathbf{z}_2 = \sum_{i \in \text{SS}} \mathbf{z}_{2,i}$, following the technique in [KRT24]. Observe that the above changes allow us to simulate signing with our non-threshold signature BS_B . In particular, when the last honest signer $i \in \text{HS}$ opens its commitment cmt_i , the game proceeds as follows:

- It sets $\mathbf{t}^* = \mathbf{t}' - \sum_{j \in \text{SS} \setminus \{i\}} \mathbf{w}_j$, where $\mathbf{w}_j$ are the perturbations of the other users.
- It queries $\mathbf{t}^*$ to the BS_B oracle and obtains $(\mathbf{x}^*, \mathbf{z}_2^*, \mathbf{c}_1^*, \tilde{\mathbf{e}}^*)$ and auxiliary information $\mathbf{w}^*$.
- It sets $\mathbf{w}_i := \mathbf{w}^*$ and embeds $(\mathbf{x}^*, \tilde{\mathbf{e}}^*)$ into the oracle H_x .
- The response $\mathbf{z}_2^*$ is employed to simulate the aggregated response $\mathbf{z}_2$.

We ensure that the BS_B oracle is queried at most once per session identifier ssid , which allows to break OMUF of BS_B . $\square$

Note that this proof strategy requires the reduction to already query the BS_B oracle in the second round in order to simulate the aggregated $\mathbf{z}_2$ and program H_x correctly. That is the reason why in our OMUF definition (Theorem B.8) we increment the counter ℓ already in round 2 and not in round 3 (cf. Section 2.2.1).

6 Instantiation

We propose two instantiations of our TBS, making different trade-offs between theoretical rigor and practical efficiency.

Our construction requires a NIZK proof system Π for lattice relations. While formal security demands straightline extractability (SLE), most efficient lattice-based systems like LNP [LNP22] provide only adaptive knowledge soundness (and the witness is typically extracted via rewinding). We could achieve SLE via encryption-to-the-sky (*i.e.* we append an encryption of the witness to the proof, and prove consistency) as in [AKSY22], but this adds roughly 150 KB to signature sizes. Instead, both our instantiations assume *heuristically* that LNP and LaBRADOR [BS23] are online-extractable for practical efficiency. We note that

both LNP and LaBRADOR are proven secure under standard lattice assumptions, namely MLWE and MSIS. We believe this is plausible for the following reasons.

- The group-based proof systems on which LNP and LaBRADOR are based on (*i.e.*, Schnorr-style Σ -protocols [Sch90] and Bulletproofs [BBB⁺18]) are online-extractable [FPS20, GOP⁺22]. While the analysis requires the algebraic group model (AGM) and no counterpart exists in the lattice setting, it is plausible that the lattice-based analogues have similar security properties.
- The requirement of online-extraction seems to be a proof artifact, and we cannot employ rewinding-based proof systems as too many OM-ISIS queries are made to rewind the adversary in the security proof. However, Katsumata et al. [KRS23] show that rewinding-based arguments are possible in Fischlin-style blind signatures, if the proof *statistically* determines the extracted witness. While their technique does not apply in our work, this provides evidence that online-extraction may not be strictly required.

Compact Instantiation This instantiation targets minimal signature size using LaBRADOR [BS23] combined with LNP [LNP22]. We follow provable statistical arguments from our theoretical analysis. While this requires larger lattice parameters, LaBRADOR’s superior compression efficiency compensates for this overhead, achieving the most compact proof size among existing lattice-based systems; however, an efficient ZK version of LaBRADOR remains an open problem.

Easy to Deploy Instantiation This instantiation prioritizes practical deployability using LNP [LNP22] with the LaZer library [LSS24]. We heuristically replace statistical arguments with computational RLWE hardness (as in [AKSY22]), allowing smaller parameters necessary for reasonable proof sizes without LaBRADOR’s compression. This is detailed in Section A.

6.1 Parameter Selection and Evaluation

We select and evaluate our parameters following our security proofs. This can be reproduced using the script `blindsig_sec.py` provided in our artifact.

- **Blindness:** Relies on plain MLWE hardness defined by the blinding noise standard deviation ($\sigma_{u,x}$), c.f. Theorem 4.2.
- **One-More Unforgeability:** We distinguish the terms corresponding to key recovery attacks and forgery attacks.
 - **Key Recovery:** Corresponds to solving the Hint-RLWE problem, c.f. Theorem 4.3. We use the reduction from RLWE in Section G to bound the effective standard deviation of the secret after revealing up to Q_5 signatures.
 - **Forgery:** Based on the OM-ISIS problem, c.f. Theorem 4.3, with bounds derived from the correctness analysis of our scheme (see Theorem 5.1). We also verify that attacks described in [AKSY22], which exploit the one-more nature of the problem, are ineffective for our parameters, confirming that the best attack reduces to standard MSIS.
 - **Statistical vs Computational:** The security proof further relies on statistical arguments to move between hybrids. For the *Compact Instantiation*, we follow the proof as is, which requires larger parameters to ensure negligible statistical distance. For the *Easy to Deploy Instantiation*, we heuristically replace these statistical arguments with computational RLWE hardness.

We evaluate the concrete hardness of MLWE and MSIS instances using the lattice estimator [APS15] and compute proof sizes⁵ using the Lazer library [LSS24], without encryption-to-the-sky. An overview of our instantiations’ signature sizes is given in Table 1 for $T = 16$ signers. Complete parameter sets for both instantiations are provided in Table 4.

⁵Our script can be found in the folder *proof* of our artifact.

6.2 Practical Evaluation

We implemented⁶ our threshold blind signature protocol using Python and the Lazer library [LSS24], evaluating performance in both distributed (EC2 t2.large signers in Ohio and Paris regions, with a user on Mac M3 in New York) and local settings. Table 3 shows practical performance with total execution times in the order of seconds, even in a distributed setting. This demonstrates that our scheme is not only theoretically sound but also practically deployable in real-world distributed environments.

Table 3: Perf. evaluation. D = distributed, L = local.

Phase	2-out-of-3		4-out-of-16	
	D	L	D	L
Total Protocol Time (s)	1.502	0.123	1.933	0.166
Sig. Finalization (s)			2.263	
Sig. Verification (s)			1.270	

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⁶The source code for our implementations is made publicly available at <https://github.com/GuilhemN/Blind-Threshold-Signatures-Lattices>.

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A Detailed Proof System Construction

In this appendix, we provide the detailed construction of the LNP-based NIZK system used in our deployable instantiation. We employ the Lyubashevsky, Nguyen, and Plançon (LNP) proof system [LNP22] directly.

A.1 Proof System

Recall that Π is a NIZK with random oracle H_Π for the relation $\mathcal{R}$ defined in Eq. (4). For our deployable instantiation, we use the LNP proof system [LNP22] directly, making heuristic assumptions about extractability properties. The scheme by Lyubashevsky, Nguyen, and Plançon is efficient, suffices for our relation, and proofs are easy to implement thanks to the recent LaZer library for Π_{LNP} introduced by Lyubashevsky, Seiler, and Steuer [LSS24].

A.1.1 NIZK Construction

The NIZK Π proceeds as follows:

- $\Pi.\text{Setup}^{H_\Pi}(1^\lambda)$: Samples $\text{crs} \leftarrow \Pi_{\text{LNP}}.\text{Setup}^{H_\Pi}(1^\lambda)$ and outputs crs .
- $\Pi.\text{Prove}^{H_\Pi}(\text{crs}, \mathbb{x}, \mathbb{w})$: Given crs and $(\mathbb{x}, \mathbb{w}) \in \mathcal{R}$, outputs $\pi \leftarrow \Pi_{\text{LNP}}.\text{Prove}^{H_\Pi}(\text{crs}, \mathbb{x}, \mathbb{w})$.

- $\Pi.\text{Verify}^{\text{H}\Pi}(\text{crs}, \mathbb{x}, \pi)$: Given crs and statement $\mathbb{x}$, outputs $\Pi_{\text{LNP}}.\text{Verify}^{\text{H}\Pi}(\text{crs}, \mathbb{x}, \pi)$.

Let ν be a parameter of the scheme. To lower the parameter requirements for correctness of the NIZK proof, we decompose $\mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4$ into their ν highest bits $\mathbf{x}_\top$ (resp. low bits $\mathbf{x}_\perp$).

Let $B_{\mathbf{r}} := e^{1/4} \sigma_{\text{rnd}} \sqrt{2n}$. Then, we use LNP to prove the following relation $\mathcal{R}_{\text{E}}$ as

$$\mathcal{R}_{\text{E}} = \left\{ (\mathbb{x}_{\text{E}}, \mathbb{w}_{\text{E}}) \left| \begin{array}{l} \mathbf{A}(2^\nu \mathbf{x}_\top + \mathbf{x}_\perp) = \mathbf{u}, \mathbf{c}_1 = \mathbf{a}\mathbf{r} + \mathbf{r}_1, \\ \mathbf{c}_2 = \mathbf{A}_2 \mathbf{r} + \mathbf{r}_2 + \tilde{p} \cdot [\mathbf{x}_\top \ \mathbf{x}_\perp], \\ \|\mathbf{x}_\perp\|_2 \leq \sqrt{4n}2^\nu, \|\mathbf{x}_\top\|_2 \leq B/2^\nu, \\ \|\mathbf{r}, \mathbf{r}_1, \mathbf{r}_2\|_2 \leq B_{\mathbf{r}} \end{array} \right. \right\}.$$

Note that this relation does no longer prove an exact norm bound on $\mathbf{x}$, and this would incur an increase $\sqrt{4n}2^\nu$ in the OM-ISIS bound in Theorem 4.3, but this has virtually no effect on security for our parameters. Interestingly, we can rewrite $\mathcal{R}_{\text{E}}$ in the form of $\tilde{\mathbf{A}} \cdot \tilde{\mathbf{x}} = \tilde{\mathbf{u}}$ with $\tilde{\mathbf{A}} \in \mathcal{R}_q^{1 \times 8}$.

B Deferred Preliminaries

B.1 Pseudorandom Functions

We only give an informal overview of pseudorandom functions (PRFs) and refer the reader to, *e.g.*, [BS25] for a formal treatment. A PRF f with key-space $\mathcal{K}$, domain $\mathcal{X}$, and range $\mathcal{Y}$ is an efficiently computable, deterministic function $f : \mathcal{K} \times \mathcal{X} \rightarrow \mathcal{Y}$. For a uniformly random key $k \in \mathcal{K}$, a polynomial time adversary cannot distinguish whether it interacts with an oracle that takes inputs x and returns evaluations $f_k(x)$ or values from a truly random function.

B.2 Shamir's Secret Sharing

We recap the secret sharing technique of Shamir [Sha79]. Let $\mathbf{s} \in \mathcal{R}_q$ be a secret to be shared. Let $N < q$ and $S \subseteq [N]$ such that $|S| \geq T$. Further, we require that for all $i, j \in [N], i \neq j : (i - j) \in \mathbb{Z}_q^\times$.

Definition B.1. A T -out-of- N secret sharing of s is the set $(\vec{P}(i))_{i \in [N]}$, where $\vec{P}(\cdot) \in \mathcal{R}_q[X]$ is sampled uniformly at random conditioned on $\deg(\vec{P}) = T - 1$ and $\vec{P}(0) = \mathbf{s}$. We write $([\mathbf{s}])_{i \in [N]}$ to denote the shares of $\mathbf{s}$.

We further define for $i \in S$ the Lagrange coefficients

$$L_{S,i} := \prod_{j \in S \setminus \{i\}} \frac{-j}{i - j}.$$

Note that $\sum_{i \in S} [\mathbf{s}]_i \cdot L_{S,i} = \mathbf{s}$.

B.3 Blind Signatures

We define Blind Signatures [Cha82].

Definition B.2 (). A round-optimal Blind Signature is a tuple $\text{BS} = (\text{Setup}, \text{KeyGen}, \text{Signer}, \text{User}_0, \text{User}_1, \text{Vfy})$, where

- $\text{pp} \leftarrow \text{Setup}(1^\lambda)$: Given the security parameter 1^λ , it generates public parameters pp . We implicitly assume that all other algorithms take pp as input.
- $(\text{vk}, \text{sk}) \leftarrow \text{KeyGen}(\text{pp})$: Given the public parameters pp , it outputs a key pair (vk, sk) .
- $(\text{pm}^{\text{U}}, \text{st}^{\text{U}}) \leftarrow \text{User}_0(\text{vk}, \mu)$: Given the public key vk and a message μ , it outputs the user message pm^{U} and a new user state st^{U} .

$\text{Exp}_{\mathcal{A}}^{\text{blind}}(\lambda)$
1 : $\text{pp} \leftarrow \text{Setup}(1^\lambda)$
2 : $(\text{vk}, \mu_0, \mu_1) \leftarrow \mathcal{A}(\text{pp})$
3 : $b \leftarrow \{0, 1\}$
4 : $(\text{pm}_0^{\text{U}}, \text{st}_0^{\text{U}}) \leftarrow \text{User}_0(\text{vk}, \mu_0)$
5 : $(\text{pm}_1^{\text{U}}, \text{st}_1^{\text{U}}) \leftarrow \text{User}_0(\text{vk}, \mu_1)$
6 : $(\text{pm}_b^{\text{S}}, \text{pm}_{1-b}^{\text{S}}) \leftarrow \mathcal{A}(\text{pm}_b^{\text{U}}, \text{pm}_{1-b}^{\text{U}})$
7 : $\text{sig}_0 \leftarrow \text{User}_1(\text{st}_0^{\text{U}}, \text{pm}_0^{\text{S}})$
8 : $\text{sig}_1 \leftarrow \text{User}_1(\text{st}_1^{\text{U}}, \text{pm}_1^{\text{S}})$
9 : if $\text{Vfy}(\text{vk}, \mu_0, \text{sig}_0) = 0 \vee \text{Vfy}(\text{vk}, \mu_1, \text{sig}_1) = 0$ then
10 : $(\text{sig}_0, \text{sig}_1) \leftarrow (\perp, \perp)$
11 : $b' \leftarrow \mathcal{A}(\text{sig}_0, \text{sig}_1)$
12 : return $b = b'$

Figure 3: The blindness experiment $\text{Exp}_{\mathcal{A}}^{\text{blind}}$ for (non-threshold) blind signatures.

- $(\text{pm}^{\text{S}}, \text{aux}_{\text{sig}}) \leftarrow \text{Signer}(\text{sk}, \text{pm}^{\text{U}})$: Given the signing key sk and the user's first message pm^{U} , it outputs the signer's message pm^{S} with auxiliary information aux_{sig} . The auxiliary information is only needed in the security proof of the threshold version of the protocol and unforgeability holds even when the adversary sees aux_{sig} .
- $\text{sig}/\perp \leftarrow \text{User}_1(\text{st}^{\text{U}}, \text{pm}^{\text{S}})$: Given the user's state st^{U} and the signer's message pm^{S} , it outputs $\perp$ if a failure occurred and else the signature sig .
- $1/0 \leftarrow \text{Vfy}(\text{vk}, \mu, \text{sig})$: Given the public key vk , a message μ and a signature sig , it outputs 1 if the signature verifies for that message and otherwise 0.

We define correctness, one-more unforgeability and blindness under malicious keys.

Definition B.3 (Correctness). A blind signature BS is perfectly correct if for all $\text{pp} \leftarrow \text{Setup}(1^\lambda)$, $(\text{vk}, \text{vk}) \in \text{KeyGen}(\text{pp})$ and all messages μ it holds that $\text{Vfy}(\text{vk}, \mu, \text{sig}) = 1$ except with negligible probability, where $(\text{pm}^{\text{U}}, \text{st}^{\text{U}}) \leftarrow \text{User}_0(\text{vk}, \mu)$, $\text{pm}^{\text{S}} \leftarrow \text{Signer}(\text{sk}, \text{pm}^{\text{U}})$, and $\text{sig} \leftarrow \text{User}_1(\text{st}^{\text{U}}, \text{pm}^{\text{S}})$.

Definition B.4 (Blindness under Malicious Keys). Let BS be a blind signature scheme. Let $\mathcal{A}$ be a stateful PPT adversary. We define $\mathcal{A}$'s advantage in the blindness experiment of Figure 3 as

$$\text{Adv}_{\text{BS}, \mathcal{A}}^{\text{blind}}(\lambda) := 2 \left| \Pr[\text{Exp}_{\mathcal{A}}^{\text{blind}}(\lambda) = 1] \right|.$$

We say that BS is blind if $\text{Adv}_{\text{BS}, \mathcal{A}}^{\text{blind}}(\lambda) = \text{negl}(\lambda)$ for all PPT $\mathcal{A}$.

Definition B.5 (One-more Unforgeability). Let BS be a blind signature scheme. Let $\mathcal{A}$ be a stateful PPT adversary. We define $\mathcal{A}$'s advantage in the one more-unforgeability experiment of Figure 4 as

$$\text{Adv}_{\text{BS}, \mathcal{A}}^{\text{bs-omuf}}(\lambda) := \Pr[\text{Exp}_{\mathcal{A}}^{\text{omuf}}(\lambda) = 1].$$

We say that BS is one-more unforgeable (OMUF) if $\text{Adv}_{\text{BS}, \mathcal{A}}^{\text{bs-omuf}}(\lambda) = \text{negl}(\lambda)$ for all PPT $\mathcal{A}$.

$\text{Exp}_{\mathcal{A}}^{\text{omuf}}(\lambda)$
1 : $\text{pp} \leftarrow \text{Setup}(1^\lambda)$ 2 : $(\text{vk}, \text{sk}) \leftarrow \text{KeyGen}(\text{pp})$ 3 : $Q \leftarrow 0$ 4 : $((\mu_1, \text{sig}_1), \dots, (\mu_{\ell+1}, \text{sig}_{\ell+1})) \leftarrow \mathcal{A}^{\text{O}^{\text{Signer}}}(\text{vk})$ 5 : req $Q \leq \ell \wedge \{\mu_1, \dots, \mu_{\ell+1}\} = \ell + 1$ 6 : return $\forall i \in [\ell + 1] : \text{Vfy}(\text{vk}, \mu_i, \text{sig}_i) = 1$ $\text{O}^{\text{Signer}}(\text{pm}^U)$ 1 : $Q \leftarrow Q + 1$ 2 : $(\text{pm}^S, \text{aux}_{\text{sig}}) \leftarrow \text{Signer}(\text{sk}, \text{pm}^U)$ 3 : return pm^S

Figure 4: One-more unforgeability experiment $\text{Exp}_{\mathcal{A}}^{\text{omuf}}$ for (non-threshold) blind signatures.

B.4 Threshold Blind Signatures

We introduce threshold blind signatures following the definition of Lehmann et al. [LNÖ25] for $r = 3$ rounds of interaction.

Definition B.6. A three-round Threshold Blind Signature is a tuple of PPT algorithms $\text{TBS} = (\text{Setup}, \text{KeyGen}, \text{User}_0, \text{User}_1, \text{User}_2, \text{User}_3, \text{Signer}_1, \text{Signer}_2, \text{Signer}_3)$, as follows:

- $\text{pp} \leftarrow \text{Setup}(1^\lambda)$: Given the security parameter as input, it outputs public parameters pp . We implicitly assume that all other algorithms take pp as input.
- $(\text{vk}, (\text{sk}_i)_{i \in [N]}, \text{aux}) \leftarrow \text{KeyGen}(N, T)$: Given the threshold parameters $N, T \in \mathbb{N}$ with $T < N$, it outputs a public key vk , N secret keys $(\text{sk}_i)_{i \in [N]}$ and some auxiliary information aux .
- $(\text{st}_0^U, \text{pm}_0^U) \leftarrow \text{User}_0(\text{vk}, \text{aux}, \mu, \text{SS})$: Given a public key vk , auxiliary information aux , a message to be signed μ and the set of signers SS that are going to sign the message, it outputs a first message of the user pm_0^U and a new user state st_0^U . The auxiliary information comes from key generation and is public.
- $(\text{st}_1^{(S,i)}, \text{pm}_1^{(S,i)}) \leftarrow \text{Signer}_1(i, \text{sk}_i, \text{aux}, \text{ctx}, \text{pm}_0^U)$: Given the index $i \in [N]$ of the signer, the i -th secret key sk_i , the auxiliary information aux , the context information ctx and the user's first message pm_0^U , it outputs a new state for the i -th signer $\text{st}_1^{(S,i)}$ and a message from that signer $\text{pm}_1^{(S,i)}$.
- $(\text{st}_1^U, \text{pm}_1^U) \leftarrow \text{User}_1(\text{st}_0^U, (\text{pm}_1^{(S,i)})_{i \in \text{SS}})$: Given the user's state from the previous round st_0^U and the messages $(\text{pm}_1^{(S,i)})_{i \in \text{SS}}$ of all signers in the signer set SS , it outputs a new state st_1^U and a new message pm_1^U .
- $(\text{st}_2^{(S,i)}, \text{pm}_2^{(S,i)}, \text{ssid}_i, \text{SS}_i) \leftarrow \text{Signer}_2(i, \text{st}_1^{(S,i)}, \text{pm}_1^U)$: Given the old state of the i -th signer $\text{st}_1^{(S,i)}$ and the second message from the user pm_1^U , it outputs a new state $\text{st}_2^{(S,i)}$, a next message $\text{pm}_2^{(S,i)}$, a subsession identifier⁷ ssid_i and a signer set SS_i that the i -th signer thinks it is running the protocol for.
- $(\text{st}_2^U, \text{pm}_2^U) \leftarrow \text{User}_2(\text{st}_1^U, (\text{pm}_2^{(S,i)})_{i \in \text{SS}})$: Given the previous state of the user st_1^U and the messages from all the signers $(\text{pm}_2^{(S,i)})_{i \in \text{SS}}$, it outputs an updated state st_2^U and a next user message pm_2^U .

⁷The ssid will typically contain a part of the transcript together with context information that identifies the running session.

- $(\text{pm}_3^{(S,i)}, \text{aux}_{\text{sig}}) \leftarrow \text{Signer}_3(i, \text{st}_2^{(S,i)}, \text{pm}_2^U)$: Given an index $i \in [N]$, the previous state of the signer $\text{st}_2^{(S,i)}$ and the user's message pm_2^U it outputs the final message $\text{pm}_2^{(S,i)}$ of the signer with auxiliary information aux_{sig} . The auxiliary information is only needed in the security proof of the protocol and unforgeability holds even when the adversary sees aux_{sig} .
- $\text{sig}/\perp \leftarrow \text{User}_3(\text{st}_2^U, (\text{pm}_3^{(S,i)})_{i \in \text{SS}})$: Given the previous state of the user st_2^U and the messages $\text{pm}_3^{(S,i)}$ from all signers $i \in \text{SS}$ it outputs either $\perp$ to indicate failure or a signature sig if no failure occurred.
- $1/0 \leftarrow \text{Vfy}(\text{vk}, \mu, \text{sig})$: Given the public key vk , a message μ and a signature sig , it outputs 1 if the signature verifies for that message and otherwise 0.

We remark that in our definition Signer_2 outputs $(\text{ssid}_i, \text{SS}_i)$ instead of Signer_3 . The values $(\text{ssid}_i, \text{SS}_i)$ are relevant for the definition of one-more unforgeability which differs from the definition in [LNÖ25]. We discuss this below.

We recall correctness which states that if the signing protocol is followed, then the signature output by User_3 is valid. We note that we allow for (negligible) correctness error ε in our definition.

Definition B.7 (Correctness). Let $\lambda, N, T \in \mathbb{N}$. Let ε be a negligible function in λ . A threshold blind signature scheme TBS is ε -correct if for all $\text{PP} \leftarrow \text{Setup}(1^\lambda)$, for all $(\text{vk}, (\text{sk}_i)_{i \in [N]}, \text{aux}) \leftarrow \text{TBS.KeyGen}(\text{PP}, N, T)$, for all μ in the message space, and all signer sets $\mathcal{S} \subseteq [N]$ such that $|\mathcal{S}| = T$, the signing protocol $\langle \text{User}, \text{Signer} \rangle$ outputs a signature sig such that $\text{TBS.Vfy}(\text{vk}, \text{sig}, \mu) = 1$ with probability $1 - \varepsilon$.

Defining unforgeability for threshold blind signatures is rather complex since the message is hidden and the signing protocol is distributed amongst T signers, it is non-trivial to count the number of issued signatures. We recall one-more unforgeability following the strongest definition OMUF-3 within the hierarchy given in [LNÖ25] except for some small modifications. Intuitively, OMUF-3 as defined in [LNÖ25] states that a malicious user cannot produce more than ctr valid signatures for distinct signature-message pairs, where ctr is the number of signatures that can be obtained from honest runs of the protocol. In OMUF-3, the game keeps track of ctr through a counter⁸: The counter is incremented whenever $T - C$ honest shares are issued through the signing oracle for the same session ssid and intended set SS of co-signers, where C is the number of corrupted users.

Let us discuss our modification:

1. In [LNÖ25], the session ssid_i and set of co-signers SS_i with respect to which signer i signs is fixed in the final round. If $T - C$ signers ran the last-round signing protocol for identical ssid and SS , then ctr is incremented. In our definition, these steps happen in round 2 instead of round 3. While this weakens the definition slightly, this is a common limitation in classical and post-quantum threshold signatures (e.g., [DKM⁺24, CKM23a]). We remark that our definition still ensures *session agreement* (i.e., which blind signing sessions go together) and *signer set agreement* (i.e., signers agree on the signing set SS and can choose to avoid co-signing with suspicious parties). We refer to [LNÖ25] for more details.
2. Our definition ensures that it is hard to produce more than ctr valid signatures for *distinct messages*, but not distinct signature-message pairs. This definition is standard for lattice-based blind signatures [AKSY22, dPK22, BLNS23, JS24]. That is, we do achieve standard unforgeability but not strong unforgeability.
3. Our definition allows the signers to output some auxiliary information aux_{sig} along with their last message. This value is only for the security proof. The underlying Plover [EEN⁺24] signature has the same property. Unforgeability holds even when the adversary sees aux_{sig} .

⁸More precisely, the value of the counter ctr is implicitly represented by the number of elements in ES in Fig. 5.

$\text{Exp}_{\text{TBS}, \mathcal{A}, n, t}^{\text{OMUF-3}}(1^\lambda)$
<pre> 1 : $\text{pp} \leftarrow \text{Setup}(1^\lambda)$ 2 : $S_1, \dots, S_r := \emptyset$ 3 : $\text{ES} := \emptyset$ 4 : $\text{El}[\cdot, \cdot] := \emptyset$ 5 : $\text{CS} \leftarrow \mathcal{A}(\text{pp}, N, T)$ 6 : if $\text{CS} \geq T$ then return false 7 : $(\text{vk}, (\text{sk}_i)_{i \in [N]}, \text{aux}) \leftarrow \text{KeyGen}(\text{pp}, N, T)$ 8 : $(\ell, (\mu_k^*, \text{sig}_k^*)_{k \in [\ell]}) \leftarrow \mathcal{A}^{\text{Signer}(\cdot)}(\text{vk}, (\text{sk}_i)_{i \in \text{CS}}, \text{aux})$ 9 : return $(\ell > \text{ES}) \wedge \forall k \in [\ell] : \left(\begin{array}{l} \text{Vfy}(\text{vk}, \text{sig}_k^*, \mu_k^*) = 1 \\ \wedge \forall j \in [\ell] \setminus \{k\} : \mu_k^* \neq \mu_j^* \end{array} \right)$ </pre>
$\mathcal{O}^{\text{Signer}}(j, \text{sid}, i, \text{ctx}_{\text{id}}^{\text{sid}}, \text{pm}_{j-1}^{\text{U}})$
<pre> 1 : return $\perp$ if $(i, \text{sid}) \in S_j$ 2 : return $\perp$ if $j > 1 \wedge (i, \text{sid}) \notin S_1, \dots, S_{j-1}$ 3 : $S_j := S_j \cup \{(i, \text{sid})\}$ 4 : if $j = 1$ then $(\text{st}_1^{(S, i)}, \text{pm}_1^{(S, i)}) \leftarrow \text{Signer}_1(i, \text{sk}_i, \text{aux}, \text{ctx}_{\text{id}}^{\text{sid}}, \text{pm}_0^{\text{U}})$ 5 : else if $j = r$: $(\text{st}_j^{(S, i)}, \text{pm}_j^{(S, i)}, \text{aux}_{\text{sig}}) \leftarrow \text{Signer}_j(i, \text{st}_{j-1}^{(S, i)}, \text{pm}_{j-1}^{\text{U}})$ 6 : return $(\text{pm}_j^{(S, i)}, \text{aux}_{\text{sig}})$ 7 : else : $(\text{st}_j^{(S, i)}, \text{pm}_j^{(S, i)}) \leftarrow \text{Signer}_j(i, \text{st}_{j-1}^{(S, i)}, \text{pm}_{j-1}^{\text{U}})$ 8 : if $j = r - 1$ then 9 : $(\text{ssid}_i^{\text{sid}}, \text{SS}_i^{\text{sid}}) := \text{st}_{r-1}^{(S, i)}$ 10 : $\mathcal{S}_{\text{hon}} := \text{SS}_i^{\text{sid}} \setminus \mathcal{C}$ 11 : $\text{El}[\text{ssid}_i^{\text{sid}}, \mathcal{S}_{\text{hon}}] := \text{El}[\text{ssid}_i^{\text{sid}}, \mathcal{S}_{\text{hon}}] \cup \{i\}$ 12 : if $\mathcal{S}_{\text{hon}} \subseteq \text{El}[\text{ssid}_i^{\text{sid}}, \mathcal{S}_{\text{hon}}]$ then $\text{ES} := \text{ES} \cup \{\text{ssid}_i^{\text{sid}}\}$ 13 : return $(\text{pm}_{r-1}^{(S, i)}, \text{ssid}_i^{\text{sid}}, \text{SS}_i)$ 14 : return $\text{pm}_j^{(S, i)}$ </pre>

Figure 5: One-More Unforgeability Definition OMUF-3 based on [LNÖ25]. Our modifications are discussed before Theorem B.8.

Definition B.8. Let $\lambda, N, T \in \mathbb{N}$ such that $T \leq N$. A threshold blind signature scheme TBS is 3-one more unforgeable if for all PPT adversaries $\mathcal{A}$:

$$\Pr \left[\text{Exp}_{\text{TBS}, \mathcal{A}, n, t}^{\text{OMUF-3}}(\lambda) = 1 \right] \leq \text{negl}(\lambda),$$

where the experiment is defined in Figure 5.

We recall blindness which ensures that an adversary cannot link signatures (obtained through the threshold blind signing protocol) to its issuance session.

Definition B.9. Let $\lambda \in \mathbb{N}$. A threshold blind signature scheme TBS is blind if for all PPT adversaries $\mathcal{A}$:

$$\left| \Pr \left[\text{Exp}_{\text{TBS}, \mathcal{A}}^{\text{blind}}(\lambda) = 1 \right] - \frac{1}{2} \right| \leq \text{negl}(\lambda),$$

where the experiment is defined in Figure 6.

$\text{Exp}_{\text{TBS}, \mathcal{A}}^{\text{blind}}(1^\lambda)$
1: $\text{PP} \leftarrow \text{Setup}(1^\lambda)$ 2: $b \leftarrow \{0, 1\}$ 3: $S_1, \dots, S_r := \emptyset$ 4: $b' \leftarrow \mathcal{A}^{\text{User}_0, \dots, \text{User}_r}$ 5: return $b = b'$
$\mathcal{O}^{\text{User}_0}(\text{sid}, \text{vk}^{\text{sid}}, \text{aux}^{\text{sid}}, \mu_0^{\text{sid}}, \mu_1^{\text{sid}}, \text{SS}_0^{\text{sid}}, \text{SS}_1^{\text{sid}})$ 1: return $\perp$ if $\text{sid} \in S_1$ 2: $S_1 := S_1 \cup \{\text{sid}\}$ 3: $(\text{st}_{0,0}^{\text{U}}, \text{pm}_{0,0}^{\text{U}}) \leftarrow \text{User}_0(\text{vk}^{\text{sid}}, \text{aux}^{\text{sid}}, \mu_b^{\text{sid}}, \text{SS}_b^{\text{sid}})$ 4: $(\text{st}_{1,0}^{\text{U}}, \text{pm}_{1,0}^{\text{U}}) \leftarrow \text{User}_0(\text{vk}^{\text{sid}}, \text{aux}^{\text{sid}}, \mu_{1-b}^{\text{sid}}, \text{SS}_{1-b}^{\text{sid}})$ 5: return $(\text{pm}_{0,0}^{\text{U}}, \text{pm}_{1,0}^{\text{U}})$
$\mathcal{O}^{\text{User}_j}(\text{sid}, (\text{pm}_{0,j}^{(\text{S},i)})_{i \in \text{SS}_0^{\text{sid}}}, (\text{pm}_{1,j}^{(\text{S},i)})_{i \in \text{SS}_1^{\text{sid}}})$ 1: return $\perp$ if $\text{sid} \in S_j \wedge \text{sid} \notin S_{j-1}$ 2: $S_j := S_j \cup \{\text{sid}\}$ 3: $(\text{st}_{0,j}^{\text{U}}, \text{pm}_{0,j}^{\text{U}}) \leftarrow \text{User}_j(\text{st}_{0,j-1}^{\text{U}}, (\text{pm}_{0,j}^{(\text{S},i)})_{i \in \text{SS}_0^{\text{sid}}})$ 4: $(\text{st}_{1,j}^{\text{U}}, \text{pm}_{1,j}^{\text{U}}) \leftarrow \text{User}_j(\text{st}_{1,j-1}^{\text{U}}, (\text{pm}_{1,j}^{(\text{S},i)})_{i \in \text{SS}_1^{\text{sid}}})$ 5: return $(\text{pm}_{0,j}^{\text{U}}, \text{pm}_{1,j}^{\text{U}})$
$\mathcal{O}^{\text{User}_r}(\text{sid}, (\text{pm}_{0,r}^{(\text{S},i)})_{i \in \text{SS}_0^{\text{sid}}}, (\text{pm}_{1,r}^{(\text{S},i)})_{i \in \text{SS}_1^{\text{sid}}})$ 1: return $\perp$ if $\text{sid} \in S_r \wedge \text{sid} \notin S_{r-1}$ 2: $S_r := S_r \cup \{\text{sid}\}$ 3: $\text{sig}_b^{\text{sid}} \leftarrow \text{User}_r(\text{st}_{0,r-1}^{\text{U}}, (\text{pm}_{0,r}^{(\text{S},i)})_{i \in \text{SS}_0^{\text{sid}}})$ 4: $\text{sig}_{1-b}^{\text{sid}} \leftarrow \text{User}_r(\text{st}_{1,r-1}^{\text{U}}, (\text{pm}_{1,r}^{(\text{S},i)})_{i \in \text{SS}_1^{\text{sid}}})$ 5: if $\text{sig}_0^{\text{sid}} = \perp \vee \text{sig}_1^{\text{sid}} = \perp$ then return $(\perp, \perp)$ 6: return $(\text{sig}_0^{\text{sid}}, \text{sig}_1^{\text{sid}})$

Figure 6: Blindness experiment for TBS from [LNÖ25], where $j \in \{1, \dots, r-1\}$.

C Full Proofs for BS_B

In this section, we give full proofs for blindness (Theorem 4.2) and one-more unforgeability (Theorem 4.3) of the blind signature scheme BS_B .

C.1 Proof of Correctness (Theorem 4.1)

Proof. We first recall a useful tail bound lemma from [DKM⁺24, Lemma 3.4], derived from [Lyu12, Lemma 4.4].

Lemma C.1. *Let $\mathbf{s} \leftarrow \mathcal{D}_{\mathcal{R}, \sigma}^\ell$, and $\mathbf{c} \in \mathcal{R}$. Then, we have*

$$\Pr[\|\mathbf{c} \cdot \mathbf{s}\|_2 \geq e^{1/4} \cdot \|\mathbf{c}\|_1 \sigma \sqrt{n\ell}] \leq 2^{-n\ell/10}$$

Let us check that the witness $\mathbf{w} = \mathbf{y} - (\mathbf{x}' + \mathbf{x})$ multiplied by $\mathbf{A}$ correctly recovers the target $\mathbf{u}$.

$$\begin{aligned} \mathbf{A} \cdot (\mathbf{y} - (\mathbf{x} + \mathbf{x}')) &= \mathbf{z}'_1 + \mathbf{a}\mathbf{z}_2 + \mathbf{b}\mathbf{c}_1 - \mathbf{A}(\mathbf{x} + \mathbf{x}') \\ &= \mathbf{u} + \mathbf{A}(\mathbf{x}' + \mathbf{x}) - \mathbf{a}\mathbf{z}_2 - \mathbf{b}\mathbf{c}_1 + \mathbf{a}\mathbf{z}_2 + \mathbf{b}\mathbf{c}_1 - \mathbf{A}(\mathbf{x} + \mathbf{x}') \\ &= \mathbf{u} \end{aligned}$$

Now, we would like to bound the norm of $\mathbf{w}$. First recall that $\mathbf{y} = (\mathbf{z}'_1, \mathbf{z}_2, \mathbf{c}_1, 0)^\top$. In particular, we have

$$\begin{aligned} \mathbf{z}'_1 &= \mathbf{t} - \tilde{\mathbf{e}} - \mathbf{a}\mathbf{z}_2 - \mathbf{b}\mathbf{c}_1 \\ &= \mathbf{t} - \mathbf{w} + \mathbf{w} - \tilde{\mathbf{e}} - \mathbf{a}(\mathbf{c}_1\mathbf{s} + \mathbf{r}) - \mathbf{b}\mathbf{c}_1 \\ &= \beta\mathbf{c}_1 + \mathbf{c}_2 + \mathbf{a}\mathbf{r} + \mathbf{p} - \tilde{\mathbf{e}} - \mathbf{a}\mathbf{r} - \mathbf{a}\mathbf{c}_1\mathbf{s} - (\beta - \mathbf{a}\mathbf{s} + \mathbf{e})\mathbf{c}_1 \\ &= \mathbf{c}_2 + \mathbf{p} + \mathbf{e}\mathbf{c}_1 - \tilde{\mathbf{e}} = \mathbf{z}_1 + \mathbf{c}_2 - \tilde{\mathbf{e}} \end{aligned}$$

Now, let's bound elements in $\mathbf{w}$. $\mathbf{w}$ can be decomposed as the sum of:

1. $\mathbf{c}_1 \cdot [\mathbf{e} \ \mathbf{s} \ 0 \ 0]^\top$, whose norm can be bounded overwhelmingly by

$$\|\mathbf{c}_1\|_1 \cdot \|(\mathbf{e}, \mathbf{s})\|_2 \leq \sqrt{2} \cdot n^{3/2} \cdot \left\lceil \frac{q-1}{2\beta} \right\rceil \cdot e^{1/4} \sigma_{\text{sk}}.$$

2. $\mathbf{x}'$ whose norm can be bounded overwhelmingly by

$$\sqrt{2n} \cdot e^{1/4} \cdot \sigma_{\mathbf{u}, \mathbf{x}}$$

3. $\mathbf{x} + [\mathbf{p} \ \mathbf{r} \ 0 \ 0]$ whose norm can be bounded overwhelmingly by $\sqrt{n} \cdot e^{1/4} \cdot \sqrt{2\sigma_{\text{rnd}}^2 + 4\sigma_{\mathbf{s}, \mathbf{x}}^2}$ by first observing that the convolution of these two vectors is indistinguishable from a Gaussian with larger parameters applying Theorem 2.3, and then applying the tail bound lemma.

4. $\tilde{\mathbf{e}} + \mathbf{c}_2$ whose norm can be bounded overwhelmingly by $\sqrt{n} \frac{\beta + \tilde{p}}{2}$.

We obtain the expression of B applying the triangular inequality to the above terms, ensuring that $\mathbf{w}$ has a norm lower than B with overwhelming probability. $\square$

C.2 Full Proof of Blindness (Theorem 4.2)

Proof. We show a sequence of hybrid experiments such that the view of $\mathcal{A}$ is independent of μ_0 and μ_1 .

Game $\mathbf{G}_0$: This is the original blindness experiment.

Game $\mathbf{G}_1$: In this game, we simulate the random oracle $\mathbf{H}_\Pi$ and zero-knowledge proofs π . That is, denote by $\text{Sim} = (\text{Sim}_0, \text{Sim}_1)$ the zero-knowledge simulator of Π . We simulate $\mathbf{H}_\Pi$ via Sim_0 and in both sessions, we set $\pi_i \leftarrow \text{Sim}_1(\text{crs}, \mathbf{x}_i)$ in $\mathcal{O}^{\text{User}_1}$ for $i \in \{0, 1\}$ when sig_b and sig_{1-b} are derived, where the statement $\mathbf{x}_i = (\mathbf{A}, \mathbf{u}_i)$. Note that π_i is only computed if $(\mathbf{x}_i, \mathbf{w}_i) \in \mathcal{R}$ due to the check in line 7, else $\mathcal{O}^{\text{User}_1}$ outputs $(\perp, \perp)$. Therefore, we have that

$$|\Pr[\mathbf{G}_1] - \Pr[\mathbf{G}_0]| \leq \text{Adv}_{\Pi, B}^{\text{ZK}}(\lambda).$$

Game $\mathbf{G}_2$: In this game, we replace $\mathbf{t}'_i = \mathbf{A}\mathbf{x}'_i + \mathbf{u}_i$ with a random $\mathbf{t}'_i \leftarrow \mathcal{R}_q$. An adversary that can distinguish $\mathbf{A}\mathbf{x}'_i$ from a random value can easily be turned into an adversary $\mathcal{B}$ breaking RLWE with respect to the matrix $[1 \ \mathbf{a}]$. A simple hybrid argument yields

$$|\Pr[\mathbf{G}_2] - \Pr[\mathbf{G}_1]| \leq 2\text{Adv}_{\mathcal{B}, q, \chi_{\mathbf{u}, x}}^{\text{RLWE-}\times}(\lambda).$$

We can conclude that in $\mathbf{G}_2$, the view of $\mathcal{A}$ is independent of the bit b . Therefore, a malicious signer can guess b at most with probability $1/2$. □

C.3 Full Proof of Unforgeability (Theorem 4.3)

Before presenting the full proof, we introduce a useful lemma on the spectral norm of a matrix $\mathbf{T} \in \mathcal{R}^4$ with Gaussian coefficients. It was communicated to us privately and we do not claim ownership of it.

Lemma C.2. *Let $\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3 \leftarrow \mathcal{D}_{\mathcal{R}, \sigma}$, and define $\mathbf{T} = [\mathbf{s}_1 \ \mathbf{s}_2 \ \mathbf{s}_3 \ 1]$. We have $s_1(\mathbf{T})^2 \leq 6n\sigma(\lambda \log 2 + \log(6n)) + 1$ with all but probability $2^{-\lambda}$.*

Proof. We start by bounding the spectral norm of a single polynomial $\mathbf{f} \leftarrow \mathcal{D}_{\mathcal{R}, \sigma}$. Let's note $\mathbf{f} = \sum_{i=1}^n f_i x^i$. Let us also note $\Omega = (\zeta)_{\zeta \in \omega}$ the set of complex roots of $x^n + 1$, and $\mathbf{V} = (\zeta^j)_{\zeta \in \Omega, 1 \leq j \leq n}$ the Vandermonde matrix associated to Ω . Since $\mathcal{M}(\mathbf{f})$ can be diagonalized as $\mathcal{M}(f) = \mathbf{V} \cdot \text{Diag}((f(\zeta))_{\zeta \in \Omega}) \cdot \mathbf{V}^\top$, this also implies that the set of singular values of $\mathcal{M}(f)$ is $\{|f(\zeta)|, \zeta \in \Omega\}$. Our goal is to upper bound the $|f(\zeta)|$'s.

We observe that $f(\zeta) = \langle (f_i)_{1 \leq i \leq n}, (\zeta^i)_{1 \leq i \leq n} \rangle$. This inner product can be bounded using Lemma 4.3 from [Lyu12] which states that $|\langle (f_i)_{1 \leq i \leq n}, (\zeta^i)_{1 \leq i \leq n} \rangle| \leq r$ except with probability at most $\frac{2^{-\lambda}}{3n}$, as long as

$$r \geq \sigma \sqrt{2n} \sqrt{\lambda \log 2 + \log(6n)}$$

Now, let us compute the spectral norm of $\mathbf{T} \cdot \mathbf{T}^\top = \sum_{i=1}^3 s_i^2 + 1 := \mathbf{a}$. Using the union bound, we have $|f(\zeta)|$ for every polynomial entry f in $\mathbf{T}$ and every $\zeta \in \Omega$. This implies $|\mathbf{a}(\zeta)| \leq 3r^2 + 1$, which concludes the proof as $s_1(\mathbf{T}) = \sqrt{s_1(\mathbf{a})}$. □

We now move to the actual proof of unforgeability of our blind signature.

Proof. We show a sequence of hybrid experiments such that after $\mathbf{G}_6$ we can construct an adversary $\tilde{\mathcal{B}}$ against the OM-ISIS assumption with advantage roughly that of $\mathcal{A}$ in $\mathbf{G}_6$. We write $\Pr[\mathbf{G}_i]$ as shorthand for the probability that the i -th hybrid experiment outputs 1.

Game $\mathbf{G}_0$: This is the original unforgeability experiment.

Game $\mathbf{G}_1$: In this game, we change how the crs of Π is computed. Instead of running $\Pi.\text{Setup}$, we set $(\text{crs}, \text{td}_\Pi) \leftarrow \Pi.\text{ExtSetup}(1^\lambda)$ such that we know an extraction trapdoor.

Setup $^{\mathbf{G}_1}(1^\lambda)$

```

1:  $(\text{crs}, \text{td}_\Pi) \leftarrow \Pi.\text{ExtSetup}(1^\lambda)$ 
2:  $\mathbf{a} \leftarrow \mathcal{R}_q^\times, \ \mathbf{d} \leftarrow \mathcal{R}_q$ 
3:  $\text{pp} := (\text{crs}, \mathbf{a}, \mathbf{d})$ 
4: return pp

```

We get

$$|\Pr[\mathbf{G}_1] - \Pr[\mathbf{G}_0]| \leq \text{Adv}_{\Pi, \mathcal{B}}^{\text{CRS}}(\lambda).$$

Game G_2 : In this game, we replace $\mathbf{d}$ by a gadget extending the matrix $[1 \ \mathbf{a} \ \mathbf{b}]$ from [YJW23] for which we know a trapdoor. Concretely, in the setup, we run $\mathbf{d} = \tilde{p} - (\mathbf{a} \cdot \mathbf{s}'_1 + \mathbf{b} \cdot \mathbf{s}'_2 + \mathbf{s}'_3)$ for $\mathbf{s}'_i \leftarrow \chi_{\mathbf{s}'}$ and use $\mathbf{d}$ in the $\mathbf{pp}$, where $\tilde{p} \in \mathcal{R}_q \cap \mathbb{N}$ is such that there exists a $\tilde{q} \in \mathcal{R}_q \cap \mathbb{N}$ with $\tilde{p}\tilde{q} = 0$. Note that now $\mathbf{A}$ and $\mathbf{T}_\mathbf{d} := [\mathbf{s}'_3 \ \mathbf{s}'_1 \ \mathbf{s}'_2 \ 1]$ are a matrix-approximate trapdoor pair, *i.e.*, $\mathbf{A}\mathbf{T}_\mathbf{d} = \tilde{p} \bmod q$ (cf. Section 2.1.1).

Setup $^{G_2}(1^\lambda)$

```

1 : (crs, td $_\Pi$ )  $\leftarrow$   $\Pi$ .ExtSetup( $1^\lambda$ )
2 :  $\mathbf{a} \leftarrow \mathcal{R}_q^\times$ 
3 :  $(\mathbf{s}'_1, \mathbf{s}'_2, \mathbf{s}'_3) \leftarrow \chi_{\mathbf{s}'}$ 
4 :  $\mathbf{d} := \tilde{p} - (\mathbf{a} \cdot \mathbf{s}'_1 + \mathbf{b} \cdot \mathbf{s}'_2 + \mathbf{s}'_3)$ 
5 : return  $\mathbf{pp} := (\text{crs}, \mathbf{a}, \mathbf{d})$ 

```

By the RLWE assumption with respect to the matrix $[1 \ \mathbf{a}]$, $\mathbf{d}$ is indistinguishable from uniform. Formally, we can define an adversary $\mathcal{B}''$ such that

$$|\Pr[G_2] - \Pr[G_1]| \leq \text{Adv}_{\mathcal{B}'', q, \chi_{\mathbf{s}'}}^{\text{RLWE-}\times}(\lambda).$$

Game G_3 : The goal of this game is to extract a portion $\tilde{\mathbf{x}} \leftarrow \chi_{\mathbf{s}'}$ of the randomness in $\mathbf{x} \leftarrow \chi_{\mathbf{s}, \mathbf{x}}$. We will later leverage it to ensure that pre-image samplers are called over a uniform target. To this end, we introduce the distribution $\chi_{\tilde{\mathbf{s}}, \mathbf{x}} = \mathcal{D}_{\mathcal{R}_q, \sqrt{\sigma_{\mathbf{s}, \mathbf{x}}^2 - \sigma_{\mathbf{s}'}^2}}$. In this game, we sample $\tilde{\mathbf{x}} \leftarrow \chi_{\tilde{\mathbf{s}}, \mathbf{x}}$ and $\tilde{\mathbf{x}}' \leftarrow \chi_{\mathbf{s}'}$ and use their sum in the signer's procedure.

Signer $^{G_3}(\text{sk}, \text{pm}^U)$

```

1 :  $\tilde{\mathbf{x}} \leftarrow \chi_{\tilde{\mathbf{s}}, \mathbf{x}}^4$ 
2 :  $\tilde{\mathbf{x}}' \leftarrow \chi_{\mathbf{s}'}$ 
3 :  $\mathbf{x} = \tilde{\mathbf{x}} + \tilde{\mathbf{x}}'$ 
4 :  $\tilde{\mathbf{e}} \leftarrow \chi_{\mathbf{e}}$ 
5 :  $\mathbf{t} := \mathbf{A}\mathbf{x} + \tilde{\mathbf{e}} + \mathbf{t}'$ 
6 :  $(\mathbf{r}, \mathbf{p}) \leftarrow \chi_{\text{rnd}}^2$ 
7 :  $\mathbf{w} := \mathbf{a}\mathbf{r} + \mathbf{p}$ 
8 :  $(\mathbf{c}_1, \mathbf{c}_2) \leftarrow \text{Decomp}_\beta(\mathbf{t} - \mathbf{w})$ 
9 :  $\mathbf{z}_2 := \mathbf{c}_1\mathbf{s} + \mathbf{r}$ 
10 :  $\text{aux}_{\text{sig}} := \mathbf{w}$ 
11 :  $\text{pm}^S := (\mathbf{x}, \mathbf{z}_2, \mathbf{c}_1, \tilde{\mathbf{e}})$ 
12 : return  $(\text{pm}^S, \text{aux}_{\text{sig}})$ 

```

By the convolution Theorem 2.3 and as $1/(\sigma_{\mathbf{s}, \mathbf{x}}^2 - \sigma_{\mathbf{s}'}^2) + 1/\sigma_{\mathbf{s}'}^2 \leq \eta'_\varepsilon(\mathbb{Z}^n)$, the statistical distance with the previous game is bounded by $8 \cdot 4 \cdot \varepsilon$,

$$|\Pr[G_3] - \Pr[G_2]| \leq 32\varepsilon.$$

Game G_4 : The goal of this game is to prepare to simulate signing via Hint-RLWE in the next game, G_5 . To that end, we employ the strategy from [EEN⁺24] to simulate a short preimage of $[1 \ \mathbf{a} \ \mathbf{b}]$ for a random element $\tilde{\mathbf{t}}$. As the user expects a random preimage for $\mathbf{t} = \mathbf{t}' + \mathbf{A}\mathbf{x} + \tilde{\mathbf{e}}$, we employ the trapdoor $\mathbf{T}_\mathbf{d}$ introduced in G_3 to sample $\mathbf{x}$ such that $\mathbf{t} = \tilde{\mathbf{t}}$. We define the pre-image sampling parameter $r = (q/\tilde{p}) \cdot \eta_\varepsilon(\mathbb{Z}^n)$.

In more detail, on every signing request $\mathbf{t}'$, we sample $\mathbf{u}' \leftarrow \mathcal{R}_q$ at random. Then, we decompose this $\mathbf{u}'$ to obtain $(\mathbf{c}_1, \mathbf{c}_2) \leftarrow \text{Decomp}_\beta(\mathbf{u}')$ instead of decomposing $\mathbf{t} - \mathbf{w}$ with $\mathbf{w} = \mathbf{a}\mathbf{r} + \mathbf{p}$. This ensures that the short secrets $\mathbf{r}$ and $\mathbf{p}$ are only used to compute $\mathbf{z}_1$ and $\mathbf{z}_2$, enabling a reduction to Hint-RLWE in the next game. To ensure that $\mathbf{z} = [\mathbf{z}_1 \ \mathbf{z}_2 \ \mathbf{c}_1]^\top$ is correctly distributed, $\mathbf{z}_2$ is further shifted by $\mathbf{c}_2$. Note that $\mathbf{z}$ is a short preimage for $\bar{\mathbf{t}} = [1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z} + \mathbf{A}\tilde{\mathbf{x}}'$. Finally, we use the trapdoor $\mathbf{T}_d$ to sample a short $\tilde{\mathbf{x}}$ such that

$$\mathbf{A}\tilde{\mathbf{x}} + \tilde{\mathbf{e}} = \bar{\mathbf{t}} - \mathbf{t}'. \quad (5)$$

In summary, the signing oracle is simulated as follows:

Signer^{G₄}(sk, pm^U)

```

1 :  $\mathbf{u}' \leftarrow \mathcal{R}_q$ 
2 :  $(\mathbf{c}_1, \mathbf{c}_2) \leftarrow \text{Decomp}_\beta(\mathbf{u}')$ 
3 :  $(\mathbf{r}, \mathbf{p}) \leftarrow \chi_{\text{rnd}}^2$ 
4 :  $\mathbf{w} := \mathbf{a}\mathbf{r} + \mathbf{p}$ 
5 :  $\mathbf{z}_1 := \mathbf{c}_1\mathbf{e} + \mathbf{p} + \mathbf{c}_2, \ \mathbf{z}_2 := \mathbf{c}_1\mathbf{s} + \mathbf{r}, \ \mathbf{c}_1 := \mathbf{c}_1$ 
6 :  $\mathbf{z} = [\mathbf{z}_1 \ \mathbf{z}_2 \ \mathbf{c}_1]^\top$ 
7 :  $\tilde{\mathbf{x}}' \leftarrow \chi_{\mathbf{s}'}$ 
8 :  $\bar{\mathbf{t}} := [1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z} - \mathbf{t}' - \mathbf{A} \cdot \tilde{\mathbf{x}}'$ 
9 :  $\tilde{\mathbf{x}} \leftarrow \text{ApproxPreSamp}(\mathbf{A}, \mathbf{T}_d, \bar{\mathbf{t}}, r, \sqrt{\sigma_{\mathbf{S},x}^2 - \sigma_{\mathbf{s}'}^2})$ 
10 :  $\mathbf{x} = \tilde{\mathbf{x}} + \tilde{\mathbf{x}}'$ 
11 :  $\tilde{\mathbf{e}} := \bar{\mathbf{t}} - \mathbf{A}\mathbf{x} \pmod q$ 
12 :  $\text{aux}_{\text{sig}} := \mathbf{w}$ 
13 :  $\text{pm}^S := (\mathbf{x}, \mathbf{z}_2, \mathbf{c}_1, \tilde{\mathbf{e}})$ 
14 : return  $(\text{pm}^S, \text{aux}_{\text{sig}})$ 
```

Let us show that the view of the user did not change. We have in $\mathbf{G}_4$ due to $\mathbf{b} = \beta - (\mathbf{a}\mathbf{s} + \mathbf{e})$ that

$$\begin{aligned}
[1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z} &= \mathbf{c}_1\mathbf{e} + \mathbf{p} + \mathbf{c}_2 + \mathbf{a}(\mathbf{c}_1\mathbf{s} + \mathbf{r}) + \mathbf{b}\mathbf{c}_1 \\
&= \mathbf{c}_1(\mathbf{a}\mathbf{s} + \mathbf{e}) + \mathbf{c}_2 + \mathbf{a}\mathbf{r} + \mathbf{p} + \mathbf{b}\mathbf{c}_1 \\
&= \mathbf{c}_1\beta + \mathbf{c}_2 + \mathbf{w} \\
&= \mathbf{u}' + \mathbf{w}
\end{aligned}$$

Therefore, $\bar{\mathbf{t}}$ is distributed uniformly over $\mathcal{R}_q$. By the properties of **ApproxPreSamp**, Theorem 2.7, the distribution of $(\mathbf{A}, \tilde{\mathbf{x}}, \mathbf{u}, \tilde{\mathbf{e}})$ in $\mathbf{G}_4$ has statistical distance $\varepsilon_{\text{samp}}$ to that tuple in $\mathbf{G}_3$, *i.e.*, where $\tilde{\mathbf{x}}$ was sampled from $\chi_{\tilde{\mathbf{S},x}}$ and $\tilde{\mathbf{e}}$ from $\chi_{\mathbf{e}}$.

Note that we can apply Theorem 2.7 because our parameter constraints from Section 4 verify the condition $\sigma_{\tilde{\mathbf{S},x}}^2 - \sigma_{\mathbf{s}'}^2 \geq ((q/\tilde{p})^2 + 1)\eta'_\epsilon(\mathbb{Z}^n) \cdot (s_1(\mathbf{T}_d)^2 + 1)$ with overwhelming probability over the randomness of $\mathbf{T}_d$ as proved by Theorem C.2.

Further, as $\bar{\mathbf{t}} = [1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z} - \mathbf{t}' - \mathbf{A} \cdot \tilde{\mathbf{x}}'$ and $\bar{\mathbf{t}} = \mathbf{A}\tilde{\mathbf{x}} + \tilde{\mathbf{e}}$, we have in $\mathbf{G}_4$ that

$$\begin{aligned}
\mathbf{t} - \mathbf{w} &= \mathbf{A}(\tilde{\mathbf{x}}' + \tilde{\mathbf{x}}) + \mathbf{t}' + \tilde{\mathbf{e}} - \mathbf{w} \\
&= \mathbf{A}\tilde{\mathbf{x}}' + \bar{\mathbf{t}} + \mathbf{t}' - \mathbf{w} \\
&= [1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z} - \mathbf{w} \\
&= \mathbf{u}'.
\end{aligned}$$

Hence, $\mathbf{z}_2$ and $\mathbf{z}_3 = \mathbf{c}_1$ are also distributed as in the previous game. In total, we obtain

$$|\Pr[\mathbf{G}_4] - \Pr[\mathbf{G}_3]| \leq \varepsilon_{\text{samp}}.$$

Game $\mathbf{G}_5$: In this game, we exchange the LWE sample $\mathbf{as} + \mathbf{e}$ in the public key by a uniformly random value $\mathbf{b}'$. Note that each signing query still would contain some information about the secret values $\mathbf{s}$ and $\mathbf{e}$. To that end, we can use the “hints” $\mathbf{h}_i = (\mathbf{h}_{1,i}, \mathbf{h}_{2,i})$, for $i \in [Q_S]$ that are computed exactly like the Hint-RLWE challenger would compute them. That is, in the i -th signing query, we now set $\mathbf{z}_1 := \mathbf{h}_{1,i} + \mathbf{c}_2$ and $\mathbf{z}_2 := \mathbf{h}_{2,i}$.

KeyGen $^{\mathbf{G}_5}(1^\lambda)$	Signer $^{\mathbf{G}_5}(\text{sk}, \text{pm}^U)$
1 : $(\mathbf{s}, \mathbf{e}) \leftarrow \chi_{\text{sk}}^2$	1 : $\mathbf{u}' \leftarrow \mathcal{R}_q$
2 : $\mathbf{b}' \leftarrow \mathcal{R}_q^\times$	2 : $(\mathbf{c}_1, \mathbf{c}_2) \leftarrow \text{Decomp}_\beta(\mathbf{u}')$
3 : $\mathbf{b} := \beta - \mathbf{b}'$	3 : $(\mathbf{r}, \mathbf{p}) \leftarrow \chi_{\text{rnd}}^2$
4 : $\mathbf{A} := [1 \ \mathbf{a} \ \mathbf{b} \ \mathbf{d}]$	4 : $\mathbf{w} := \mathbf{ar} + \mathbf{p}$
5 : $\text{vk} := \mathbf{A}, \text{sk} := \perp$	5 : $\mathbf{h}_1 = \mathbf{c}_1\mathbf{e} + \mathbf{p}; \mathbf{h}_2 = \mathbf{c}_1\mathbf{s} + \mathbf{r}$
	6 : $\mathbf{z}_1 := \mathbf{h}_1 + \mathbf{c}_2$
	7 : $\mathbf{z}_2 := \mathbf{h}_2$
	8 : $\mathbf{z}_3 := \mathbf{c}_1$
	9 : $\tilde{\mathbf{x}}' \leftarrow \chi_{\text{s}'}^2$
	10 : $\tilde{\mathbf{t}} := [1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z} - \mathbf{t}' - \mathbf{A} \cdot \tilde{\mathbf{x}}$
	11 : $\tilde{\mathbf{x}} \leftarrow \text{ApproxPreSamp}(\mathbf{A}, \mathbf{T}_d, \tilde{\mathbf{t}}, r, \sqrt{\sigma_{S,x}^2 - \sigma_{s'}^2})$
	12 : $\mathbf{x} := \tilde{\mathbf{x}} + \tilde{\mathbf{x}}'$
	13 : $\tilde{\mathbf{e}} := \tilde{\mathbf{t}} - \mathbf{Ax} \pmod{q}$
	14 : $\text{aux}_{\text{sig}} := \mathbf{w}$
	15 : $\text{pm}^S := (\mathbf{x}, \mathbf{z}_2, \mathbf{c}_1, \tilde{\mathbf{e}})$
	16 : return $(\text{pm}^S, \text{aux}_{\text{sig}})$

Note that the Hint-RLWE challenger computes $\mathbf{h}_{1,i} = \mathbf{c}_{1,i}\mathbf{e} + \mathbf{p}_i$ and $\mathbf{h}_{2,i} = \mathbf{c}_{1,i}\mathbf{s} + \mathbf{r}_i$, where $\mathbf{r}_i, \mathbf{p}_i \leftarrow \chi_{\text{rnd}}, \mathbf{s}, \mathbf{e} \leftarrow \chi_{\text{sk}}, \mathbf{c}_{1,i} \leftarrow \mathcal{C}_1$ for $i \in [Q_S]$. The reduction would further choose $\mathbf{c}_2 \leftarrow \mathcal{C}_2^{\|\mathbf{c}_1\|}$, *i.e.*, the conditional distribution of $\mathbf{c}_2$ conditioned on the event that the challenger outputs $\mathbf{c}_1$. Observe, that $\mathbf{c}_2$ fixes $\text{aux}_{\text{sig}} = \mathbf{w} = \mathbf{t} - \beta\mathbf{c}_1 + \mathbf{c}_2$ because $\beta, \mathbf{c}_1, \mathbf{t}$ are known to the user. Therefore, we get

$$|\Pr[\mathbf{G}_5] - \Pr[\mathbf{G}_4]| \leq \text{Adv}_{B', q, Q_S, \chi_{\text{sk}}^2, \chi_{\text{rnd}}^2, \mathcal{C}_1}^{\text{Hint-RLWE} \times}(\lambda),$$

where Q_S is the number of $\mathcal{O}^{\text{Signer}}$ queries that $\mathcal{A}$ makes.

Game $\mathbf{G}_6$: In this game, after the adversary $\mathcal{A}$ outputs its forgeries $(\mu_i, \text{sig}_i)_{i \in [Q_S+1]}$, the game does not only check if all signatures verify, but additionally runs the extractor on the NIZK proofs, *i.e.*,

$$\forall i \in [Q_S + 1] : w_i \leftarrow \text{Ext}(\overline{\text{crs}}, \text{td}, x_i, \text{sig}_i).$$

If the extraction fails, *i.e.*, there exists $i^* \in [Q_S + 1]$ such that $w_{i^*} \leftarrow \text{Ext}(\overline{\text{crs}}, \text{td}, x_{i^*}, \pi_{i^*})$ but $(x_{i^*}, w_{i^*}) \notin \mathcal{R}$, then the game outputs $\perp$.

One can turn the adversary $\mathcal{A}$ in $\mathbf{G}_6$ and $\mathbf{G}_5$ into an adversary in the straightline extractability experiment from Theorem 2.13. Note that in Equation (2) the winning condition for the adversary is like in $\mathbf{G}_5$, *i.e.*, all proofs must verify, and in Equation (3) the winning condition is like in $\mathbf{G}_6$, *i.e.*, all proofs must verify and the extraction must be successful. Then we get by Theorem 2.13 that

$$\Pr[\mathbf{G}_6] \geq \frac{\Pr[\mathbf{G}_5] - \text{negl}(\lambda)}{\text{poly}(\lambda, Q_H)}.$$

Reduction from OM-ISIS. Let $\mathcal{A}$ be an adversary that wins in $\mathbf{G}_6$ with probability $\Pr[\mathbf{G}_6]$. We construct the adversary $\bar{\mathcal{B}}$ that plays the OM-ISIS game and internally runs $\mathcal{A}$. To generate a public key vk , the reduction $\bar{\mathcal{B}}$ gets $\bar{\mathbf{A}} = [1 \ \mathbf{a} \ \mathbf{b} \ \mathbf{d}]$, *i.e.*, the random matrix provided by the OM-ISIS challenger, and checks if $\mathbf{a} \in \mathcal{R}_q^\times$ and $\mathbf{b} \in \mathcal{R}_q^\times$. If either $\mathbf{a}$ or $\mathbf{b}$ is not invertible, $\bar{\mathcal{B}}$ aborts and outputs $\perp$. Else $\bar{\mathcal{B}}$ gives $\bar{\mathbf{A}}$ as public key to $\mathcal{A}$.

Further, instead of answering to RO calls with a uniformly random value $\mathbf{u}$, $\bar{\mathcal{B}}$ sends a *syndrome query* to the OM-ISIS challenger. The challenger responds with a uniformly random value $\mathbf{u}$ and outputs this as the hash.

Finally, $\bar{\mathcal{B}}$ uses *preimage queries* to respond to signing queries. This means, that on input $\mathbf{t}'$ from the $\mathcal{A}$, $\bar{\mathcal{B}}$ simulates $\mathbf{z}$ as in $\mathbf{G}_6$ and forwards $\bar{\mathbf{t}} = [1 \ \mathbf{a} \ \mathbf{b}] \mathbf{z} - \mathbf{t}' - \mathbf{A} \tilde{\mathbf{x}}' - \tilde{\mathbf{e}}$ to the challenger and receives back a short vector $\tilde{\mathbf{x}}$ such that $\bar{\mathbf{t}} = \bar{\mathbf{A}} \tilde{\mathbf{x}}$, where $\tilde{\mathbf{e}} \leftarrow \chi_{\mathbf{e}}$. Finally, $\bar{\mathcal{B}}$ outputs $(\mathbf{x}, \mathbf{z}_2, \mathbf{c}_1, \tilde{\mathbf{e}})$ as signature.

KeyGen $^{\bar{\mathcal{B}}}(1^\lambda)$	Signer $^{\bar{\mathcal{B}}}(\text{sk}, \mathbf{t}')$
1 : $(\mathbf{s}, \mathbf{e}) \leftarrow \chi_{\text{sk}}^2$	1 : $\mathbf{u}' \leftarrow \mathcal{R}_q$
2 : Obtain $\bar{\mathbf{A}} = [1 \ \mathbf{a} \ \mathbf{b} \ \mathbf{d}]$ from $\mathcal{C}$	2 : $(\mathbf{c}_1, \mathbf{c}_2) \leftarrow \text{Decomp}_\beta(\mathbf{u}')$
3 : if $\mathbf{a} \notin \mathcal{R}_q^\times \vee \mathbf{b} \notin \mathcal{R}_q^\times$:	3 : $(\mathbf{r}, \mathbf{p}) \leftarrow \chi_{\text{rnd}}^2$
4 : abort	4 : $\mathbf{w} := \mathbf{a}\mathbf{r} + \mathbf{p}$
5 : $\text{vk} := \bar{\mathbf{A}}, \text{sk} := \perp$	5 : $\mathbf{h}_1 = \mathbf{c}_1 \mathbf{e} + \mathbf{p}; \mathbf{h}_2 = \mathbf{c}_1 \mathbf{s} + \mathbf{r}$
$\text{H}_\mu^{\bar{\mathcal{B}}}(\mu, \text{vk})$	6 : $\mathbf{z}_1 := \mathbf{h}_1 + \mathbf{c}_2$
1 : if $\text{Q}_{\text{H}_\mu}[\mu, \text{vk}] = \perp$ then	7 : $\mathbf{z}_2 := \mathbf{h}_2$
2 : Send <i>syndrome</i> to $\mathcal{C}$	8 : $\mathbf{z}_3 := \mathbf{c}_1$
3 : Obtain $\mathbf{u}$ from $\mathcal{C}$	9 : $\tilde{\mathbf{x}}' \leftarrow \chi_{\mathbf{s}'}$
4 : $\text{Q}_{\text{H}_\mu}[\mu, \text{vk}] \leftarrow \mathbf{u}$	10 : $\bar{\mathbf{t}} := [1 \ \mathbf{a} \ \mathbf{b}] \mathbf{z} - \mathbf{t}' - \mathbf{A} \cdot \tilde{\mathbf{x}}'$
5 : return $\text{Q}_{\text{H}_\mu}[\mu, \text{vk}]$	11 : $\tilde{\mathbf{e}} \leftarrow \chi_{\mathbf{e}}$
	12 : Send $\bar{\mathbf{t}} - \tilde{\mathbf{e}}$ to $\mathcal{C}$
	13 : Obtain $\tilde{\mathbf{x}} \in \mathcal{R}_q^4$ from $\mathcal{C}$
	14 : $\mathbf{x} := \tilde{\mathbf{x}} + \tilde{\mathbf{x}}'$
	15 : $\text{aux}_{\text{sig}} := \mathbf{w}$
	16 : $\text{pm}^S := (\mathbf{x}, \mathbf{z}_2, \mathbf{c}_1, \tilde{\mathbf{e}})$
	17 : return $(\text{pm}^S, \text{aux}_{\text{sig}})$

The challenger outputs a uniformly random matrix $\bar{\mathbf{A}} \in \{1\} \times \mathcal{R}_q^3$. Let's call the event that $\mathbf{a}$ or $\mathbf{b}$ is not invertible E . Theorem 3.2 says that $\Pr[E] \leq 1 - e^{-2}$. Conditioned on $\neg E$, $\mathcal{A}'$'s view is statistically close to $\mathcal{A}'$'s view in $\mathbf{G}_6$:

First, the distribution of Signer's output is statistically close to the output in $\mathbf{G}_6$. This is obtained by applying Theorem 2.8 which shows that sampling $\tilde{\mathbf{x}}$ from $\mathcal{D}_{\mathcal{R}_q, \sqrt{\sigma_{\mathbf{s}, \mathbf{x}}^2 - \sigma_{\mathbf{s}'}^2}}^4$ conditioned on $\mathbf{A} \tilde{\mathbf{x}} = \bar{\mathbf{t}} - \tilde{\mathbf{e}}$ is distributed as in $\mathbf{G}_6$. Indeed, our choice of parameters ensure that $\mathbf{A} \cdot \tilde{\mathbf{x}}$ is close to uniform for $\tilde{\mathbf{x}} \leftarrow \mathcal{D}_{\mathcal{R}_q, \sqrt{\sigma_{\mathbf{s}, \mathbf{x}}^2 - \sigma_{\mathbf{s}'}^2}}^4$. Additionally, we need to show that the target $\bar{\mathbf{t}}$ is close to uniform, which is guaranteed by the term $\mathbf{A} \cdot \tilde{\mathbf{x}}'$ by the regularity Theorem 3.1.

Third, note that since $\mathbf{G}_5$, the elements $\mathbf{a}, \mathbf{b}$ of the public key vk are already distributed as uniformly random in $\mathcal{R}_q^\times$. Importantly, because $\mathbf{a}$ and $\mathbf{b}$ are invertible, we can use the regularity bound from Theorem 3.1 with $m = 2$. This shows that $\mathbf{d}$ is statistically close to uniform.

Therefore, conditioned on $\mathbf{a}, \mathbf{b} \in \mathcal{R}_q^\times$, the view of $\mathcal{A}$ in $\mathbf{G}_6$ differs from $\mathcal{A}'$'s view in the reduction at most with probability $2\delta + 2^{\ell n} (5/4)^{3n\ell} q^{-3n\varepsilon}$.

Bounding $\bar{\mathcal{B}}$'s advantage. If $\bar{\mathcal{B}}$ did not abort and $\mathcal{A}$ wins the game, then $\mathcal{A}$ outputs $Q_S + 1$ message-signature pairs $((\mu_i, \text{sig}_i))_{i \in [Q_S + 1]}$ that pass verification. Additionally, $\mathbf{G}_6$ ensures that the straightline extraction Ext on all submitted signatures $\text{sig}_i = \pi_i$ succeeds in obtaining $Q_S + 1$ witnesses $(\mathbf{x}_i)_{i \in [Q_S + 1]}$ such that for all $i \in [Q_S + 1]$: $\mathbf{u}_i = \bar{\mathbf{A}}\mathbf{x}_i$ and $\|\mathbf{x}_i\| \leq B$. Because the hash values $\mathbf{u}_i$ come from *syndrome* queries, $\bar{\mathcal{B}}$ found $Q_S + 1$ short $\bar{\mathbf{A}}$ -preimages of syndrome queries, which breaks OM-ISIS.

$$\Pr[\mathbf{G}_6] \leq e^2 \text{Adv}_{\bar{\mathcal{B}}, q, \chi_{S, x}^4, Q_S, B}^{\text{OM-ISIS}} + 2\delta + 2^{\ell n} (5/4)^{3n\ell} q^{-3n\varepsilon} + \text{negl}(\lambda).$$

□

D Full Proofs for TBS_B

In this section, we give full proofs for blindness (Theorem 5.2) and one-more unforgeability (Theorem 5.3) of the threshold blind signature scheme TBS_B .

D.1 Proof of Correctness (Theorem 5.1)

Sketch. The proof of correctness in the distributed setting goes as in the non-threshold setting for Theorem 4.1, except that each signer now contributes some randomness $(\mathbf{r}_i, \mathbf{p}_i)$ to the final server response $\mathbf{z}$. We can apply the convolution lemma Theorem 2.3 T times to bound the norm of $\mathbf{x} + \sum_{i \in \text{SS}} [\mathbf{p}_i \quad \mathbf{r}_i \quad 0 \quad 0]$ by $\sqrt{n} \cdot e^{1/4} \cdot \sqrt{2T\sigma_{\text{rnd}}^2 + 4\sigma_{S, x}^2}$.

The correctness of the zero share computation, and correct Shamir sharing of the secret also ensures the correct retrieval of the value $\mathbf{z}$, and the rest of the proof follows.

We thus obtain as overwhelming bound on w : $B = e^{1/4} \cdot \sqrt{n} \cdot \left(n\sqrt{2} \left\lceil \frac{q-1}{2\beta} \right\rceil \sigma_{\text{sk}} + \sigma_{\text{U}, x} + \sqrt{2T\sigma_{\text{rnd}}^2 + 4\sigma_{S, x}^2} \right) + \sqrt{n} \frac{\beta + \bar{p}}{2}$.

□

D.2 Full Proof of Blindness (Theorem 5.2)

The proof follows the general strategy from Section C.2.

Proof. We show a sequence of hybrid experiments such that the view of $\mathcal{A}$ is independent of μ_0 and μ_1 .

Game $\mathbf{G}_0$: This is the original blindness experiment.

Game $\mathbf{G}_1$: In this game, we simulate the random oracle $\mathbf{H}_\Pi$ and zero-knowledge proofs π . That is, denote by $\text{Sim} = (\text{Sim}_0, \text{Sim}_1)$ the zero-knowledge simulator of Π . We simulate $\mathbf{H}_\Pi$ via Sim_0 . Next, we change the response of the oracle $\mathcal{O}^{\text{User}_3}$. On a query $(\text{sid}, (\text{pm}_{0,r}^{(S,i)})_{i \in \text{SS}_0^{\text{sid}}}, (\text{pm}_{1,r}^{(S,i)})_{i \in \text{SS}_1^{\text{sid}}})$, the oracle computes the two signatures as $\pi_i \leftarrow \text{Sim}_1(\text{crs}, \mathbb{x}_i)$, where the statement $\mathbb{x}_i = (\mathbf{A}, \mathbf{u}_i)$. Note that π_i is only computed if $(\mathbb{x}_i, \mathbb{w}_i) \in \mathcal{R}$ due to the check in line 8, else $\mathcal{O}^{\text{User}_3}$ outputs $(\perp, \perp)$. Therefore, we have that

$$|\Pr[\mathbf{G}_1] - \Pr[\mathbf{G}_0]| \leq \text{Adv}_{\Pi, B}^{\text{ZK}}(\lambda).$$

Game $\mathbf{G}_2$: In this game, we change $\mathcal{O}^{\text{User}_0}$. On a query $(\text{sid}, \text{vk}^{\text{sid}}, \text{aux}^{\text{sid}}, \mu_0^{\text{sid}}, \mu_1^{\text{sid}}, \text{SS}_0^{\text{sid}}, \text{SS}_1^{\text{sid}})$, the oracle chooses a random $\mathbf{t}'_i \leftarrow \mathcal{R}_q$, for $i \in \{0, 1\}$, and outputs $(\text{pm}_{0,0}^{\text{U}}, \text{pm}_{1,0}^{\text{U}}) = ((\mathbf{t}'_0, \text{SS}), (\mathbf{t}'_1, \text{SS}))$ instead of using $\mathbf{t}'_i = \mathbf{A}\mathbf{x}'_i + \mathbf{H}_\mu(\mu_i^{\text{sid}}, \text{vk})$. An adversary that can distinguish $\mathbf{A}\mathbf{x}'_i$ from a random value can easily be turned into an adversary $\mathcal{B}'$ breaking RLWE with respect to the matrix $[1 \ \mathbf{a}]$. A simple hybrid argument yields

$$|\Pr[\mathbf{G}_2] - \Pr[\mathbf{G}_1]| \leq 2\text{Adv}_{\mathcal{B}', q, \chi_{\text{U}, x}}^{\text{RLWE-}\times}(\lambda).$$

Observe that the oracles $\mathcal{O}^{\text{User}_1}$ and $\mathcal{O}^{\text{User}_2}$ just broadcast their inputs to all servers in SS . Therefore, these outputs give $\mathcal{A}$ no additional information. We can conclude that in $\mathbf{G}_2$, the view of $\mathcal{A}$ is independent of the bit b . Therefore, a malicious signer can guess b at most with probability $1/2$. $\square$

D.3 Full Proof of Unforgeability (Theorem 5.3)

In our proof, we reduce one-more unforgeability of TBS with norm bound B to one-more unforgeability of BS_B . Roughly, we show that signing leaks no more information than a non-threshold BS_B signing session (*i.e.*, we can simulate signing via a signing oracle for BS_B) thanks to the masking of the response in Signer_3 . As signatures have the same form as BS_B signatures, finding $Q_S + 1$ valid signatures is hard by Theorem 4.3.

Proof. We argue via hybrid experiments and write $\Pr[\mathbf{G}_i]$ for the probability that the i -th hybrid experiment outputs 1. The proof follows the structure in [KRT24] to argue that the response $\mathbf{z}_{2,i}$ is uniform conditioned on summing to (part of) a valid signature $\mathbf{z}_2$. Then, we show that solving TBS is as hard as BS_B employing the above fact. We denote the set of honest signers by $\text{HS} = [N] \setminus \text{CS}$, the set of honest signers within SS by $\text{sHS} = \text{SS} \cap \text{HS}$, and the set of corrupted signers within SS by $\text{sCS} = \text{SS} \cap \text{CS}$. We denote by Q_S the number of $\mathcal{O}^{\text{Signer}}$ queries and by $Q_{\text{H}_{xyz}}$ the number of queries to random oracle H_{xyz} , where $xyz \in \{\text{x}, \text{mask}, \text{com}, \mu\}$.

Game $\mathbf{G}_0$: This is the original unforgeability experiment.

Game $\mathbf{G}_1$: The game delays programming the random oracle H_{com} until Signer_2 is executed for user i . The changes are depicted below.

$\text{Signer}_1(i, \text{sk}_i, \text{aux}, \text{ctx}, \text{pm}_0^{\text{U}})$	<pre> 1: parse ($\mathbf{t}', \text{SS}$) $\leftarrow \text{pm}_0^{\text{U}}$ 2: $\text{cmt}_i \leftarrow \{0, 1\}^{2\lambda}$ 3: $\text{st}_1^{(S,i)} \leftarrow (\text{SS}, \mathbf{t}', \text{cmt}_i)$ 4: return ($\text{st}_1^{(S,i)}, \text{pm}_1^{(S,i)} := \text{cmt}_i$) </pre>
$\text{Signer}_2(i, \text{st}_1^{(S,i)}, \text{pm}_1^{\text{U}})$	<pre> 1: parse (cmt_j)$_{j \in \text{SS}} \leftarrow \text{pm}_1^{\text{U}}$ 2: $\text{SS}_i := \text{SS}$ 3: $\text{ssid}_i := (\mathbf{t}', \text{SS}_i, (\text{cmt}_j)_{j \in \text{SS}}, \text{ctx})$ 4: $(\mathbf{r}_i, \mathbf{p}_i) \leftarrow \chi_{\text{rnd}}^2, \mathbf{w}_i := \mathbf{a}\mathbf{r}_i + \mathbf{p}_i$ 5: if $\text{Q}_{\text{H}_{\text{com}}}[\mathbf{w}_i] \neq \perp$ then 6: Abort the game 7: $\text{Q}_{\text{H}_{\text{com}}}[\mathbf{w}_i] := \text{cmt}_i$ 8: $\text{st}_2^{(S,i)} \leftarrow (\text{ssid}_i, \mathbf{w}_i, \mathbf{r}_i)$ 9: return ($\text{st}_2^{(S,i)}, \text{pm}_2^{(S,i)} := \mathbf{w}_i, \text{ssid}_i, \text{SS}_i$) </pre>

If the game does not abort, then both games are identically distributed as cmt_i and $\mathbf{w}_i$ are generated in the same manner. To upper bound the abort probability, observe that an abort occurs if and only if H_{com} is queried on input $(i, \mathbf{w}_i)$ before H_{com} is programmed in Signer_2 . Here, we depart from the proof strategy in [KRT24] as we cannot argue that $\mathbf{w}_i$ has high min-entropy due to some regularity lemma (without noticeable impact on our parameters). Instead, we argue that provoking such an abort is computationally hard under RLWE.

Let us call the event that this occurs in the j -th H_{com} query $E_{k,j}$, *i.e.*, the value committed in cmt_i generated by the game in the k -th $\mathcal{O}^{\text{Signer}}$ is queried to H_{com} in the j -th query. Note that $E = \cup_{j \in [Q_{H_{\text{com}}}], k \in [Q_S]} E_{k,j}$ is the event that $\mathbf{G}_1$ aborts.

We can upper bound $\Pr[E_{k,j}]$ under RLWE as follows. The reduction $\mathcal{B}$ obtains a RLWE sample $\mathbf{w}^* = \mathbf{a}\mathbf{s}_1^* + \mathbf{s}_2^*$ or a random element $\mathbf{w}^* \leftarrow \mathcal{R}_q$ as challenge. Then, $\mathcal{B}$ proceeds as in $\mathbf{G}_0$ except that it commits to $(i, \mathbf{w}^*)$ in cmt_i in the k -th $\mathcal{O}^{\text{Signer}}$ query if the request is for Signer_1 . In case $E_{k,j}$ occurs, it outputs 1, else it outputs 0.

If $\mathbf{w}^*$ is an RLWE sample, then the view of $\mathcal{A}$ is as in $\mathbf{G}_0$, therefore it outputs 1 with $\Pr[E_{k,j}]$. If $\mathbf{w}^*$ is uniform, it outputs 1 with probability $1/|\mathcal{R}_q| = q^{-n}$ as $\mathbf{w}^*$ is information-theoretically hidden. Therefore, we have

$$\begin{aligned} \Pr[\mathbf{G}_0] &= \Pr[\mathbf{G}_0 \wedge \neg E] + \Pr[\mathbf{G}_0 \wedge E] \\ &\leq \Pr[\mathbf{G}_1] + \Pr[E] \\ &\leq \Pr[\mathbf{G}_1] + \sum_{j \in [Q_{H_{\text{com}}}], k \in [Q_S]} \Pr[E_{k,j}] \\ &\leq \Pr[\mathbf{G}_1] + Q_{H_{\text{com}}} \cdot Q_S \cdot (\text{Adv}_{\mathcal{B}, q, \chi_{\text{rnd}}}^{\text{RLWE-}\times}(\lambda) + q^{-n}) \end{aligned}$$

Game $\mathbf{G}_2$: Now the game aborts in case there are collisions in H_{com} . For this, it keeps track of the H_{com} outputs cmt in a set Cmt that is initially empty.

$\text{Signer}_1(i, \text{sk}_i, \text{aux}, \text{ctx}, \text{pm}_0^U)$

```

1 : parse ( $\mathbf{t}', \text{SS}$ )  $\leftarrow \text{pm}_0^U$ 
2 :  $\text{cmt}_i \leftarrow \{0, 1\}^{2\lambda}$ 
3 : if  $\text{cmt}_i \in \text{Cmt}$  then Abort the game
4 :  $\text{Cmt} \leftarrow \text{Cmt} \cup \{\text{cmt}_i\}$ 
5 :  $\text{st}_1^{(S,i)} \leftarrow (\text{SS}, \mathbf{t}', \text{cmt}_i)$ 
6 : return  $(\text{st}_1^{(S,i)}, \text{pm}_1^{(S,i)} := \text{cmt}_i)$ 

```

$H_{\text{com}}(i, \mathbf{w})$

```

1 : if  $\text{Q}_{H_{\text{com}}}[i, \mathbf{w}] = \perp$  then
2 :    $\text{cmt} \leftarrow \{0, 1\}^{2\lambda}$ 
3 :   if  $\text{cmt}_i \in \text{Cmt}$  then Abort the game
4 :    $\text{Cmt} \leftarrow \text{Cmt} \cup \{\text{cmt}_i\}$ 
5 :    $\text{Q}_{H_{\text{com}}}[i, \mathbf{w}] \leftarrow \text{cmt}$ 
6 : return  $\text{Q}_{H_{\text{com}}}[i, \mathbf{w}]$ 

```

Observe that an H_{com} output is chosen in Signer_1 and each H_{com} query. As H_{com} outputs 2λ bits, a birthday bound yields

$$|\Pr[\mathbf{G}_2] - \Pr[\mathbf{G}_1]| \leq \frac{(H_{\text{com}} + Q_S)^2}{2^{2\lambda}}$$

Game $\mathbf{G}_3$: The game introduces the following tables for book keeping. The tables are indexed by ssid to keep track of the responses of honest signers in session ssid .

- $\text{UnOpnHS}[\text{ssid}]$: stores the set of honest signers that have not executed Signer_2 with session ssid yet.

- $\text{UnSgnHS}[\text{ssid}]$: stores the set of honest signers that have not executed Signer_3 with session ssid yet.
- $\text{SumRnd}[\text{ssid}]$: stores the sum $\sum_{i \in \text{sHS}} \mathbf{r}_i$ of randomness employed by the honest signers $i \in \text{sHS}$ in Signer_2 .
- $\text{Mask}[\text{ssid}, i]$: stores the zero share Δ_i .
- $\text{MaskedRsp}[\text{ssid}, i]$ stores the masked response $z_{2,i}$.

$\text{Signer}_2(i, \text{st}_1^{(S,i)}, \text{pm}_1^U)$

```

1 : parse  $(\text{cmt}_j)_{j \in \text{SS}} \leftarrow \text{pm}_1^U$ 
2 :  $\text{SS}_i := \text{SS}$ 
3 :  $\text{ssid}_i := (\mathbf{t}', \text{SS}_i, (\text{cmt}_j)_{j \in \text{SS}}, \text{ctx})$ 
4 : if  $\text{UnOpnHS}[\text{ssid}_i] = \perp$  then
5 :    $\text{UnOpnHS}[\text{ssid}_i] \leftarrow \text{sHS}$ 
6 :    $\text{SumRnd}[\text{ssid}_i] \leftarrow 0$ 
7 :  $(\mathbf{r}_i, \mathbf{p}_i) \leftarrow \chi_{\text{rnd}}^2, \mathbf{w}_i := \mathbf{a}\mathbf{r}_i + \mathbf{p}_i$ 
8 :  $\text{SumRnd}[\text{ssid}_i] \leftarrow \text{SumRnd}[\text{ssid}_i] + \mathbf{r}_i$ 
9 : if  $\text{Q}_{\text{Hcom}}[i, \mathbf{w}_i] \neq \perp$  then
10 :   Abort the game
11 :  $\text{Q}_{\text{Hcom}}[i, \mathbf{w}_i] := \text{cmt}_i$ 
12 :  $\text{UnOpnHS}[\text{ssid}_i] \leftarrow \text{UnOpnHS}[\text{ssid}_i] \setminus \{i\}$ 
13 :  $\text{st}_2^{(S,i)} \leftarrow (\text{ssid}_i, \mathbf{w}_i, \mathbf{r}_i)$ 
14 : return  $(\text{st}_2^{(S,i)}, \text{pm}_2^{(S,i)} := \mathbf{w}_i, \text{ssid}_i, \text{SS}_i)$ 

```

$\text{Signer}_3(i, \text{st}_2^{(S,i)}, \text{pm}_2^U)$

```

1 : parse  $(\mathbf{w}_j)_{j \in \text{SS}} \leftarrow \text{pm}_2^U$ 
2 : req  $(\forall j \in \text{SS}, \text{cmt}_j = \text{Hcom}(j, \mathbf{w}_j))$ 
3 :  $\mathbf{w} := \sum_{j \in \text{SS}} \mathbf{w}_j$ 
4 :  $(\mathbf{x}, \tilde{\mathbf{e}}) := \text{H}_x(\text{vk}, \mathbf{w}); \quad \mathbf{t} := \mathbf{t}' + \mathbf{A}\mathbf{x} + \tilde{\mathbf{e}}$ 
5 :  $(\mathbf{c}_1, \mathbf{c}_2) \leftarrow \text{Decomp}_\beta(\mathbf{t} - \mathbf{w})$ 
6 : if  $\text{UnSgnHS}[\text{ssid}_i] = \perp$  then  $\text{UnSgnHS}[\text{ssid}_i] \leftarrow \text{sHS}$ 
7 :  $\Delta_i := \text{ZeroShare}(\text{seed}_i[\text{SS}], \text{ssid}_i)$ 
8 :  $\mathbf{z}_{2,i} := \mathbf{c}_1 \cdot L_{\text{SS},i} \cdot \llbracket \mathbf{s} \rrbracket_i + \mathbf{r}_i + \Delta_i$ 
9 :  $\text{Mask}[\text{ssid}_i, i] \leftarrow \Delta_i$ 
10 :  $\text{MaskedRsp}[\text{ssid}_i, i] \leftarrow \mathbf{z}_{2,i}$ 
11 :  $\text{UnSgnHS}[\text{ssid}_i] \leftarrow \text{UnSgnHS}[\text{ssid}_i] \setminus \{i\}$ 
12 :  $\text{pm}_3^{(S,i)} := \mathbf{z}_{2,i}, \quad \text{aux}_{\text{sig}} := \mathbf{w}$ 
13 : return  $(\text{pm}_3^{(S,i)}, \text{aux}_{\text{sig}})$ 

```

This game is purely syntactical, and we have

$$\Pr[\mathbf{G}_3] = \Pr[\mathbf{G}_2].$$

Let us discuss some consequences of this game.

First, note that all ssid output by an honest signer $i \in \text{HS}$ are distinct as ssid contains cmt_i which is collision-free due to the abort condition added in $\mathbf{G}_2$.

Therefore, each $\mathcal{O}^{\text{Signer}}$ with ssid updates UnOpnHS and UnSgnHS exactly once to remove the honest signer. Furthermore, $\text{SumRnd}[\text{ssid}]$ contains the sum if $\text{UnOpnHS}[\text{ssid}] = \emptyset$. Also, $\text{Mask}[\text{ssid}, i]$ and $\text{MaskedRsp}[\text{ssid}, i]$ contain the zero-share Δ_i and response $\mathbf{z}_{2,i}$, respectively, employed by honest signer $i \in \text{HS}$ in session ssid if $i \notin \text{UnSgnHS}[\text{ssid}]$.

Note that due to the opening check of cmt_i in Signer_3 , it must hold that $\text{UnOpnHS}[\text{ssid}] = \emptyset$, else the check fails due to the abort condition added in $\mathbf{G}_2$.

We employ these facts below (without explicit mention).

Game $\mathbf{G}_4$: The game aborts if $\mathcal{A}$ queries a seed $\text{seed}_{i,j}$ with $i, j \in \text{HS}$ in H_{mask} . Furthermore, we unroll ZeroShare within Signer_3 as defined in Eq. (1). Also, the game samples the partial masks $\mathbf{m}_{i,j}$ for $i, j \in \text{HS}$ at random in Signer_3 instead of via $\mathbf{m}_{i,j} = \text{H}_{\text{mask}}(\text{seed}_{i,j}, \text{ssid}_i)$.

$\text{Signer}_3(i, \text{st}_2^{(S,i)}, \text{pm}_2^U)$

```

    // identical to lines 1 to line 6 in  $\mathbf{G}_3$ 
7 : for  $j \in \text{sCS}$  do
8 :    $\mathbf{m}_{i,j} := \text{H}_{\text{mask}}(\text{seed}_{i,j}, \text{ssid}_i)$ 
9 :    $\mathbf{m}_{j,i} := \text{H}_{\text{mask}}(\text{seed}_{j,i}, \text{ssid}_i)$ 
10 :  $\text{sUS} \leftarrow \text{UnSgnHS}[\text{ssid}_i]$ 
11 : for  $j \in \text{sHS} \setminus \text{sUS}$  do
12 :    $\mathbf{m}_{i,j} := \text{Q}_{\text{H}_{\text{mask}}}[\text{seed}_{i,j}, \text{ssid}_i]$ 
13 :    $\mathbf{m}_{j,i} := \text{Q}_{\text{H}_{\text{mask}}}[\text{seed}_{j,i}, \text{ssid}_i]$ 
14 : for  $j \in \text{sUS} \setminus \{i\}$  do
15 :    $\mathbf{m}_{i,j} \leftarrow \mathcal{R}_q, \text{Q}_{\text{H}_{\text{mask}}}[\text{seed}_{i,j}, \text{ssid}_i] := \mathbf{m}_{i,j}$ 
16 :    $\mathbf{m}_{j,i} \leftarrow \mathcal{R}_q, \text{Q}_{\text{H}_{\text{mask}}}[\text{seed}_{j,i}, \text{ssid}_i] := \mathbf{m}_{j,i}$ 
17 :  $\Delta_i := \sum_{j \in \text{SS}} (\mathbf{m}_{i,j} - \mathbf{m}_{j,i})$ 
18 :  $\mathbf{z}_{2,i} := \mathbf{c}_1 \cdot L_{\text{SS},i} \cdot \llbracket \mathbf{s} \rrbracket_i + \mathbf{r}_i + \Delta_i$ 
19 :  $\text{Mask}[\text{ssid}_i, i] \leftarrow \Delta_i$ 
20 :  $\text{MaskedRsp}[\text{ssid}_i, i] \leftarrow \mathbf{z}_{2,i}$ 
21 :  $\text{UnSgnHS}[\text{ssid}_i] \leftarrow \text{UnSgnHS}[\text{ssid}_i] \setminus \{i\}$ 
22 :  $\text{pm}_3^{(S,i)} := \mathbf{z}_{2,i}, \text{aux}_{\text{sig}} := \mathbf{w}$ 
23 : return  $(\text{pm}_3^{(S,i)}, \text{aux}_{\text{sig}})$ 

```

$\text{H}_{\text{mask}}(\text{seed}, \text{ssid})$

```

1 : if  $\exists (i, j) \in \text{HS}^2 : \text{seed} = \text{seed}_{i,j}$ 
2 :   Abort the game
3 : if  $\text{Q}_{\text{H}_{\text{mask}}}[\text{seed}, \text{ssid}] = \perp$  then
4 :    $\mathbf{m} \leftarrow \mathcal{R}_q$ 
5 :    $\text{Q}_{\text{H}_{\text{mask}}}[\text{seed}, \text{ssid}] \leftarrow \mathbf{m}$ 
6 : return  $\text{Q}_{\text{H}_{\text{mask}}}[\text{seed}, \text{ssid}]$ 

```

First, we argue that Δ_i is distributed as in the previous game. Observe that for $j \in \text{sHS} \setminus \text{sUS}$, the table entries $\text{Q}_{\text{H}_{\text{mask}}}[\text{seed}_{i,j}, \text{ssid}_i]$ and $\text{Q}_{\text{H}_{\text{mask}}}[\text{seed}_{j,i}, \text{ssid}_i]$ were initialized in the Signer_2 query for signer

j in session ssid_i . On the other hand, the table entries $Q_{H_{\text{mask}}}[\text{seed}_{i,j}, \text{ssid}_i]$ and $Q_{H_{\text{mask}}}[\text{seed}_{j,i}, \text{ssid}_i]$ for $j \in \text{sUS}$ were not initialized yet, as the table $Q_{H_{\text{mask}}}$ is exclusively programmed by the game for seeds $\text{seed}_{k,\ell}$ with $k, \ell \in \text{HS}$ due the abort is introduced in H_{mask} . Therefore, H_{mask} is distributed as in the previous game, and Δ_i is computed according to its definition in Eq. (1).

Next, let us upper bound the abort probability in H_{mask} . As $\text{seed}_{i,j} = i \| j \| \text{rand}_{i,j}$ with $\text{rand}_{i,j} \in \{0, 1\}^\lambda$ chosen at random, we have⁹

$$|\Pr[\mathbf{G}_4] - \Pr[\mathbf{G}_3]| \leq \frac{Q_{H_{\text{mask}}}}{2^\lambda}.$$

Game $\mathbf{G}_5$: The game samples zero share Δ_i at random if $i \in \text{sHS}$ is not the last signer with respect to ssid_i in Signer_3 . Otherwise, if i is the last signer with respect to ssid_i , *i.e.*, $\text{sUS} = \{i\}$ for $\text{sUS} = \text{UnSgnHS}[\text{ssid}_i]$, then the game samples the zero share Δ_i *consistently* with the other shares, *i.e.*, as in line 13.

$\text{Signer}_3(i, \text{st}_2^{(S,i)}, \text{pm}_2^U)$

```

    // identical to lines 1 to line 6 in  $\mathbf{G}_3$ 
7 :  $\text{sUS} \leftarrow \text{UnSgnHS}[\text{ssid}_i]$ 
8 : if  $\text{sUS} \neq \{i\}$  then
9 :    $\Delta_i \leftarrow \mathcal{R}_q$ 
10 : else
11 :   for  $j \in \text{sCS}$  do
12 :      $\Delta_j := \text{ZeroShare}(\vec{\text{seed}}_i[\text{SS}], \text{ssid}_i)$ 
13 :    $\Delta_i := - \sum_{j \in \text{sHS} \setminus \{i\}} \text{Mask}[\text{ssid}_i, j] - \sum_{j \in \text{sCS}} \Delta_j$ 
14 :  $\mathbf{z}_{2,i} := \mathbf{c}_1 \cdot L_{\text{SS},i} \cdot \llbracket \mathbf{s} \rrbracket_i + \mathbf{r}_i + \Delta_i$ 
15 :  $\text{Mask}[\text{ssid}_i, i] \leftarrow \Delta_i$ 
16 :  $\text{MaskedRsp}[\text{ssid}_i, i] \leftarrow \mathbf{z}_{2,i}$ 
17 :  $\text{UnSgnHS}[\text{ssid}_i] \leftarrow \text{UnSgnHS}[\text{ssid}_i] \setminus \{i\}$ 
18 :  $\text{pm}_3^{(S,i)} := \mathbf{z}_{2,i}, \text{aux}_{\text{sig}} := \mathbf{w}$ 
19 : return  $(\text{pm}_3^{(S,i)}, \text{aux}_{\text{sig}})$ 

```

Let us prove that both $\mathbf{G}_5$ and $\mathbf{G}_4$ are identically distributed. Observe that in $\mathbf{G}_4$, if $\text{sUS} \neq \{i\}$, then there remains an honest signer $j \in \text{sUS} \setminus \{i\}$ that did not execute Signer_3 for ssid_i yet. Therefore, Δ_i is distributed uniform over $\mathcal{R}_q$ as the zero-shares $\mathbf{m}_{i,j}$ and $\mathbf{m}_{j,i}$ are drawn uniformly at random. Else, all zero-shares $\mathbf{m}_{i,j}$ and $\mathbf{m}_{j,i}$ are determined by $Q_{H_{\text{mask}}}$ and it as $\sum_{j \in \text{SS}} \Delta_i = 0$, the zero-share Δ_i is determined by

$$\Delta_i = - \sum_{j \in \text{SS} \setminus \{i\}} \Delta_j.$$

Therefore, Δ_i in $\mathbf{G}_5$ is distributed as in $\mathbf{G}_4$ and we have

$$\Pr[\mathbf{G}_5] = \Pr[\mathbf{G}_4].$$

Game $\mathbf{G}_6$: The game samples responses $\mathbf{z}_{2,i}$ at random if $i \in \text{sHS}$ is not the last signer with respect to ssid_i in Signer_3 . Otherwise, the game samples the response $\mathbf{z}_{2,i}$ *consistently* with the other responses, *i.e.*, as in line 13.

⁹Note that no birthday bound is needed as the queries are domain-separated due to the indices i, j in $\text{seed}_{i,j}$. The formal argument is as in [KRT24, Section E, Game 7].

$\text{Signer}_3(i, \text{st}_2^{(S,i)}, \text{pm}_2^U)$

```

    // identical to lines 1 to line 6 in  $\mathbf{G}_3$ 
7:  $\text{sUS} \leftarrow \text{UnSgnHS}[\text{ssid}_i]$ 
8: if  $\text{sUS} \neq \{i\}$  then
9:    $\mathbf{z}_{2,i} \leftarrow \mathcal{R}_q$ 
10: else
11:   for  $j \in \text{sCS}$  do
12:      $\Delta_j := \text{ZeroShare}(\vec{\text{seed}}_i[\text{SS}], \text{ssid}_i)$ 
13:      $\mathbf{z}_{2,i} := \mathbf{c}_1 \cdot \mathbf{s} - \sum_{j \in \text{sCS}} (\mathbf{c}_1 \cdot L_{\text{SS},j} \cdot \mathbf{s}_j - \Delta_j)$ 
         $+ \text{SumRnd}[\text{ssid}_i] - \sum_{j \in \text{sHS} \setminus \{i\}} \text{MaskedRsp}[\text{ssid}_i, j]$ 
14:    $\text{MaskedRsp}[\text{ssid}_i, i] \leftarrow \mathbf{z}_{2,i}$ 
15:    $\text{UnSgnHS}[\text{ssid}_i] \leftarrow \text{UnSgnHS}[\text{ssid}_i] \setminus \{i\}$ 
16:    $\text{pm}_3^{(S,i)} := \mathbf{z}_{2,i}, \text{aux}_{\text{sig}} := \mathbf{w}$ 
17: return  $(\text{pm}_3^{(S,i)}, \text{aux}_{\text{sig}})$ 

```

Let us consider an intermediate game that proceeds as $\mathbf{G}_5$, except that instead of sampling $\Delta_i \leftarrow \mathcal{R}_q$ in Signer_3 if $\text{sUS} \neq \{i\}$, the game samples $\Delta_i^* \leftarrow \mathcal{R}_q$ and set $\Delta_i = \Delta_i^* - \mathbf{c}_1 \cdot L_{\text{SS},i} \llbracket \mathbf{s} \rrbracket_i + \mathbf{r}_i$. The intermediate game is distributed as $\mathbf{G}_5$.

Observe that if $\text{sUS} \neq \{i\}$, then we have $\mathbf{z}_{2,i} = \Delta_i^*$ which is distributed uniform over $\mathcal{R}_q$. Else, we have $\text{sUS} = \{i\}$. In this case, observe that $\mathbf{c}_1$ is identical across all honest signers in session ssid_i , as ssid_i determines $\mathbf{t}'$ and $\mathbf{w}_j$ for $j \in \text{SS}$ (indirectly through cmt_j). These values determine which value is decomposed in Signer_3 , yielding $\mathbf{c}_1$ and $\mathbf{c}_2$. Therefore, we have

$$\begin{aligned}
\mathbf{z}_{2,i} &= \mathbf{c}_1 \cdot L_{\text{SS},i} \cdot \llbracket \mathbf{s} \rrbracket_i + \mathbf{r}_i + \Delta_i \\
&= \mathbf{c}_1 \cdot L_{\text{SS},i} \cdot \llbracket \mathbf{s} \rrbracket_i + \mathbf{r}_i \\
&\quad - \sum_{j \in \text{sHS} \setminus \{i\}} \underbrace{\text{Mask}[\text{ssid}_i, j]}_{\Delta_j^* - \mathbf{c}_1 \cdot L_{\text{SS},j} \llbracket \mathbf{s} \rrbracket_j + \mathbf{r}_j} - \sum_{j \in \text{sCS}} \Delta_j \\
&= \underbrace{\sum_{j \in \text{SS}} \mathbf{c}_1 \cdot L_{\text{SS},j} \cdot \llbracket \mathbf{s} \rrbracket_j}_{\mathbf{c}_1 \cdot \mathbf{s}} + \underbrace{\sum_{j \in \text{sHS}} \mathbf{r}_j}_{\text{SumRnd}[\text{ssid}_i]} - \underbrace{\sum_{j \in \text{sHS} \setminus \{i\}} \Delta_j^*}_{\text{MaskedRsp}[\text{ssid}_i, j]} \\
&\quad - \sum_{j \in \text{sCS}} (\mathbf{c}_1 \cdot L_{\text{SS},j} \cdot \llbracket \mathbf{s} \rrbracket_j - \Delta_j),
\end{aligned}$$

where the identity for $\text{Mask}[\text{ssid}_i, j]$ follows as $\mathbf{c}_1$ is identical for all honest signers in session ssid_i as argued above, and the final equality follows from the properties of Shamir's sharing. This is exactly the distribution in $\mathbf{G}_6$. Consequently, we have

$$\Pr[\mathbf{G}_6] = \Pr[\mathbf{G}_5].$$

Game $\mathbf{G}_7$: In Signer_2 , when the last honest signer opens its commitment with respect to ssid_i , the game extracts the values $(\text{ctx}, \text{SS}, j, \mathbf{x}_j, \tilde{\mathbf{e}}_j, \mathbf{w}_j)$ for malicious commitments cmt_j with $j \in \text{sCS}$ via the query table $\mathbf{Q}_{\text{H}_{\text{com}}}$ for H_{com} . It stores the values in a table $\text{ExtComCS}[\text{ssid}_i, j] = (\mathbf{x}_j, \tilde{\mathbf{e}}_j, \mathbf{w}_j)$ (unless no preimage is found). Further, if for all $j \in \text{sCS}$ a H_{com} preimage was found, the game sets $\mathbf{w} = \sum_{j \in \text{SS}} \mathbf{w}_j$ and aborts if H_x is already set on input $(\text{vk}, \mathbf{w})$. Also, the game aborts in Signer_3 if $\text{ExtComCS}[\text{ssid}_i, j] \neq \mathbf{w}_j$, but the opening checks for cmt_j passes.

$\text{Signer}_2(i, \text{st}_1^{(S,i)}, \text{pm}_1^U)$

```

    // identical to lines 1 to line 6 in  $\mathbf{G}_3$ 
7 :  $(\mathbf{r}_i, \mathbf{p}_i) \leftarrow \chi_{\text{rnd}}^2, \mathbf{w}_i := \mathbf{a}\mathbf{r}_i + \mathbf{p}_i$ 
8 : if  $\forall j \in \text{sCS}, \exists \mathbf{w}_j, Q_{\text{Hcom}}[j, \mathbf{w}_j] = \text{cmt}_j$ 
     $\wedge \text{UnOpnHS}[\text{ssid}_i] = \{i\}$  then
9 :   for  $j \in \text{sCS}$  do  $\text{ExtComCS}[\text{ssid}_i, j] \leftarrow \mathbf{w}_j$ 
10 :  for  $j \in \text{sHS} \setminus \{i\}$  do choose  $\mathbf{w}_j$  s.t.  $Q_{\text{Hcom}}[j, \mathbf{w}_j] = \text{cmt}_j$ 
11 :   $\mathbf{w} := \sum_{j \in \text{SS}} \mathbf{w}_j$ 
12 :  if  $Q_{\text{Hx}}[\text{vk}, \mathbf{w}] \neq \perp$  then Abort the game
13 :  $\text{SumRnd}[\text{ssid}_i] \leftarrow \text{SumRnd}[\text{ssid}_i] + \mathbf{r}_i$ 
14 : if  $Q_{\text{Hcom}}[i, \mathbf{w}_i] \neq \perp$  then
15 :   Abort the game
16 :  $Q_{\text{Hcom}}[i, \mathbf{w}_i] := \text{cmt}_i$ 
17 :  $\text{UnOpnHS}[\text{ssid}_i] \leftarrow \text{UnOpnHS}[\text{ssid}_i] \setminus \{i\}$ 
18 :  $\text{st}_2^{(S,i)} \leftarrow (\text{ssid}_i, \mathbf{w}_i, \mathbf{r}_i)$ 
19 : return  $(\text{st}_2^{(S,i)}, \text{pm}_2^{(S,i)} := \mathbf{w}_i, \text{ssid}_i, \text{SS}_i)$ 

```

$\text{Signer}_3(i, \text{st}_2^{(S,i)}, \text{pm}_2^U)$

```

    // identical to lines 1 to line 6 in  $\mathbf{G}_3$ 
7 : if  $\exists j \in \text{sCS}, \text{ExtComCS}[\text{ssid}_i, j] \neq \mathbf{w}_j$ 
8 :   Abort the game
9 :  $\text{sUS} \leftarrow \text{UnSgnHS}[\text{ssid}_i]$ 
10 : if  $\text{sUS} \neq \{i\}$  then
11 :    $\mathbf{z}_{2,i} \leftarrow \mathcal{R}_q$ 
12 : else
13 :   for  $j \in \text{sCS}$  do
14 :      $\Delta_j := \text{ZeroShare}(\vec{\text{seed}_i[\text{SS}]}, \text{ssid}_i)$ 
15 :      $\mathbf{z}_{2,i} := \mathbf{c}_1 \cdot \mathbf{s} - \sum_{j \in \text{sCS}} (\mathbf{c}_1 \cdot L_{\text{SS},j} \cdot \mathbf{s}_j - \Delta_j)$ 
     $+ \text{SumRnd}[\text{ssid}_i] - \sum_{j \in \text{sHS} \setminus \{i\}} \text{MaskedRsp}[\text{ssid}_i, j]$ 
16 :  $\text{MaskedRsp}[\text{ssid}_i, i] \leftarrow \mathbf{z}_{2,i}$ 
17 :  $\text{UnSgnHS}[\text{ssid}_i] \leftarrow \text{UnSgnHS}[\text{ssid}_i] \setminus \{i\}$ 
18 :  $\text{pm}_3^{(S,i)} := \mathbf{z}_{2,i}, \text{aux}_{\text{sig}} := \mathbf{w}$ 
19 : return  $(\text{pm}_3^{(S,i)}, \text{aux}_{\text{sig}})$ 

```

Let us upper bound the abort probability. An abort occurs if $\mathcal{A}$ finds a H_{com} preimage $(j, \mathbf{w}_j)$ for some a-priori fixed cmt_j . As cmt is a freshly sampled 2λ bitstring for every fresh H_{com} query, the abort probability is upper bounded by $Q_{\text{Hcom}}/2^{2\lambda}$, *e.g.*, via a union bound.

Further, as $\mathbf{w}_i$ is sampled fresh in Signer_2 and is computationally uniform over $\mathcal{R}_q$ under RLWE, the probability that Q_{Hx} was previously queried on input $(\text{vk}, \mathbf{w} = \sum_{j \in \text{SS}} \mathbf{w}_j)$ is upper bounded by

$\text{Adv}_{\mathcal{B}', q, \chi_{\text{rnd}}}^{\text{RLWE-}\times}(\lambda) + q^{-n}$, where $\mathcal{B}'$ is an appropriate adversary. (This argument can be formalized as in $\mathbf{G}_1$ and we omit details.) Again, a union bound yields in total

$$|\Pr[\mathbf{G}_7] - \Pr[\mathbf{G}_6]| \leq \frac{Q_{\text{H}_{\text{com}}}}{2^{2\lambda}} + Q_{\text{H}_{\text{x}}} Q_{\text{S}} (\text{Adv}_{\mathcal{B}', q, \chi_{\text{rnd}}}^{\text{RLWE-}\times}(\lambda) + q^{-n}).$$

Reduction from OMUF of BS_B . We are ready to describe our reduction $\bar{B}$ that plays the OMUF game of BS_B and internally runs $\mathcal{A}$. Our goal is to show that if $\mathcal{A}$ is successful in $\mathbf{G}_7$, then $\bar{B}$ wins its OMUF game. The reduction $\bar{B}$ proceeds as follows.

Instead of sampling $\mathbf{A}$ as in $\mathbf{G}_7$, the reduction $\bar{B}$ obtains $\mathbf{A}$ from the OMUF game and sets $\text{vk} = \mathbf{A}$. Furthermore, it sets $\mathbf{s}_i \leftarrow \mathcal{R}_q$ for $i \in \text{sCS}$. Here, we remark that $\mathbf{s}_i$ for $i \in \text{HS}$ is not required to simulate $\mathbf{G}_7$. Furthermore, when the last honest signer opens its commitment cmt_i in Signer_2 and all malicious cmt_j for $j \in \text{sCS}$ have a H_{com} preimage, then reduction $\bar{B}$ sets $\mathbf{t}^* = \mathbf{t}' - \sum_{j \in \text{SS} \setminus \{i\}} \mathbf{w}_j$ and queries $\mathbf{t}^*$ to the signer oracle $\mathcal{O}^{\text{Signer}}$ provided by the OMUF game. It obtains response $(\mathbf{x}^*, \mathbf{z}_2^*, \mathbf{c}_1^*, \tilde{\mathbf{e}}^*)$ and $\text{aux}_{\text{sig}} = \mathbf{w}_i$. (Note that the commitment randomness of $\mathbf{w}^*$ is not known to $\bar{B}$ and therefore not added to $\text{SumRnd}[\text{ssid}_i]$.) The reduction then programs $\text{H}_{\text{com}}(\text{vk}, \mathbf{w}) = (\mathbf{x}^*, \tilde{\mathbf{e}}^*)$ and stores $\mathbf{z}_2^*$ in a table $\text{SimRsp}[\text{ssid}_i] = \mathbf{z}_2^*$ to later simulate Signer_3 . In Signer_3 , the reduction employs $\mathbf{z}_2^*$ to simulate the last signer's response $\mathbf{z}_{2,i}$ as in line 16.

KeyGen(N, T)

```

1 : Obtain  $\text{vk} = \mathbf{A}$  from OMUF game
2 : for  $(i, j) \in [N]^2$  do
3 :    $\text{rand}_{i,j} \leftarrow \{0, 1\}^\lambda$ ;  $\text{seed}_{i,j} := i \| j \| \text{rand}_{i,j}$ 
4 :  $(\vec{\text{seed}}_i)_{i \in [N]} := \left( (\text{seed}_{i,j}, \text{seed}_{j,i})_{j \in [N]} \right)_{i \in [N]}$ 
5 : for  $i \in \text{sCS}$  do  $\mathbf{s}_i \leftarrow \mathcal{R}_q$ 
6 :  $\text{vk} := \mathbf{A}, \text{aux} := \perp$ 
7 :  $(\text{sk}_i)_{i \in \text{sCS}} := (\llbracket \mathbf{s} \rrbracket_i, \vec{\text{seed}}_i)_{i \in \text{sCS}}$ 
```

$\text{Signer}_2(i, \text{st}_1^{(S,i)}, \text{pm}_1^U)$

```

    // identical to lines 1 to line 6 in  $\mathbf{G}_3$ 
7 : if  $\forall j \in \text{sCS}, \exists \mathbf{w}_j, \mathbf{Q}_{\text{Hcom}}[j, \mathbf{w}_j] = \text{cmt}_j$ 
8 :    $\wedge \text{UnOpnHS}[\text{ssid}_i] = \{i\}$  then
9 :   for  $j \in \text{sCS}$  do  $\text{ExtComCS}[\text{ssid}_i, j] \leftarrow \mathbf{w}_j$ 
10 :   $\mathbf{t}^* := \mathbf{t}' - \sum_{j \in \text{SS} \setminus \{i\}} \mathbf{w}_j$ 
11 :   $(\text{pm}^S, \text{aux}_{\text{sig}}) \leftarrow \mathcal{O}^{\text{Signer}}(\mathbf{t}^*)$ 
12 :  parse  $(\mathbf{x}^*, \mathbf{z}_2^*, \mathbf{c}_1^*, \tilde{\mathbf{e}}^*) \leftarrow \text{pm}^S$ 
13 :   $\mathbf{w}_i := \text{aux}_{\text{sig}}$ 
14 :  for  $j \in \text{sHS} \setminus \{i\}$  do choose  $\mathbf{w}_j$  s.t.  $\mathbf{Q}_{\text{Hcom}}[j, \mathbf{w}_j] = \text{cmt}_j$ 
15 :   $\mathbf{w} := \sum_{j \in \text{SS}} \mathbf{w}_j$ 
16 :  if  $\mathbf{Q}_{\text{Hx}}[\text{vk}, \mathbf{w}] \neq \perp$  then Abort the game
17 :   $\mathbf{Q}_{\text{Hx}}[\text{vk}, \mathbf{w}] \leftarrow (\mathbf{x}^*, \tilde{\mathbf{e}}^*)$ 
18 :   $\text{SimRsp}[\text{ssid}_i] \leftarrow \mathbf{z}_2$ 
19 : else
20 :   $(\mathbf{r}_i, \mathbf{p}_i) \leftarrow \chi_{\text{rnd}}^2, \mathbf{w}_i := \mathbf{a}\mathbf{r}_i + \mathbf{p}_i$ 
21 :   $\text{SumRnd}[\text{ssid}_i] \leftarrow \text{SumRnd}[\text{ssid}_i] + \mathbf{r}_i$ 
22 :  if  $\mathbf{Q}_{\text{Hcom}}[i, \mathbf{w}_i] \neq \perp$  then
23 :    Abort the game
24 :   $\mathbf{Q}_{\text{Hcom}}[i, \mathbf{w}_i] := \text{cmt}_i$ 
25 :   $\text{UnOpnHS}[\text{ssid}_i] \leftarrow \text{UnOpnHS}[\text{ssid}_i] \setminus \{i\}$ 
26 :   $\text{st}_2^{(S,i)} \leftarrow (\text{ssid}_i, \mathbf{w}_i, \mathbf{r}_i)$ 
27 :  return  $(\text{st}_2^{(S,i)}, \text{pm}_2^{(S,i)} := \mathbf{w}_i, \text{ssid}_i, \text{SS}_i)$ 

```

$\text{Signer}_3(i, \text{st}_2^{(S,i)}, \text{pm}_2^U)$

```

    // identical to lines 1 to line 6 in  $\mathbf{G}_3$ 
7 : if  $\exists j \in \text{sCS}, \text{ExtComCS}[\text{ssid}_i, j] \neq \mathbf{w}_j$ 
8 :   Abort the game
9 :  $\text{sUS} \leftarrow \text{UnSgnHS}[\text{ssid}_i]$ 
10 : if  $\text{sUS} \neq \{i\}$  then
11 :    $\mathbf{z}_{2,i} \leftarrow \mathcal{R}_q$ 
12 : else
13 :    $\mathbf{z}_2 \leftarrow \text{SimResp}[\text{ssid}_i]$ 
14 :   for  $j \in \text{sCS}$  do
15 :      $\Delta_j := \text{ZeroShare}(\vec{\text{seed}}_i[\text{SS}], \text{ssid}_i)$ 
16 :      $\mathbf{z}_{2,i} := \mathbf{z}_2 - \sum_{j \in \text{sCS}} (\mathbf{c}_1 \cdot L_{\text{SS},j} \cdot \mathbf{s}_j - \Delta_j)$ 
         $+ \text{SumRnd}[\text{ssid}_i] - \sum_{j \in \text{sHS} \setminus \{i\}} \text{MaskedRsp}[\text{ssid}_i, j]$ 
17 :    $\text{MaskedRsp}[\text{ssid}_i, i] \leftarrow \mathbf{z}_{2,i}$ 
18 :    $\text{UnSgnHS}[\text{ssid}_i] \leftarrow \text{UnSgnHS}[\text{ssid}_i] \setminus \{i\}$ 
19 :    $\text{pm}_3^{(S,i)} := \mathbf{z}_{2,i}, \text{aux}_{\text{sig}} := \mathbf{w}$ 
20 : return  $(\text{pm}_3^{(S,i)}, \text{aux}_{\text{sig}})$ 

```

Let us show that $\bar{\mathbf{B}}$ and $\mathbf{G}_7$ are distributed identically. Observe that $\bar{\mathbf{B}}$'s output only differs from $\mathbf{G}_7$ in the output of Signer_2 and Signer_3 for the last signer with respect to ssid_i by construction¹⁰.

Signer₂ : Observe that the distribution of $\text{ssid}_i, \text{SS}_i$ remains unchanged. When the last honest signer opens its commitments cmt_i in Signer_2 with respect to ssid_i , *i.e.*, $\text{UnOpnHS}[\text{ssid}_i] = \{i\}$, then the reduction sets $\mathbf{w}_i = \text{aux}_{\text{sig}}$. As $\text{aux}_{\text{sig}} = \mathbf{a}\mathbf{r}^* + \mathbf{p}^*$ with $(\mathbf{r}^*, \mathbf{p}^*) \leftarrow \chi_{\text{rnd}}^2$, the distribution of $\mathbf{w}_i$ is unchanged.

Note that $\mathbf{r}^*$ is not added to $\text{SumRnd}[\text{ssid}_i]$ yet and we have $\text{SumCom}[\text{ssid}_i] = \sum_{j \in \text{sHS} \setminus \{i\}} \mathbf{r}_i$. Therefore, $\text{SumCom}[\text{ssid}_i] + \mathbf{r}^*$ in $\bar{\mathbf{B}}$ is identical to $\text{SumCom}[\text{ssid}_i]$ in $\mathbf{G}_7$.

Signer₃ : For the last honest signer $i' \in \text{sHS}$ in Signer_3 , the reduction employs $\mathbf{z}_2^* = \mathbf{c}_1^* \mathbf{s} + \mathbf{r}^*$ to simulate $\mathbf{z}_{2,i}$ as in line 16. Observe that the $\bar{\mathbf{B}}$'s output $\mathbf{z}_{2,i}$ is identical to output $\mathbf{z}_{2,i}$ in $\mathbf{G}_7$ if $\mathbf{c}_1^* = \mathbf{c}_1$.

We have $(\mathbf{c}_1, \mathbf{c}_2) = \text{Decomp}_\beta(\mathbf{t} - \mathbf{w})$. On the other hand, we have $(\mathbf{c}_1^*, \mathbf{c}_2^*) = \text{Decomp}_\beta(\mathbf{t}^* + \mathbf{A}\mathbf{x}^* + \tilde{\mathbf{e}}^* - \text{aux}_{\text{sig}})$. Note that due to the abort condition in $\mathbf{G}_7$, the malicious commitments $\mathbf{w}_j$ for $j \in \text{sCS}$ are identical in Signer_2 and Signer_3 , thus we have $(\mathbf{x}, \tilde{\mathbf{e}}) = (\mathbf{x}^*, \tilde{\mathbf{e}}^*)$. Also, $\mathbf{w}_j$ are identical in Signer_2 and Signer_3 as H_{com} is collision-free due to the abort condition introduced in $\mathbf{G}_2$. Furthermore, as Decomp_β is deterministic¹¹ and

$$\begin{aligned}
& \mathbf{t}^* + \mathbf{A}\mathbf{x}^* + \tilde{\mathbf{e}}^* - \text{aux}_{\text{sig}} \\
&= (\mathbf{t}' - \sum_{j \in \text{SS} \setminus \{i\}} \mathbf{w}_j) + \mathbf{A}\mathbf{x}^* + \tilde{\mathbf{e}}^* - \mathbf{w}_i \\
&= \mathbf{t}' + \mathbf{A}\mathbf{x} + \tilde{\mathbf{e}} - \mathbf{w} \\
&= \mathbf{t} - \mathbf{w},
\end{aligned}$$

we have that $\mathbf{c}_1 = \mathbf{c}_1^*$ as desired.

¹⁰Note that the last signer in Signer_2 and Signer_3 can be distinct but our reduction deals with this naturally as explained below.

¹¹Recall that we implicitly assume that H_x also fixes the randomness for Decomp_β .

Bounding $\bar{\mathcal{B}}$'s advantage. If $\bar{\mathcal{B}}$ did not abort and $\mathcal{A}$ wins the game, then $\mathcal{A}$ outputs $Q_S + 1$ valid message-signature pairs $((\mu_i, \text{sig}_i))_{i \in [Q_S+1]}$ with distinct messages. Also, $\mathcal{B}$ invokes the oracle $\mathcal{O}^{\text{Signer}}(\cdot)$ if and only if Signer_2 is called for each honest signer $i \in \text{sHS}$ in SS with session identifier $\text{ssid} = (\mathbf{t}', \text{SS}, (\text{cmt}_j)_{j \in \text{SS}}, \text{ctx})$. Therefore, $\mathcal{O}^{\text{Signer}}(\cdot)$ is invoked at most Q_S times. Note that verification of TBS_B and BS_B are identical, therefore we obtain

$$\Pr[\mathbf{G}_7] \leq \text{Adv}_{\text{BS}_B, \mathcal{B}}^{\text{bs-omuf}}.$$

□

E Security Analysis under Randomized One-more ISIS

We provide a security analysis for our variants $\text{BS}_B^\$$ and $\text{TBS}_B^\$$ from Sections 4 and 5, respectively.

E.1 Blind Signature $\text{BS}_B^\$$

Observe that with the addition of $\tilde{\mathbf{y}}$, the user obtains a valid witness the relation $\mathcal{R}$. Correctness and blindness follow as in the security analysis for BS_B (cf. Section C). Let us discuss how to adapt the proof of one-more unforgeability.

Games $\mathbf{G}_0$ to $\mathbf{G}_3$: We perform the same adaptations, *i.e.*, setup Π in extraction mode, embed trapdoor $\mathbf{T}_d$ into $\mathbf{A}$ and extract randomness $\tilde{\mathbf{x}}$ from $\mathbf{x}$.

Game $\mathbf{G}_4$: The game proceeds as before, except that the game computes the target $\bar{\mathbf{t}}$ for preimage sampling via $\bar{\mathbf{t}} := [1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z} - \mathbf{t}' - \mathbf{A} \cdot \tilde{\mathbf{x}}' - \mathbf{B}\tilde{\mathbf{y}}$. This accounts for the additional term $\mathbf{B}\tilde{\mathbf{y}}$ in the definition of $\mathbf{t}$.

Game $\mathbf{G}_5$ to $\mathbf{G}_6$: We perform the same adaptations, *i.e.*, simulate the Plover preimage via Hint-MLWE and extract the witness from the proofs in $\mathcal{A}$'s forgeries.

It can be shown as before that $|\Pr[\mathbf{G}_0] - \Pr[\mathbf{G}_6]| = \text{negl}(\lambda)$ with appropriate parameters. Then, the reduction to randomized OM-ISIS proceeds as before, except that

- during setup, it embeds the matrix $\mathbf{B}$ obtained from the rOM-ISIS challenger into the parameters pp ; and
- during signing, it sets $\bar{\mathbf{t}} := [1 \ \mathbf{a} \ \mathbf{b}]\mathbf{z} - \mathbf{t}' - \mathbf{A} \cdot \tilde{\mathbf{x}}'$ and sends $\bar{\mathbf{t}} - \tilde{\mathbf{e}}$ to the rOM-ISIS challenger, and obtains $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$.

After extracting the witnesses from $\mathcal{A}$'s forgeries, the game outputs $\perp$. Note that as

$$\mathbf{A}\tilde{\mathbf{x}} + \mathbf{B}\tilde{\mathbf{y}} = \bar{\mathbf{t}} - \tilde{\mathbf{e}},$$

where $\tilde{\mathbf{y}} \leftarrow \{\pm 1\}^{4n}$ and $\tilde{\mathbf{x}} \leftarrow \mathcal{D}_{\mathcal{R}_q, \sqrt{\sigma_{\mathbf{s}, \mathbf{x}}^2 - \sigma_{\mathbf{s}'}^2}}^4$ is chosen conditioned on the above equation, we can show as before that both games have a negligible statistical distance. By definition of the relation for Π , the extracted witnesses are valid solution for rOM-ISIS which concludes the proof.

E.2 Blind Signature $\text{TBS}_B^\$$

Similar to $\text{BS}_B^\$$, it is easy to see that correctness and blindness follow as in the analysis for TBS_B (cf. Section D). Also, one-more unforgeability follows without much trouble. As in the proof for TBS_B , we can move towards a game where the signing oracles can be simulated given a valid $\text{BS}_B^\$$ signature. For the reduction to OMUF of $\text{BS}_B^\$$, we embed $\mathbf{B}$ into pp and when simulating signing oracles, we embed $\tilde{\mathbf{y}}$ from the $\text{BS}_B^\$$ signatures into H_x (in addition to $\mathbf{x}, \tilde{\mathbf{e}}$) as in the proof of TBS_B . The adaptations are straightforward and we omit details.

F Additional results on lattices

In this section, we prove the generalized Theorem 3.1 from [SS11] to compose q . The core observation is that, in [SS11], only Theorem 3.2 critically relies on q being prime to lower bound the norm of the smallest vector of a lattice related to $\mathbf{a}$.

F.1 Notations

Before proving our result, we introduce additional lattice notations.

We denote $U(S)$ the uniform distribution over a finite set S .

The minimum $\lambda_1(\Lambda)$ (resp $\lambda_1^\infty(\Lambda)$) is the Euclidean (resp. infinity) norm of any shortest non-zero vector of Λ . More generally, we define the k -th successive minimum $\lambda_k(\Lambda)$ for any $k \leq n$ as the smallest r such that Λ contains at least k linearly independent vectors of norm $\leq r$.

Lemma F.1 ([Pei07, Lemma 3.5]). *For any full-rank lattice $\Lambda \subset \mathbb{R}^n$ and $\delta \in (0, 1)$, $\eta'_\delta(\Lambda) \leq \frac{1}{\sqrt{2\pi}} \sqrt{\frac{\ln(2n(1+1/\delta))}{\pi}}$. $\min(\lambda_n(\Lambda), 1/\lambda_1^\infty(\hat{\Lambda}))$.*

Lemma F.2 ([GPV08, Corollary 2.8]). *Let $\Lambda \subseteq \Lambda' \subset \mathbb{R}^n$ be two full-rank lattices. For any $\mathbf{c} \in \mathbb{R}^n$, $\delta \in (0, 1/2)$ and $\sigma \geq \eta'_\delta(\Lambda')$, we have $\Delta(\mathcal{D}_{\Lambda-\mathbf{c}, \sigma} + \mathbf{c} \bmod \Lambda', U(\Lambda/\Lambda')) \leq 2\delta$.*

Module lattices. Recall that in this work we are in particular interested in module lattices of the form $\mathcal{R} = \mathbb{Z}_q[x]/(x^n + 1)$, with n a power-of-two.

Let $\zeta \in \mathbb{C}$ be a $2n$ -th root of unity. For $\mathbf{a} \in \mathcal{R}$, we define its algebraic norm as $\mathcal{N}(\mathbf{a}) = \prod_{i=1}^n |\mathbf{a}(\zeta^{2i+1})|$. For the specific cyclotomic fields we are considering, the algebraic norm relates to the Euclidean norm of $\mathbf{a}$: $\|\mathbf{a}\|_2 \geq \mathcal{N}(\mathbf{a})^{1/n}$, see for instance [SS11, Section 2.2]. Additionally, we have $\mathcal{N}(\mathbf{a}) = \det(\mathbf{a} \cdot \mathcal{R})$.

We also define the module

$$\Lambda(\mathbf{a}) := \{(t_1, \dots, t_m) \in \mathcal{R}^m \mid \exists s \in \mathcal{R}, \forall i \in [m], t_i = a_i s \bmod q\}$$

For $\mathbf{a} \in \mathcal{R}$, we denote $a^*(x) := a(x^{-1}) \in \mathcal{R}$, where $x^{-1} = -x^{n-1}$.

F.2 Proof of regularity lemma for composite q

We first adapt intermediate lemmas 3.1 and 3.2 in [SS11] for a composite q .

Lemma F.3 (Adaptation of [SS11, Lemma 3.1]). *Let $\mathbf{a} \in \mathcal{R}^m$. Let $\mathbf{a}^*$ be defined by $a_i^* = a_i(x^{-1})$ for all $i \in [m]$. Then,*

$$\widehat{\mathbf{a}^\perp} = \frac{1}{q} \Lambda(\mathbf{a}^*)$$

We now show that for a uniformly chosen $\mathbf{a} \in (\mathcal{R}_q^\times)^m$, the lattice $\Lambda(\mathbf{a})$ is extremely unlikely to contain unusually short vectors for the infinity norm, i.e., much shorter than the Minkowski upper bound $\det(\Lambda(\mathbf{a}))^{1/(mn)} = 1$ on $\lambda_1^\infty(\Lambda(\mathbf{a}))$. Our bound matches the Minkowski bound up to a factor $\frac{1}{\sqrt{n}} q^{-\varepsilon}$ for an arbitrarily small constant ε .

Lemma F.4 (Generalization of [SS11, Lemma 3.2] to q composite). *Let $n \geq 8$ a power-of-two, and $q = q_1 \cdot \dots \cdot q_\ell$, where $q_1, \dots, q_\ell$ are distinct primes ≥ 5 .*

$$\lambda_1^\infty(\Lambda(\mathbf{a})) \geq \frac{1}{\sqrt{n}} \cdot q^{1 - \frac{1}{m} - \varepsilon}$$

except with probability $\leq 2^{\ell n} \cdot (5/4)^{nm\ell} \cdot q^{-nm\varepsilon}$ over the uniformly random choice of $\mathbf{a} \in \mathcal{R}_q^\times$.

Proof. The proof is similar to the one in [SS11]. We bound the probability p that $\Lambda(\mathbf{a})$ contains a non-zero vector of infinite norm $< B$ by using the union bound over the probabilities $p(\mathbf{t}, s) = \Pr_{\mathbf{a}}[\forall i, t_i = a_i s_i]$ over all possible values for $\mathbf{t}$ of infinite norm $< B$ and $s \in \mathcal{R}_q$. Since all the a_i 's are independent, we have $p(\mathbf{t}, s) = \prod_{i=1}^m p_i(t_i, s)$, where $p_i(t_i, s) = \Pr_{a_i}[t_i = a_i s]$.

Without loss of generality, we can assume that for any $j \in [\ell]$ we have $g_j := \gcd(s \bmod q_j, x^n + 1 \bmod q_j) = \gcd(t_i \bmod q_j, x^n + 1 \bmod q_j)$. Indeed, if this is not the case, there exists $i \in [\ell]$ and a polynomial $g \in \mathcal{R}_{q_i}$ such that $g \mid t_i \bmod q_i$ but $g \nmid s \bmod q_i$, or $g \nmid t_i \bmod q_i$ but $g \mid s \bmod q_i$. As $a_i \bmod \mathcal{R}_{q_i}^\times$, the same divisibility holds for $a_i \cdot s \bmod q_i$ and $t_i \neq a_i s \bmod q_i$. Consequently, $t_i \neq a_i s$, i.e. $p_i(t_i, s) = 0$.

Now let's count the number of solutions to $t_i = a_i s$. We recall that $x^n + 1$ splits into distinct irreducible factors of same degree d_j modulus a prime, we thus will consider each factor of $x^n + 1 \bmod p_i$ individually. Fix $j \in [\ell]$ and an irreducible factor ϕ of g_j . We have $(t_i \bmod q_j) \bmod \phi = (a_i s \bmod q_j) \bmod \phi$ regardless of the value of $a_i \bmod q_j$. This implies that $(a_i \bmod q_j) \bmod \phi$ can be arbitrary in $\mathbb{F}_{q_j}^\times$. For remaining irreducible ϕ' dividing g_j , we have $(s \bmod q_j) \neq 0 \bmod \phi'$ and there exists a unique $(a_i \bmod q_j) \bmod \phi'$ such that $(t_i \bmod q_j) = (a_i s \bmod q_j) \bmod \phi'$. Overall, there are $(q_j^{d_j} - 1)^{\deg(g_j)/d_j}$ distinct $a_i \bmod q_j$ such that $t_i = (a_i \bmod q_j)s \bmod q_j$. By the Chinese Remainder theorem, we extend these results to q : $p_i(t_i, s) = \prod_{j=1}^\ell (q_j^{d_j} - 1)^{(\deg(g_j) - n)/d_j - n}$.

We can then bound p by considering the different factors g_j which are possible:

$$p \leq \sum_{\substack{g_1, \dots, g_\ell \in \mathcal{R} \\ \forall j, g_j | x^n + 1 \bmod q_j \\ \forall j, g_j \text{ monic}}} \sum_{\substack{\mathbf{s} \in \mathcal{R}_q \\ \forall j, g_j | s}} \sum_{\substack{\mathbf{t} \in \mathcal{R}_q^m \\ \forall i, \|\mathbf{t}_i\|_\infty < B \\ \forall j, i, g_j | t_i \bmod q_j}} \prod_{j=1}^\ell (q_j^{d_j} - 1)^{(\deg(g_j) - n)/d_j}$$

We will evaluate the number of terms in the above sum depending on the number of irreducible polynomials in $(g_j)_{j \in [\ell]}$, that we note $(k_j = \deg(g_j)/d_j)_{j \in [\ell]}$. Let $N(B, (g_j)_{j \in [\ell]})$ denote the number of $t \in \mathcal{R}_q$ such that $\|t\|_\infty \leq B$ and $\forall j \in [\ell], t = g_j \cdot t_j \bmod q_j$ for some $t_j \in \mathcal{R}_{q_j}$ of degree $< n - \deg(g_j)$.

In order to bound $N(B, (g_j)_{j \in [\ell]})$, observe that it corresponds to the number of points of the lattice $L := \sum_{j \in [\ell]} \frac{q}{q_j} (g_j \cdot \mathcal{R}_{q_j} + q_j \mathbb{Z}^n)$ that intersects with the hypercube $C(2B)$ of sidelength $2B$, where a hypercube of sidelength S is defined by $C(S) = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\|_\infty < S/2\}$. Let $\lambda := \lambda_1^\infty(L)$. As leveraged in [SS11, Lemma 3.2], we can bound $N(B, (g_j)_{j \in [\ell]}) \leq \frac{\text{vol}(C(2B+\lambda))}{\text{vol}(C(\lambda))} = (\frac{2B}{\lambda} + 1)^n$. This is because if we center an hypercube of sidelength λ on each of the $N(B, (g_j)_{j \in [\ell]})$ points of L , they do not intersect, and yet are all contained in $C(2B + \lambda)$.

To derive a lower bound on λ , consider $t \in L$. We have $\mathcal{N}(t) = \det(\langle t \rangle) \geq \det(L) = \prod_{j \in [\ell]} q_j^{\deg(g_j)}$ because $\langle t \rangle$ is a sub-ideal of L . It follows from the arithmetic-geometric inequality that $\|t\|_2 \geq \mathcal{N}(t)^{1/n} \geq \prod_{j \in [\ell]} q_j^{\deg(g_j)/n}$. By equivalence of norm, we conclude $\|t\|_\infty \geq \lambda \geq \frac{1}{\sqrt{n}} \prod_{j \in [\ell]} q_j^{\deg(g_j)/n}$. Hence, using $B = \frac{1}{\sqrt{n}} q^\beta$, if $B \leq \frac{1}{\sqrt{n}} \prod_{j \in [\ell]} q_j^{-\deg(g_j)/n}$, we have $B \leq \lambda$ and $N(B, (g_j)_{j \in [\ell]}) = 0$. Otherwise $N(B, (g_j)_{j \in [\ell]}) \leq (\frac{2B}{\lambda} + 1)^n \leq (2q^\beta \prod_{j \in [\ell]} q_j^{-\deg(g_j)/n} + 1)^n \leq 3^n q^{n\beta} \prod_{j \in [\ell]} q_j^{-\deg(g_j)}$.

Using the fact that each g_j is a product of irreducibles of $x^n + 1 \bmod q_j$, we deduce that there are at most $2^{n/d_j}$ choices for it. Also, given g_j there are at most $q_j^{n - \deg(g_j)}$ possible s such that $g_j \mid s \bmod q_j$. This implies on p :

$$\begin{aligned} p &\leq 2^{\sum_{j \in \ell} n/d_j} \cdot \max_{(g_j)_j} \prod_{j \in [\ell]} q_j^{n - \deg(g_j)} \cdot \frac{N(B, (g_j)_{j \in [\ell]})^m}{\prod_{j \in [\ell]} (q_j^{d_j} - 1)^{m(n - \deg(g_j))/d_j}} \\ &\leq 2^{\ell n} \cdot \max_{(g_j)_j} \prod_{j \in [\ell]} q_j^{n - \deg(g_j)} \cdot \frac{N(B, (g_j)_{j \in [\ell]})^m}{\prod_{j \in [\ell]} (\frac{4}{5} q_j^{d_j})^{m(n - \deg(g_j))/d_j}} \\ &\leq 2^{\ell n} \cdot (5/4)^{nm\ell} \cdot \max_{(g_j)_j} \frac{N(B, (g_j)_{j \in [\ell]})^m}{\prod_{j \in [\ell]} q_j^{(m-1)(n - \deg(g_j))}} \end{aligned}$$

Injecting the bound on $N(B, (g_j)_{j \in [\ell]})$, we get

$$\begin{aligned} p &\leq 2^{\ell n} \cdot (5/4)^{nm\ell} \cdot \max_{(g_j)_j} \frac{3^{nm} q^{nm\beta} \prod_{j \in [\ell]} q_j^{-m \deg(g_j)}}{\prod_{j \in [\ell]} q_j^{(m-1)(n - \deg(g_j))}} \\ &= 2^{\ell n} \cdot (5/4)^{nm\ell} \cdot 3^{nm} \cdot \max_{(g_j)_j} q^{nm\beta} \cdot \prod_{j \in [\ell]} q_j^{(m-1)(\deg(g_j) - n) - m \deg(g_j)} \end{aligned}$$

The above max is minimized for $\deg(g_j) = 0$ and we deduce: $p \leq 2^{\ell n} \cdot (5/4)^{nm\ell} \cdot q^{nm\beta} \cdot \prod_{j \in [\ell]} q_j^{-(m-1)n} \leq 2^{\ell n} \cdot (5/4)^{nm\ell} \cdot q^{nm\beta - (m-1)n}$.

Setting $\beta = 1 - \frac{1}{m} - \varepsilon$, the above power of q becomes $q^{-nm\varepsilon}$, which concludes this proof. $\square$

We finally move on to the proof of Theorem 3.1, which follows the one written in [SS11, Theorem 3.1].

Proof. For each $\mathbf{a} \in (\mathcal{R}_q^\times)^m$, let $D_{\mathbf{a}}$ denote the distribution of $\sum_{i \in [m]} t_i a_i$, where $\mathbf{t}$ is distributed from $\mathcal{D}_{\mathbb{Z}^{nm}, \sigma}$. Note that the statistical distance in Theorem 3.1 is exactly $\sum_{\mathbf{a} \in (\mathcal{R}_q^\times)^m} \Delta_{\mathbf{a}}$, where $\Delta_{\mathbf{a}}$ is the distance to uniformity of $D_{\mathbf{a}}$. To prove the theorem, it thus suffices to prove a bound $\Delta_{\mathbf{a}} \leq 2\delta$ for all but a proportion $\delta' = 2^{\ell n} (5/4)^{nm\ell} q^{-nm\varepsilon}$.

Now the mapping $\mathbf{t} \mapsto \sum_{i \in [m]} t_i a_i$ induces an isomorphism from the quotient group $\mathbb{Z}^{nm}/\mathbf{a}^\perp$ to its range, the latter being $\mathcal{R}_q$ thanks to the invertibility of the a_i 's. Therefore the statistical $\Delta_{\mathbf{a}}$ is equal to the distance to uniformity of $\mathbf{t} \bmod \mathbf{a}^\perp$.

By Theorem F.1, we have $\Delta_{\mathbf{a}} \leq 2\delta$ if σ is greater than the scaled smoothing parameter $\eta'_\delta(\mathbf{a}^\perp) \subseteq \mathbb{Z}^{nm}$. To bound $\eta'_\delta(\mathbf{a}^\perp)$ from above, we apply Theorem F.2 which reduces to bounding λ_1^∞ for the dual lattice $\widehat{\mathbf{a}^\perp}$. By Theorem F.3, we have $\widehat{\mathbf{a}^\perp} = \frac{1}{q} \Lambda(\mathbf{a}^*)$ (where $\mathbf{a}^*$ is one-to-one correspondence with $\mathbf{a}$). Bounding λ_1^∞ for the latter lattice has been addressed in Theorem F.4.

Finally, except with probability δ' , we obtain that $\Delta_{\mathbf{a}}$ is bounded by $\frac{1}{q} \cdot \frac{1}{\sqrt{n}} q^{1 - \frac{1}{m} - \varepsilon}$, which concludes this proof. $\square$

G Hardness of Hint-RLWE under RLWE

Our parameter selection is based on a tight reduction of Hint-RLWE from RLWE. This was formalized in [KLSS23] and extended in [DKM⁺24] to probabilistic challenges. We recall their result below, and analyze the hardness of Hint-RLWE for the challenges that are used in our work.

The spectral norm $s_1(\mathbf{M})$ of a matrix $\mathbf{M}$ is defined as the value $\max_{\mathbf{x} \neq 0} \frac{\|\mathbf{M}\mathbf{x}\|_2}{\|\mathbf{x}\|_2}$. Given a polynomial $\mathbf{c} \in \mathcal{R}$, we may abusively use the term “spectral norm $s_1(\mathbf{c})$ of $\mathbf{c}$ ” when referring to the spectral norm of the anti-circulant matrix $\mathcal{M}(\mathbf{c})$ associated to $\mathbf{c}$. We extend it to matrices of elements in $\mathcal{R}$. Finally, if $\mathbf{c} = \sum_{0 \leq i < n} c_i x^i$, then the Hermitian adjoint of $\mathbf{c}$, which we denote by $\mathbf{c}^\circ$, is defined as $\mathbf{c}^\circ = c_0 - \sum_{0 \leq i < n} c_{n-i} x^i$. Note that $\mathcal{M}(\mathbf{c})^\top = \mathcal{M}(\mathbf{c}^\circ)$.

Theorem G.1 (Hardness of Hint-RLWE, as in [DKM⁺24]). *Let $\mathcal{C}$ be a distribution over $\mathcal{R}$, and let B_{hint} be a real number such that $s_1(D) \leq B_{\text{hint}}$ with overwhelming probability, where $D = \sum_Q \mathbf{c}_i \mathbf{c}_i^\circ$. Let $\sigma, \sigma_{\text{sk}}, \sigma_{\mathbf{r}} > 0$ such that $\frac{1}{\sigma^2} = 2 \left(\frac{1}{\sigma_{\text{sk}}^2} + \frac{B_{\text{hint}}}{\sigma_{\mathbf{r}}^2} \right)$. If $\sigma \geq \sqrt{2} \eta'_\epsilon(\mathbb{Z}^n)$ for $0 < \epsilon \leq 1/2$, where $\eta'_\epsilon(\mathbb{Z}^n)$ is the smoothing parameter of $\mathbb{Z}^n$, then there exists an efficient reduction from $\text{RLWE}_{q, \sigma}$ to $\text{Hint-RLWE}_{q, Q, \sigma_{\text{sk}}, \sigma_{\mathbf{r}}, \mathcal{C}}$ that reduces the advantage by at most 4ϵ .*

Now, let's compute the quantity B_{hint} for the challenge distribution used in this work. Recall that the challenges used in our unforgeability Theorem 4.3 are sampled from the distribution $\mathbf{c}_1$, with $(\mathbf{c}_1, \mathbf{c}_2) \leftarrow \text{Decomp}_\beta(\mathbb{Z}^n)$. We rely on the analysis from [EEN⁺24, Lemma 2]. Note however that their work assumed $\mathbf{E}[\mathbf{c}_1] = 0$ which wasn't true in their case as they did not randomize Decomp_β with $\varepsilon \in \{-1, 1\}$, introducing a

slight bias in the distribution of $\mathbf{c}_1$. This is corrected in this work, and their proof works as this assumption holds after randomizing Decomp_β .

Lemma G.2. *For $j \in [Q_S]$, let $\mathbf{c}^{[j]} \leftarrow \mathcal{C}$. Let $D = \sum_{j \in [Q_S]} \mathbf{c}^{[j]} (\mathbf{c}^{[j]})^\circ$. Let $M = 2 \left\lceil \frac{q-1}{2\beta} \right\rceil + 1$. We then have $\Pr[s_1(D) \geq B_{\text{hint}}] \leq 2^{-\lambda}$, where $B_{\text{hint}} = Q_S n \cdot \frac{M^2}{12} \left(1 + \frac{O(\lambda n \log n)}{\sqrt{Q_S}}\right)$. Specifically, when $Q_S = \omega(\lambda n \log n)^2$, then B_{hint} is equivalent to $Q_S n \cdot \frac{M^2}{12}$.*

Proof. For completeness, we also recall their proof.

Recall that for $\mathbf{c} \leftarrow \mathbb{Z}_q$ and $(\mathbf{c}_1, \mathbf{c}_2) := \text{Decomp}_\beta(\mathbf{c})$, we have $|\mathbf{c}_1| \leq \left\lceil \frac{q-1}{2\beta} \right\rceil$, $\mathbf{E}[\mathbf{c}_1] = 0$ and $\mathbf{E}[\mathbf{c}_1^2] \leq \frac{M^2-1}{12}$ for $M = 2 \left\lceil \frac{q-1}{2\beta} \right\rceil + 1$ (c.f. Section 3). First, let us consider a single $\mathbf{c} \leftarrow \mathcal{C}$. Let $d(x) = \mathbf{c} \mathbf{c}^\circ = \sum_{0 \leq i < n} d_i x^i$. For each $k \in \{0, \dots, n-1\}$, d_k can be expressed explicitly as:

$$d_k = \sum_{0 \leq i < n-k} c_i c_{i+k} - \sum_{n-k \leq i < n} c_i c_{i+k-n}. \quad (6)$$

From Eq. (6), it is clear that:

1. $d_0 = \|\mathbf{c}_1\|_2^2$
2. If $k \neq 0$, d_k is a random variable satisfying:

$$\|d_k\| \leq \|c\|_2^2 \quad (7)$$

$$\mathbf{E}[d_k] = 0. \quad (8)$$

While Eq. (7) is immediate from Eq. (6), Eq. (8) is a bit more subtle. The mapping $(k, i) \in \{0, \dots, n-1\}^2 \mapsto (i+k \bmod n)$ is a group action. As such, its orbits form a partition of $\{0, \dots, n-1\}$, and we note that each orbit has an even number of elements. Each orbit can be written as $\{i_0 + kx \bmod n \mid x \in n/\gcd(k, n)\}$, where i_0 is (say) the smallest element in this orbit. We define the function $\delta_k : \{0, \dots, n-1\} \rightarrow \{-1, 1\}$ as follows: for each $0 \leq i < n$, we determine its orbit, write $i = i_0 + kx$ in this orbit, and set $\delta_k(i) = (-1)^x$. Now consider the mapping $\varphi_k : \mathcal{C}_1 \rightarrow \mathcal{C}_1$ defined as

$$\varphi_k : \left(\mathbf{c} = \sum_{0 \leq i < n} c_i x^i \right) \longrightarrow \left(\mathbf{c}' = \sum_{0 \leq i < n} \delta_k(i) c_i x^i \right). \quad (9)$$

φ_k maps $c \in \text{Supp}(\mathcal{C}_1)$ to a new $c' \in \text{Supp}(\mathcal{C}_1)$ such that $\mathcal{C}_1(c) = \mathcal{C}_1(c')$, and one can check from Eqs. (6) and (9) that $(\mathbf{c} \mathbf{c}^\circ)_k = -(\mathbf{c}' \mathbf{c}'^\circ)_k$. Since φ_k is an involution over its support, this implies that $\mathbf{E}[d_k] = 0$.

Let us note $d^{[j]} = \mathbf{c}^{[j]} (\mathbf{c}^{[j]})^\circ$. For $k \neq 0$, we can bound the sum $D_k = \sum_{j \in [Q_S]} d_k^{[j]}$ by combining Hoeffding's inequality with Eqs. (7) and (8):

$$|D_k| \leq n \left\lceil \frac{q-1}{2\beta} \right\rceil^2 \sqrt{2Q_S} ((\lambda+1) \log(2) + \log(n))$$

except with probability at most $2^{-\lambda}/n$. From the union bound, the above inequality is true for all $k \neq 0$, except with probability $\leq 2^{-\lambda}$.

We can now bound the spectral norm of D . Since D is self-adjoint, $s_1(D)$ is the largest eigenvalue of D , that is $D(\zeta)$ for some primitive root of unity $\zeta, |\zeta| = 1$. By applying Hoeffding's inequality on $\sum_{Q_S} \|\mathbf{c}^{[j]}\|^2$, with probability $\geq 1 - 2^{-\lambda}$, we have:

$$s_1(D) = D(\zeta) \leq D_0 + \sum_{k \neq 0} |D_k|$$

$$\begin{aligned}
&\leq \sum_{Q_S} \|c^{[j]}\|^2 + (n-1) \max_{1 \leq k \leq n-1} |D_k| \\
&\leq Q_S n \cdot \frac{M^2}{12} + \sqrt{\lambda \cdot Q_S \cdot n} \left\lceil \frac{q-1}{2\beta} \right\rceil \\
&\quad + (n-1)n \left\lceil \frac{q-1}{2\beta} \right\rceil^2 \sqrt{2Q_S} ((\lambda+1) \log(2) + \log(n))
\end{aligned}$$

Setting $Q_S = \omega(\lambda^2 \log^2 n)$ concludes the proof. $\square$

H Extended Parameter Sets

This appendix provides detailed parameter sets in Table 4 for both instantiations of our threshold blind signature scheme. The parameters are computed based on the security analysis and concrete instantiation requirements described in Section 6.

Parameter Set Approaches. The two parameter sets represent different NIZK instantiations and security methodologies. Note that in both instantiations, we assume that LaBRADOR and LNP are online-extractable.

- **Compact (provable):** Uses LaBRADOR [BS23] combined with the Lyubashevsky-Nguyen-Plancon (LNP) zero-knowledge proof system [LNP22] as the underlying NIZK. This approach strictly follows the statistical arguments presented in our security proofs. While this results in worse underlying lattice parameters (larger moduli and standard deviations), the superior efficiency of the LaBRADOR NIZK compensates for this overhead, ultimately achieving the most compact signatures. However, implementing a zero-knowledge version of LaBRADOR remains an open research question.
- **Easy to deploy (practical):** Directly uses the LNP zero-knowledge proof system [LNP22] as implemented by the LaZer library [LSS24]. In this approach, we replace certain statistical arguments with computational ones based on RLWE hardness, similar to the prior work [AKSY22]. This enables practical deployment with reasonable performance.

Table 4: Complete parameter sets for both instantiations of our threshold blind signature. Security analysis shows Core-SVP security, a metric which ignores several polynomial factors.

Parameter	Compact (<i>provable</i>)	Easy to Deploy (<i>practical</i>)
Basic Parameters		
n	2048	2048
q	$\approx 2^{60}$	$\approx 2^{47.5}$
Q_S	2^{50}	2^{50}
$T_{\max}$	16	16
κ	128	128
$\tilde{p}$	$\approx 2^{50}$	$\approx 2^{31.5}$
β	2^{49}	2^{40}
Standard Deviations		
σ_{sk}	32768	32768
σ_{rnd}	2^{47}	$2^{37.5}$
$\sigma_{U,x}$	256	2
$\sigma_{S,x}$	$2^{49.7}$	$2^{37.5}$
Bounds		
B_{22}	$2^{58.7}$	$2^{46.3}$
Communication		
Comm/party	30.7 KB	24.6 KB
Security Analysis (Core-SVP Bits)		
Key Recovery	120	122
Forgery	87	124
Blindness	125	120